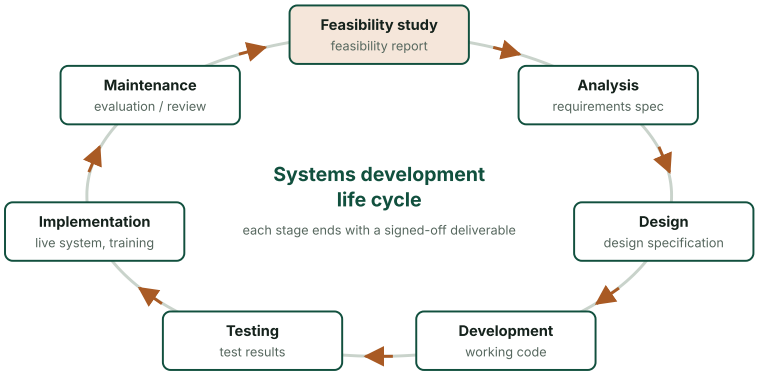
Why develop new systems? Systems, the software crisis and the life cycle

AP01 ANSWERS

I can: Identify the purpose, need and components of a system, and break a large system into sub-systems · Describe the user interface, processes and data of a given computer system and justify a suitable interface · Explain internal and external reasons for developing a new system, applied to a context · Explain the causes and consequences of the software crisis and how software engineering responded · Describe each stage of the life cycle, what it produces and why a structured approach reduces risk



Use this diagram to help you: The systems development life cycle: seven stages arranged in a cycle, each labelled with its main deliverable

A Key terms

Write the letter of the matching definition · 6 marks

E System

F Sub-system

C User interface (UI)

A Data

B Process

D Software crisis

A Raw facts and figures that have no context or meaning on their own, which are input to or stored by a system.

B An operation carried out by a computer system that manipulates data, for example calculating, sorting, searching or updating, converting data into information.

C The means by which a user interacts with a computer system, entering data into it and viewing the results of processing.

D The period from the mid-1960s when software projects were frequently late, over budget, unreliable and failed to meet user requirements because development methods could not keep pace with demand.

E A set of interdependent components that work together for a stated purpose, satisfying an identified need.

F A smaller system that forms part of a larger system and performs one part of its overall function, often exchanging data with other sub-systems.

B Fill the gaps

Use the word bank · 10 marks

analysis | systems development life cycle | system | software engineering | sub-system | feasibility study | processes | software crisis | user interface | maintenance

Every **system** has a purpose, meets a need and is made of interdependent components; large systems are split into **sub-system** s. A computer system consists of data, **processes** and a **user interface**. In the 1960s projects ran late, over budget and failed to meet user requirements, a period called the **software crisis**. The response was **software engineering**, a structured approach to building software. Today this is represented by the **systems development life cycle**, which begins with a **feasibility study**. During **analysis** the current system is investigated and the requirements are agreed, and after the system goes live, **maintenance** keeps it working.

C True or false?

Tick one box · 9 marks

	True	False
Every system has a defined purpose, meets a defined need and has interdependent components.	☑	☐
Information is raw facts and figures that have not yet been processed. *(That describes data; information is data that has been processed and given context and meaning.)*	☐	☑
The type of user interface chosen depends on the application, the environment and the characteristics of the users.	☑	☐
A change in data protection law is an example of an internal reason for a new system. *(Legislation comes from outside the organisation, so it is an external reason.)*	☐	☑
One cause of the software crisis was that developers had little past experience to base time and cost estimates on.	☑	☐
Software engineering means writing code as quickly as possible with minimal documentation. *(Software engineering applies a disciplined, systematic engineering process to designing, building and maintaining software.)*	☐	☑
The design stage produces the requirements specification. *(The requirements specification is produced during analysis; design produces the design specification.)*	☐	☑
Errors in requirements are cheapest to fix after the system has been installed. *(They are cheapest to fix early; the later an error is found, the more design, code and testing must be redone.)*	☐	☑
Evaluation can identify the need for a new system, which is why the process is called a cycle.	☑	☐

D Exam-style questions

41 marks

1 A Northern Ireland bus operator uses a real-time passenger information system at its stops. State the three characteristics that this, like every system, must have. [3]

> A defined purpose, e.g. to tell passengers when the next bus will arrive (1) / meets a defined need, e.g. passengers waiting without knowing about delays (1) / a number of interdependent components, e.g. GPS units on buses, a central server, display screens at stops (1). Accept the characteristics without examples.

2 A leisure centre run by a Northern Ireland council uses a computer system to manage memberships, class bookings and payments. Identify the purpose, the need and two components of this system. [4]

> Purpose, e.g. to manage memberships and allow customers to book classes and pay (1) / Need, e.g. high volume of bookings cannot be handled manually; avoid overbooked classes (1) / Any two components, e.g. membership database (1), online booking website/app (1), card payment terminal, access-control turnstiles, staff PCs, network

3 A newsagent in Omagh uses a computer system for stock control. For this system, describe one example of the data it holds, one example of its user interface and one example of a process it carries out. [6]

> Data (max 2): an appropriate data item, e.g. product barcode/ID, description, quantity in stock, reorder level, price (1) with description of where it comes from/why it is held (1) / User interface (max 2): e.g. barcode scanner at till / stock screen to view and edit levels / form to add new products (1) with how it is used (1) / Processes (max 2): e.g. reduce stock quantity when an item is sold, increase it when a delivery arrives, compare quantity with reorder level and produce reorder list, produce sales reports (1) with description (1)

4 Explain the difference between data and information, using an example. [2]

> Data is raw facts/figures without context or meaning (1) / Information is data that has been processed so it has context/meaning, e.g. 18 is data; 'temperature in ward 18 °C' is information (1)

5 A hospital is choosing a user interface for a self-service check-in screen in its outpatients department. Explain three factors that should influence the choice of interface. [6]

> Application (2): only a few data items needed (name/date of birth, appointment) so a simple touchscreen is suitable rather than keyboard and mouse (1+1) / Environment (2): busy, public waiting area; must be robust, easy to clean for infection control, usable quickly to avoid queues (1+1) / Users (2): members of the public of all ages and abilities, untrained, some with visual or mobility impairments, so must be intuitive, with large text, accessible height, possibly audio/multiple languages (1+1)

6 A garden centre in Enniskillen still records sales on a manual till and paper stock book. Explain two problems with the current system that could justify a new system. [4]

> Two problems, each identified (1) and explained in context (1), e.g. stock book is not updated automatically when items are sold, so stock levels are inaccurate and popular plants run out (2) / manual till cannot produce sales reports, so manager cannot see best/worst sellers to plan orders (2) / manual calculation errors at till lead to lost income or wrong change (2) / paper records can be lost or damaged and are not backed up (2)

7 A retired programmer tells your class that in the late 1960s 'nearly every big software project ran late, cost too much or didn't work properly'. Describe three causes of this situation. [3]

> Any three from: few comparable past projects to learn from, so estimates of time and cost were rough and often wrong (1) / development tools were unsophisticated so software took a long time to produce (1) / no tools to support early identification of errors so faults were slow to locate and correct and maintenance was expensive (1) / methods for capturing requirements were imprecise so systems did not meet user requirements (1) / no specialist testing staff or formal testing process (1) / demand for larger, more complex systems outgrew the industry's ability (1)

8 Software engineering grew out of the problems of the 1960s. Explain what the term means and how adopting it helped overcome those problems. [3]

> Software engineering applies a disciplined, systematic/structured engineering process to designing, building and maintaining software (1) / modelled on other engineering disciplines (1) / addressed the crisis by, any one from: structured stages/life cycle allow better planning and estimation so projects are on time and within budget (1); formal requirements specification ensures systems meet user needs (1); formal testing improves quality and reliability (1); documentation makes maintenance easier (1)

9 Name the deliverable produced at the end of the feasibility stage and the end of the design stage, and explain why the client signs off each stage before the next begins. [4]

> Feasibility: feasibility report (1) / Design: design (system) specification (1) / Sign-off means the client agrees the deliverable meets their needs before more money and work are committed (1) / misunderstandings are found early when they are cheaper to correct, rather than after coding (1) / gives a clear milestone to monitor progress / provides an agreed baseline if there is a dispute later (1)

10 Discuss the advantages and disadvantages of using a structured systems development life cycle to develop a new patient booking system for a GP practice. [6]

> Indicative content. Advantages: clear stages and milestones so the practice manager can monitor progress and cost; requirements (e.g. appointment types, GP rotas, text reminders) are fully documented and signed off before coding, reducing misunderstandings; thorough documentation aids maintenance and staff training; formal testing important as patient data and appointments are critical; legal requirements (UK GDPR, special category health data) can be identified early. Disadvantages: inflexible if requirements change, e.g. new NHS/Department of Health guidance; the practice may not see a working system until late; returning to earlier stages is slow and expensive; users may realise late that the system does not suit how receptionists actually work. Levels: L3 (5–6) balanced discussion of both sides applied to the GP context with a conclusion; L2 (3–4) several points, some applied, may be one-sided; L1 (1–2) generic points, little or no application.

E Quick quiz

Circle one answer · 9 marks

1 A school's cashless canteen uses fingerprint readers at the tills. In the three-element model of a computer system, the fingerprint reader is part of the... [1]

A data
B processes
C user interface ✓
D sub-system boundary

2 Which of these was a cause of the software crisis of the 1960s? [1]

A Too many specialist software testers
B Little past experience on which to base time and cost estimates ✓
C Development tools that were too advanced for programmers
D Users asking for systems that were too small

3 Which stage decides whether a project is worth carrying out? [1]

A Feasibility study ✓
B Analysis
C Design
D Evaluation

4 Which is an internal reason for developing a new system? [1]

A A competitor launches an app
B New legislation is passed
C Customers expect online ordering
D Staff re-enter the same data on three forms ✓

5 Which deliverable is produced at the end of analysis? [1]

A Test results
B User guide
C Requirements specification ✓
D Feasibility report

6 Why does a satnav give directions by voice while the car is moving? [1]

A Voice output is cheaper than a screen
B The driver can keep their eyes on the road ✓
C Touchscreens do not work in cars
D Voice is the only way to show a map

7 Software engineering is best defined as... [1]

A a disciplined, systematic engineering process for designing, building and maintaining software ✓
B the repair of computer hardware
C writing code without a design
D the study of computer networks

8 What is a legacy system? [1]

A A brand-new cloud system
B A test version of a system
C A system left in a will
D An old system still in use, often hard to maintain ✓

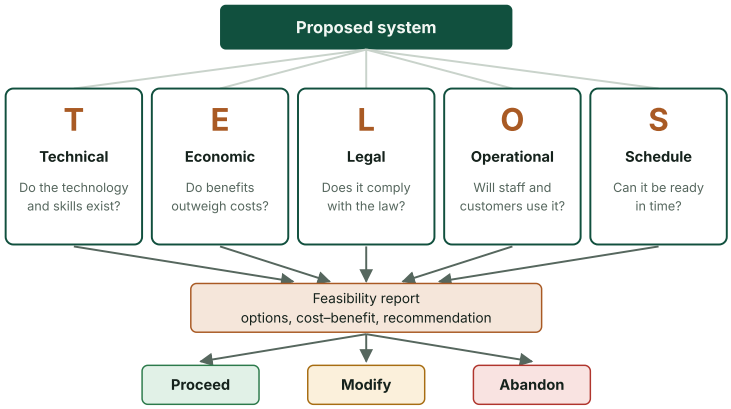
9 A main disadvantage of a strict sequential life cycle is that... [1]

A there is no documentation
B testing is never done
C it is inflexible when requirements change ✓
D the client cannot sign off stages

User needs, time and cost: the feasibility study and the development team

AP02 ANSWERS

I can: Explain why a system has to satisfy the organisation as well as its individual users · Explain how user needs, time and cost constrain what a development project can deliver · Apply the five TELOS feasibility criteria to a scenario and carry out a simple payback calculation · Describe what a feasibility report contains and who uses it · Describe the main role and typical duties of the systems analyst, project manager and programmer



Use this diagram to help you: A feasibility study tests a proposed system against five TELOS criteria and produces a feasibility report recommending to proceed, modify or abandon

A Key terms

Write the letter of the matching definition · 6 marks

E Feasibility study

F Feasibility report

C Operational feasibility

B Cost–benefit analysis

D Payback period

A Intangible benefit

A A benefit that is difficult to measure in money, such as improved customer satisfaction or staff morale.

B A comparison of the estimated costs of a proposed system with its expected tangible and intangible benefits over a period of time.

C Whether a proposed system will work in practice in the organisation and be accepted and used by its staff and customers.

D The length of time it takes for the cumulative benefits of a system to equal its cumulative costs.

E A preliminary investigation to decide whether a proposed system is technically, economically, legally, operationally and schedule feasible, ending in a recommendation.

F The document produced by a feasibility study that describes the options considered, their feasibility and a justified recommendation for management.

B Fill the gaps

Use the word bank · 10 marks

feasibility study | payback period | Technical | systems analyst | project manager | Economic | Legal | budget | user needs | programmer

A new system must meet the **user needs** of both the organisation and the individuals who use it, and it is constrained by the time available and the **budget**. A **feasibility study** decides whether the system should go ahead. **Technical** feasibility asks whether the hardware, software and skills exist. **Economic** feasibility compares costs with benefits, and the **payback period** is the time taken for savings to cover the initial cost. **Legal** feasibility checks compliance with laws such as the UK GDPR. The **systems analyst** carries out the study and defines the requirements, the **project manager** plans the schedule and controls the budget, and the **programmer** writes and tests the code.

C True or false?

Tick one box · 9 marks

	True	False
A system that suits the organisation but that customers find hard to use may still fail to achieve its purpose.	☑	☐
Time and cost have no effect on which features can be included in a new system. *(A fixed deadline and budget limit how much can be built, so features that cannot be delivered in time or within budget must be cut or delayed.)*	☐	☑
A feasibility study may recommend that the organisation does nothing.	☑	☐
Schedule feasibility is about whether the organisation can afford the system. *(That is economic feasibility; schedule feasibility asks whether it can be delivered on time.)*	☐	☑
Improved staff morale is an example of an intangible benefit.	☑	☐
A cost–benefit analysis produces exact, guaranteed figures. *(The figures are estimates based on assumptions.)*	☐	☑
The feasibility report is written for programmers to use when coding. *(It is written for management or the client to decide whether to approve the project; programmers work from the system specification.)*	☐	☑
The systems analyst uses interviews, observation, questionnaires and document inspection to find the requirements.	☑	☐
The programmer is mainly responsible for allocating resources and reporting progress to the client. *(Those are project manager duties; the programmer writes, comments, tests and debugs the code.)*	☐	☑

D Exam-style questions

36 marks

1 State the purpose of a feasibility study. [1]

> To decide whether a proposed system is possible/worthwhile/should go ahead, resulting in a recommendation to management (1)

2 A ferry company operating from Larne is developing an online booking system. Explain why the system must meet the needs of both the company and its individual customers. [4]

> Company (max 2): the volume of bookings is too large to handle by phone or in person (1), so an online system is needed to take bookings and payments efficiently and fill sailings (1) / Customers (max 2): customers must find it accessible and easy to use, e.g. clear choice of sailing, vehicle type, payment (1); if not, they may book with a competitor so the system fails its purpose (1)

3 Explain how time and cost can each limit what is delivered when a new system is developed for a Belfast secondary school. [4]

> Time (2): the system must be operational by a fixed date, e.g. the start of the school year (1), so features that cannot be built and tested by then must be dropped or delayed (1) / Cost (2): the budget is limited (1), so it restricts the number of developers, hardware purchased, licences, training or amount of testing (1)

4 Explain what is meant by operational feasibility, using an example. [2]

> Whether the system will work in practice in the organisation and be accepted/used by staff and customers (1) / Example, e.g. reception staff with little IT experience may resist an online booking system unless trained (1)

5 Distinguish between a tangible and an intangible benefit, giving an example of each. [4]

> Tangible: a benefit that can be measured in money (1), e.g. saving £8 000 a year in staff overtime (1) / Intangible: a benefit that is difficult to measure in money (1), e.g. improved customer satisfaction or staff morale (1)

6 A new system costs £30 000 to develop and £4 000 per year to run. It is expected to save £14 000 per year. Calculate the payback period, showing your working. [3]

> Net saving per year = £14 000 − £4 000 = £10 000 (1) / Payback = £30 000 ÷ £10 000 (1) / = 3 years (1)

7 Describe three items that would be included in a feasibility report. [3]

> Any three from: terms of reference/introduction (1) / description of the current system and its problems (1) / alternative solutions considered, including doing nothing (1) / technical, economic, legal, operational and schedule findings for each option (1) / outline costs and benefits/cost–benefit analysis (1) / justified recommendation (1) / outline plan and estimated timescale (1)

8 A GAA club in County Tyrone is replacing its paper membership records with an online system. State the main role of the systems analyst and describe three duties they would carry out on this project. [4]

> Main role: to establish what users need from the new system and what data processing it must do (1) / Any three duties applied to the club: carry out a feasibility study and produce a feasibility report (1) / liaise with the club committee/membership secretary (1) / fact-find using interviews, questionnaires to members, observation or inspection of the paper records and forms (1) / define the system specification for the new membership system (1)

9 Describe three duties of a project manager during the development of a new appointments system for a dental practice. [3]

> Any three from: plan the project schedule (1) / split the project into tasks/sub-tasks (1) / allocate resources (staff, hardware, software) to tasks (1) / identify risks to the project (1) / monitor progress against milestones (1) / ensure delivery on time and within budget (1) / report back to the dental practice/client (1)

10 A hotel group in Belfast is considering a facial-recognition check-in system. Evaluate the feasibility of this proposal, referring to at least three TELOS criteria. [8]

> Indicative content. Technical: requires cameras, recognition software, integration with the booking system and secure storage; accuracy in varied lighting; staff skills to support it. Economic: high cost of hardware, licences and integration v. savings in reception staff time and faster check-in at busy times; intangible benefit of a modern image; uncertain payback. Legal: facial images used to identify people are biometric, special category data under the UK GDPR and Data Protection Act 2018; needs a lawful basis and an Article 9 condition (in practice explicit consent), a data protection impact assessment, security and retention rules; an alternative check-in for guests who refuse. Operational: some guests may find it intrusive or distrust it; staff need training; must still cope with failures, children and groups; reception roles change. Schedule: can it be installed and tested before a peak season or event without disrupting trading? Conclusion: likely feasible technically but legal and operational risks are high; might recommend a pilot or an optional system. Levels: L3 (7–8) evaluates at least three criteria in depth applied to the hotel, with a justified conclusion; L2 (4–6) describes three criteria with some application, limited judgement; L1 (1–3) identifies criteria generically with little application.

E Quick quiz

Circle one answer · 9 marks

1 Which member of the development team is mainly responsible for identifying the user and data processing requirements of a new system? [1]

A Project manager
B Systems analyst ✓
C Programmer
D Network technician

2 A council system must be live before a new law takes effect in six months. Which factor affecting development does this deadline represent? [1]

A Cost
B User needs
C Time ✓
D Data

3 What does the 'O' in TELOS stand for? [1]

A Organisational
B Output
C Optimal
D Operational ✓

4 Which is a tangible benefit? [1]

A Saving £12 000 a year on paper and postage ✓
B Better company image
C Happier customers
D Improved decision-making

5 A system costs £20 000 and gives a net saving of £5 000 a year. What is the payback period? [1]

A 2 years
B 5 years
C 4 years ✓
D 25 years

6 Who uses the feasibility report to decide whether to go ahead? [1]

A Management / the client ✓
B End users only
C Testers
D Programmers

7 Which is a typical duty of a project manager? [1]

A Writing source code from module specifications
B Interviewing users to find requirements
C Splitting the project into tasks and allocating resources ✓
D Adding comments to code

8 Why does a programmer add meaningful comments to source code? [1]

A To make the program run faster
B To help other programmers understand and maintain the code later ✓
C To encrypt the code
D Because the compiler requires them

9 Checking that a new system complies with the UK GDPR is part of... [1]

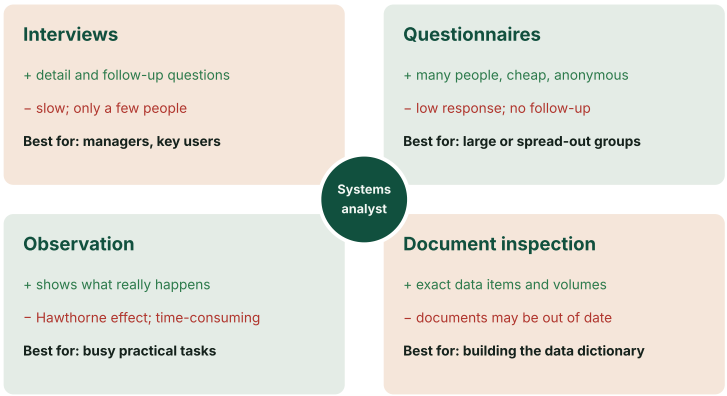
A technical feasibility
B schedule feasibility
C economic feasibility
D legal feasibility ✓

AS 1: APPROACHES TO SYSTEMS DEVELOPMENT · ANALYSIS

Analysis and fact-finding methods

AP03 ANSWERS

I can: Explain why analysts must fully understand the existing system before a new one is designed · Distinguish direct from indirect data sources using examples · Describe how interviews, questionnaires, observation and document sampling are planned and carried out · Evaluate each technique and justify a combination of methods for a given organisation



Use this diagram to help you: The four fact-finding methods with an advantage, a disadvantage and the users each suits, around the systems analyst

A Key terms

Write the letter of the matching definition · 6 marks

B Analysis

F Fact-finding

D Systems analyst

C Direct data source

A Indirect data source

E Hawthorne effect

A Data that already exists and is reused for a purpose other than the one for which it was originally collected.

B The stage of systems development in which the existing system is investigated and the requirements for its replacement are identified and agreed with the client.

C A source of data collected first-hand specifically for the current purpose, such as a survey designed for the project.

D The person who investigates the current system, establishes the requirements and specifies the new system.

E The tendency of people to change their behaviour because they know they are being observed, which can make observation unreliable.

F Gathering information about a current system, using techniques such as interviews, questionnaires, observation and document sampling, to establish how it works and what is required.

B Fill the gaps

Use the word bank · 10 marks

Hawthorne effect | closed question | indirect data source | direct data source | analysis | questionnaire | interview | document sampling | observation | triangulation

During **analysis** the systems analyst investigates the existing system. Data collected first-hand for the project, such as a survey, comes from a **direct data source**, while reused data such as census figures comes from an **indirect data source**. An **interview** allows follow-up questions but reaches only a few people. A **questionnaire** reaches many people cheaply, and its **closed question** answers are easy to analyse. During **observation** staff may act differently because they are being watched, which is called the **Hawthorne effect**. Examining invoices and forms is called **document sampling**, and checking facts from one source against another is called **triangulation**.

C True or false?

Tick one box · 9 marks

	True	False
Questionnaires are suitable for gathering views from large numbers of people who are spread across several locations.	☑	☐
Observation always gives an accurate picture because the analyst sees the work directly. *(Staff may change their behaviour when they know they are being watched (the Hawthorne effect), and infrequent tasks may not be seen.)*	☐	☑
Population statistics from a government website are an example of an indirect data source when used to plan a new shop.	☑	☐
A direct data source is usually cheaper than an indirect one. *(Direct sources are usually more expensive because they have to be designed and carried out; indirect data already exists.)*	☐	☑
An unstructured interview allows the analyst to ask follow-up questions based on the answers given.	☑	☐
Questionnaires allow the analyst to change a question if a respondent misunderstands it. *(Questions are fixed before the questionnaire is issued, so there is no chance to clarify or follow up.)*	☐	☑
Document sampling can show the volume of data that the current system processes.	☑	☐
Checkboxes allow a respondent to select only one answer from a list. *(Checkboxes allow several answers to be selected; radio buttons restrict the respondent to one answer.)*	☐	☑
Analysts usually combine several fact-finding techniques and cross-check the results.	☑	☐

D Exam-style questions

34 marks

1 Before designing a new ticketing system for a Derry theatre, the systems analyst spends three weeks investigating how the box office currently works. State three things the analyst is trying to achieve during this investigation. [3]

> Any three from: to understand the existing system's inputs, processes and outputs (1) / to identify strengths of the current system to keep (1) / to identify weaknesses or problems the new system must solve (1) / to identify opportunities for improvement or new features (1) / to establish and agree the user requirements (1) / to provide the information needed for the system specification (1)

2 A sports shop in Bangor is deciding whether to stock a new range of hurling equipment. Using this example, explain the difference between a direct and an indirect data source. [2]

> A direct data source is collected first-hand specifically for the current purpose, e.g. a survey of the shop's customers about hurling equipment (1) / an indirect data source already exists, was gathered for some other

reason and is reused here, e.g. local GAA club membership figures or the shop's past sales records (1)

3 The owner of a café in Portrush is considering opening a second café in Portstewart. Identify one direct and one indirect data source she could use, and state one drawback of each. [4]

Direct, e.g. a survey/questionnaire of people in Portstewart, interviews with local residents or a trial pop-up stall (1); drawback: expensive/time-consuming to design, carry out and analyse (1). Indirect, e.g. NISRA population statistics, council business listings, sales data from the Portrush café (1); drawback: collected for another purpose so may not contain exactly the data needed / may need filtering / may be out of date (1)

4 Describe two things a systems analyst should do when planning an interview with staff. [4]

Any two described, two marks each: decide exactly what information is needed (1) and write questions targeted at the interviewee's expertise (1) / select interviewees carefully (1) as not everyone can or needs to be interviewed, e.g. choose experienced users (1) / tell the interviewee the purpose, structure, length and use of the information (1) so they are prepared and reassured (1) / prepare a standard response sheet (1) so notes are organised and comparable (1)

5 A rugby club in Ballymena wants its 450 members' views on a new online membership system. Describe two different types of question input that could be used in an online questionnaire, giving an example of each. [4]

Two types, two marks each (type with description 1, suitable example 1): radio buttons allow only one option to be chosen (1), e.g. membership category: junior, adult, senior (1) / checkboxes allow several options (1), e.g. which teams or sections do you follow (1) / slider or rating scale for a value on a range (1), e.g. rate the current renewal process from 1 to 5 (1) / free-text box for answers in own words (1), e.g. suggest one improvement (1)

6 Explain one advantage and one disadvantage of using observation to investigate how receptionists use a hotel booking system. [4]

Advantage (1 + 1 expansion): shows how receptionists actually work rather than the official procedure (1), revealing workarounds or problems they might not mention in an interview (1). Disadvantage (1 + 1): receptionists may behave differently because they know they are being watched / Hawthorne effect (1) so the analyst sees an unrepresentative picture (1); or time-consuming / may miss infrequent tasks such as group bookings (1+1)

7 State three things an analyst could learn by sampling documents such as invoices and order forms. [3]

Any three from: the data items recorded and their formats (1) / the volume of data processed, e.g. invoices per week (1) / the key users of the system (1) / how data flows between departments (1) / the outputs the organisation requires (1) / how the system arose or has changed (1)

8 Explain how an analyst could check that the answers given in interviews and questionnaires are reliable. [2]

Use triangulation: compare answers with other sources such as documents, observation or other interviewees (1) / include related questions in a questionnaire (e.g. age and date of birth) and reject inconsistent responses (1) / paraphrase answers back to interviewees to confirm understanding (1). Max 2.

9 A district council in Northern Ireland handles requests for bulky-waste collections by telephone and paper forms at three recycling centres and a central call centre. It wants a new computerised system. Discuss the fact-finding techniques the analyst should use, evaluating their suitability and justifying the combination you recommend. [8]

Indicative content: Document sampling of existing paper request forms and call logs to find data items, volumes and peak times; low cost, no disruption, but forms may be incomplete. Interviews with the waste services manager and call-centre supervisors to understand rules, charges and problems; detailed and allow follow-up, but time-consuming and few people covered. Questionnaire to staff at all three recycling centres (and possibly residents) because they are spread out; cheap, anonymous, analysed easily, but low response and no follow-up. Observation of call handlers taking requests and centre staff processing forms to see real workarounds and timings; first-hand, but Hawthorne effect, different practices at each site, rare tasks missed. Triangulation to cross-check findings. Conclusion: a justified combination, e.g. documents first, then interviews with managers,

> questionnaires for dispersed staff, observation at the call centre. Levels: 7-8 several techniques evaluated with strengths and weaknesses applied to the council context and a justified conclusion, good technical language; 4-6 some evaluation, partly applied, limited conclusion; 1-3 techniques named or described with little evaluation or context.

E Quick quiz

Circle one answer · 9 marks

1 A systems analyst wants to watch warehouse staff picking orders to see how they really work. Which problem is most likely? [1]

A Low response rate
B Staff may work differently because they know they are being watched ✓
C Documents may be out of date
D Questions cannot be changed once issued

2 Which fact-finding technique is most suitable for collecting the views of 1,200 pupils in a large school? [1]

A Interview each pupil
B Observation
C Online questionnaire ✓
D Document sampling

3 Which of these is an indirect data source for a newsagent planning a new shop? [1]

A Published population figures for the town ✓
B A street survey the newsagent writes
C Interviews with passers-by
D A taste test of new products

4 Which technique lets the analyst ask follow-up questions? [1]

A Questionnaire
B Document sampling
C Observation
D Interview ✓

5 Which questionnaire input type allows several answers to be selected? [1]

A Radio buttons
B Checkboxes ✓
C Slider
D Drop-down list with one choice

6 Which technique is best for finding the exact data items recorded on an existing order form? [1]

A Document sampling ✓
B Questionnaire
C Interview
D Observation

7 An interviewee says she has driven lorries for six years, but HR records show she is 21. Checking one source against another like this is called... [1]

A validation
B verification by double entry
C normalisation
D triangulation ✓

8 What is the main drawback of an indirect data source? [1]

A It is always very expensive
B It may not contain exactly the data needed and may need filtering ✓
C It cannot be stored electronically
D It is only available after the system is built

9 Which is NOT a purpose of the analysis stage? [1]

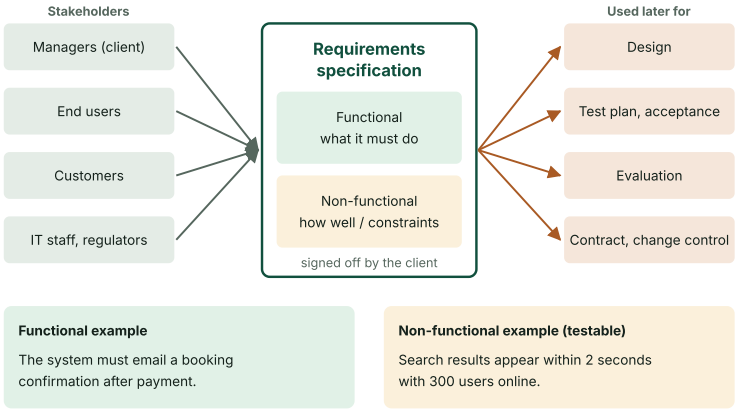
A Understand the inputs, processes and outputs of the current system
B Identify weaknesses of the current system
C Write and test the program code ✓
D Agree the user requirements

AS 1: APPROACHES TO SYSTEMS DEVELOPMENT · ANALYSIS

Stakeholders, requirements and the system specification

AP04 ANSWERS

I can: Identify internal and external stakeholders and explain what each needs from a system · Distinguish functional from non-functional requirements and classify given examples · Write clear, measurable requirements for stock control, reservation, payroll and billing systems from a stated perspective · Describe what a system specification contains and how it is used throughout the project



Use this diagram to help you: Stakeholders supply requirements to the requirements specification, which is then used for design, testing, evaluation and as a contract

A Key terms

Write the letter of the matching definition · 6 marks

B Stakeholder

F End user

A Functional requirement

C Non-functional requirement

D Quality attribute

E System specification

A A statement of something the system must do, such as an input, process, calculation or output.

B A person or group with an interest in a system or who is affected by it, such as managers, end users, customers and regulators.

C A condition or constraint on how the system behaves, such as response time, availability, reliability, security or usability, rather than a function it performs.

D A measurable characteristic used to judge a system, such as performance, reliability or security; non-functional requirements specify these.

E The document produced at the end of analysis that states precisely what the client expects the system to do, including requirements, constraints and schedule, and is agreed by the client.

F A person who operates the system directly in their day-to-day work or as a customer.

B Fill the gaps

Use the word bank · 9 marks

quality attribute | MoSCoW | stakeholder | non-functional requirement | testable | end user | functional requirement | client | system specification

Anyone with an interest in a system or affected by it is a **stakeholder**. The people who operate it each day are the **end user**s, while the **client** commissions and pays for it. A **functional requirement** states something the system must do, such as producing payslips. A **non-functional requirement** states a condition or quality it must meet, such as a response time of two seconds, and is sometimes called a **quality attribute**. Each requirement should be **testable** and can be prioritised using **MoSCoW**. All requirements are recorded in the **system specification**, which is agreed with the client at the end of analysis.

C True or false?

Tick one box · 9 marks

	True	False
'The system must email a receipt after payment' is a functional requirement.	☑	☐
'The system must be user friendly' is a well-written, testable requirement. *(It is vague; it needs a measurable criterion such as the time a new user takes to complete a task.)*	☐	☑
Customers can be stakeholders in a system even though they do not work for the organisation.	☑	☐
Security, availability and response time are examples of non-functional requirements.	☑	☐
The system specification is written after the system has been tested. *(It is produced at the end of analysis, before design, and is later used to test and evaluate the system.)*	☐	☑
A non-functional requirement describes a new feature that the system must provide. *(Features are functional requirements; non-functional requirements set conditions or quality standards for how the system behaves.)*	☐	☑
Calculating net pay after tax and National Insurance is a functional requirement of a payroll system.	☑	☐
Traditionally, the system specification formed the basis of a contract between client and supplier.	☑	☐
Agile development has made system specifications more rigid and harder to change. *(Agile approaches have led to less prescriptive specifications with built-in processes for changing requirements.)*	☐	☑

D Exam-style questions

39 marks

1 Define the term stakeholder and give two examples of stakeholders in a new school management information system. [3]

> Definition: a person or group with an interest in, or affected by, the system (1). Any two examples (1 for both): teachers / pupils / parents / senior leadership / office staff / IT technician / Education Authority / Department of Education (regulator).

2 A requirements list for a Strabane veterinary practice includes 'Record each animal's vaccinations' and 'Show any pet's record within two seconds'. Using these as examples, explain how a functional requirement differs from a non-functional requirement. [2]

> A functional requirement describes an action the system must carry out / a function, input, process or output, e.g. recording vaccinations (1) / a non-functional requirement describes a condition, constraint or quality the system must meet, such as performance, e.g. displaying a record within two seconds (1)

3 A hardware merchant in Newry is replacing its paper-based stock control system. State two functional and two non-functional requirements of the new system. [4]

> Functional, any two (1 each): record/track current stock levels including items on order / alert staff when stock reaches reorder level / record goods received and sold / produce stock reports / generate purchase orders / support FIFO stock rotation. Non-functional, any two (1 each): stock levels must be accurate/up to date / available during trading hours / accessible only to authorised staff / backed up daily / support multiple simultaneous users / respond within a stated time

4 A sports centre in Lisburn pays its part-time coaches by the hour. Give two functional and two non-functional requirements of its payroll system. [4]

> Functional, any two (1 each): add/edit/remove coaches / enter hours worked / edit hourly rate / calculate gross pay, deductions and net pay / produce payslips / send details to HMRC / back up payroll records. Non-functional, any two (1 each): only authorised payroll staff can access pay data / reliable and accurate calculations / reasonable response time or pay run completed within a set time / comply with UK GDPR and HMRC rules / backed up regularly

5 An online booking system is being developed for a Belfast concert venue. Identify two functional requirements from the venue's perspective and two from the customer's perspective. [4]

> Venue, any two (1 each): monitor ticket sales for each event / set and change ticket prices / record customer booking history / produce sales reports / limit tickets per customer. Customer, any two (1 each): search for events by date or artist / choose seats / pay online by card / receive e-ticket or confirmation email / cancel or transfer tickets

6 The requirement 'The system should be fast' has been written in a draft specification. Explain why this is a poor requirement and rewrite it. [3]

> It is vague/not measurable (1) so it cannot be tested / there is no way to show whether it has been met (1). Rewrite with a measure, e.g. 'search results must be displayed within 2 seconds with 200 users logged in' (1)

7 State the name of the document produced at the end of the analysis stage and describe three items it would contain. [4]

> System specification / requirements specification (1). Any three described (1 each): scope and objectives / description of the current system and its problems / functional requirements / non-functional requirements / data requirements / DFDs / constraints such as budget, hardware or deadline / development schedule

8 Explain two ways in which the system specification is used later in a project. [4]

> Any two, 2 marks each: used by designers to plan the system (1) so the design meets the agreed requirements (1) / used to create the test plan or acceptance tests (1) so each requirement can be checked (1) / used in evaluation (1) to judge whether the finished system meets the requirements (1) / acts as an agreement or contract (1) to settle disputes about what was promised (1)

9 Explain why system specifications today are often less rigid than those used in the past. [3]

> Any three from: many more software suppliers now exist (1) / development skills are more widespread, so clients have more flexibility (1) / Agile methods deliver in short iterations with regular feedback (1) / requirements are expected to change, so specifications include change processes (1) / rigid contracts with penalties made change costly (1)

10 A local electricity supplier is developing a new online billing system for its customers. Discuss the functional and non-functional requirements that should be included, considering the needs of different stakeholders. [8]

> Indicative content. Stakeholders: customers, billing and customer-service staff, finance managers, IT staff, regulators (Utility Regulator, Information Commission/ICO). Functional: customer account registration and login; calculate bills from meter readings and tariff; add VAT; email or post bills; online payment and direct debit; record payments; send reminders for unpaid bills; let customers view usage history; staff reports on arrears. Non-functional: security of personal and card data and compliance with UK GDPR / DPA 2018; availability 24/7;

> accuracy of calculations; capacity for peak load when bills are issued; response time; accessibility for elderly or disabled customers; regular backups. Discussion of conflicts, e.g. strong security versus ease of use for customers. Levels: 7-8 wide range of correctly classified, measurable requirements linked to stakeholders, with discussion of priorities or conflicts; 4-6 several correct requirements of both types with some link to stakeholders; 1-3 a few requirements, possibly misclassified or generic.

E Quick quiz

Circle one answer · 9 marks

1 A cinema is developing a new online booking system. Which of these is a non-functional requirement? [1]

A Customers can choose their seats
B The system prints a ticket
C Booking pages load within 2 seconds when 1,000 users are online ✓
D Staff can add new film showings

2 What is the main document produced at the end of the analysis stage? [1]

A The test plan
B The user guide
C The source code
D The system specification ✓

3 Which is a functional requirement of a payroll system? [1]

A Calculate each employee's net pay ✓
B Only payroll staff can log in
C Back up every night
D Run on the existing network

4 Which is the best-written requirement? [1]

A The system should be quick
B The system should look good
C Staff can log in
D A member can book a class in no more than three clicks ✓

5 MoSCoW is used to... [1]

A test code
B prioritise requirements ✓
C draw DFDs
D estimate payback

6 Which group are the end users of a hospital's ward system? [1]

A The nurses and doctors who use it daily ✓
B The hospital's shareholders
C The software developers
D The Information Commission

7 Why were system specifications traditionally written very precisely? [1]

A They were published to the public
B They formed the basis of a contract, and changes led to penalties or extra charges ✓
C They were used as user manuals
D They replaced testing

8 Which of these is a non-functional requirement for a stock control system? [1]

A Alert staff when stock reaches the reorder level
B Produce a slow-moving stock report
C Allow only warehouse staff to change stock levels ✓
D Record goods received

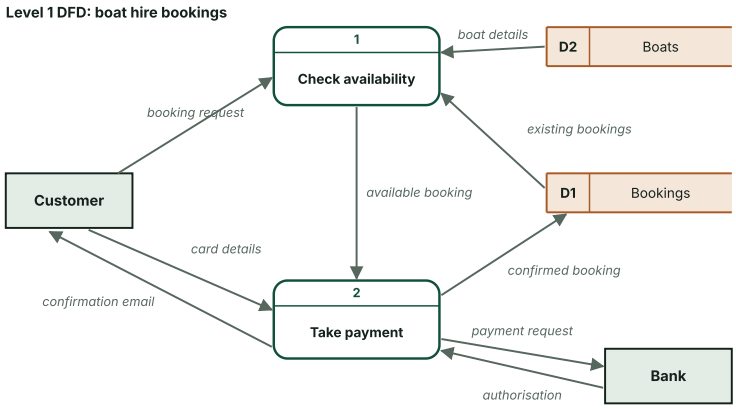
9 A question asks for requirements 'from the airline's perspective'. Which answer fits best? [1]

A Choose a seat on the plane
B Monitor current bookings on every flight ✓
C Receive a confirmation email
D Pay by debit card

Data flow diagrams

AP05 ANSWERS

I can: Explain why DFDs are used during analysis and who benefits from them · Name, draw and identify each DFD element, including duplicated entities · Draw a context diagram and a balanced Level 1 DFD from a written scenario · Spot errors in a DFD and explain how to correct them



Use this diagram to help you: A level 1 data flow diagram for a boat hire booking system with external entities, numbered processes, data stores and labelled data flows

A Key terms

Write the letter of the matching definition · 6 marks

A Data flow diagram (DFD)

F Context diagram

E Level 1 DFD

D External entity

B Source

C Sink

A A diagram that models how data moves through a system using external entities, processes, data stores and labelled data flows.

B An external entity from which data enters the system.

C An external entity to which the system sends data.

D A person, organisation or system outside the system boundary that is a source or destination of data.

E A DFD in which the single process of the context diagram is exploded into its main numbered sub-processes, with data stores and internal flows added.

F A top-level (Level 0) DFD showing the whole system as a single process with its external entities and the data flows between them, defining the system boundary.

B Fill the gaps

Use the word bank · 9 marks

context diagram | source | data flow diagram | process | sink | data store | external entity | Level 1 | balancing

A **data flow diagram** models how data moves through a system. A person or organisation outside the system boundary is an **external entity**; one that supplies data is a **source** and one that receives data is a **sink**. A **process** transforms data and must have at least one input and one output. A **data store** holds data for later use. The **context diagram** shows the whole system as one

process. It is exploded into a **Level 1** DFD, which must keep the same external flows, a rule called **balancing**.

C True or false?

Tick one box · 9 marks

	True	False
A context diagram normally shows the whole system as a single process.	☑	☐
Data may flow directly from an external entity into a data store. *(All data must pass through a process before it reaches a data store.)*	☐	☑
Every data flow on a DFD should be labelled with the data it carries.	☑	☐
A DFD shows the order in which decisions are made. *(That is a flowchart; a DFD shows the movement of data, not sequence or decisions.)*	☐	☑
The external entities on a Level 1 DFD are usually the same as those on the context diagram.	☑	☐
A process that only receives data and sends nothing out is valid. *(A process must have at least one input and at least one output because it transforms data.)*	☐	☑
An external entity can be both a source and a sink.	☑	☐
The CCEA AS specification requires candidates to draw Level 2 and Level 3 DFDs. *(Only context (Level 0) and Level 1 diagrams are required.)*	☐	☑
A duplicated external entity is drawn more than once to avoid crossing data flow lines.	☑	☐

D Exam-style questions

34 marks

1 An analyst plans to show a Newtownabbey car dealership's managers a set of data flow diagrams of their proposed servicing system. Describe two ways the diagrams will be useful. [2]

> Any two from: provides a visual/graphical representation of the system (1) / shows the system at different levels of detail (1) / shows how the system interacts with external entities / defines the boundary (1) / identifies the processes and data stores (1) / gives a shared understanding that users can check (1) / helps identify bottlenecks or problems in the current system (1)

2 In a DFD for a school library's loans system, explain how a source differs from a sink, giving an example of each from the library. [4]

> A source lies outside the system and sends data into it (1), e.g. a pupil/borrower sending a loan request or book return (1) / a sink lies outside the system and is sent data by it (1), e.g. a pupil receiving an overdue reminder, or a librarian/teacher receiving a report (1)

3 A trainee analyst at a Lisburn software firm has drawn a context diagram for a car-wash booking app and is about to draw the Level 1 DFD. State two ways in which the Level 1 diagram will differ from the context diagram. [2]

> Any two from: context diagram has a single process; Level 1 has several numbered sub-processes (1) / context diagram normally shows no data stores; Level 1 shows data stores (1) / Level 1 shows internal data flows between processes and stores; context shows only flows crossing the boundary (1) / context gives an overview for management; Level 1 gives more detail (1)

4 Explain what is meant by balancing, with reference to a Level 1 DFD and the context diagram it was exploded from. [2]

> The data flows into and out of the system (to/from external entities) on the context diagram (1) must also appear, with the same names, on the Level 1 DFD / the levels must be consistent, with no flows added or lost at the

boundary (1)

5 A DFD shows a data flow labelled 'Order' going directly from the external entity Customer to the data store D1 Orders. Explain why this is incorrect and how it should be corrected. [2]

Data cannot flow directly from an external entity to a data store / all flows must pass through a process (1). Insert a process such as 'Record order' between Customer and D1 Orders, with Order flowing into the process and order details flowing from the process to the store (1)

6 An analyst has agreed a context diagram for a dental practice's appointments system with the practice manager. Describe the steps the analyst should follow to produce the Level 1 DFD. [3]

Any three from: the single process is exploded/decomposed into several sub-processes (1) / sub-processes are numbered (1.0, 2.0...) and named (1) / the same external entities are kept (1) / the data flows to and from external entities are kept the same, so the diagrams balance (1) / data stores are identified and added (1) / internal flows between processes and stores are added (1)

7 A swimming club in Coleraine runs a gala entry system. Swimmers send entry forms to the system. The system sends each swimmer an entry confirmation and sends the gala organiser a list of entries. The club treasurer receives a fees report. Draw a context diagram for this system. [5]

One process labelled e.g. Gala Entry System (1). Entities Swimmer, Gala Organiser and Treasurer shown as external entities (1). Flows, correctly labelled and directed: Entry form, Swimmer to system (1) / Entry confirmation, system to Swimmer (1) / Entry list to Gala Organiser and Fees report to Treasurer (1). No data stores required.

8 A school canteen top-up system works as follows: a parent sends a top-up request and card details; the card details are sent to a bank, which returns an authorisation; the pupil's balance in the Accounts file is updated; the parent receives a receipt; each evening a sales report is produced for the catering manager from the Transactions file. Draw a Level 1 DFD for this system. [8]

Indicative marking (up to 8): entities Parent, Bank and Catering Manager kept outside the system (1); numbered processes with verb names, e.g. 1.0 Process top-up, 2.0 Update balance, 3.0 Produce sales report (1); data stores D1 Accounts file and D2 Transactions file (1); Top-up request and Card details from Parent to 1.0 (1); Card details/payment request from 1.0 to Bank and Authorisation from Bank to 1.0 (1); top-up amount/authorised top-up from 1.0 to 2.0, with balance read from and updated balance written to D1 and transaction details written to D2 (1); Receipt to Parent from the relevant process (1); transaction details from D2 to 3.0 and Sales report from 3.0 to Catering Manager (1). Accept other sensible process divisions that obey the rules; deduct for unlabelled flows or entity-to-store links.

9 Evaluate the use of data flow diagrams as a tool for communicating the analyst's understanding of a system to the client and end users. [6]

Indicative content. Strengths: simple set of symbols so non-technical clients and users can understand and check them; visual overview easier to discuss than text; levels allow an overview for managers (context) and detail for users (Level 1); clearly defines the system boundary and scope; helps reveal missing data, bottlenecks or duplicated work; provides agreement that feeds into the specification. Limitations: do not show timing, sequence or decision logic, so users may expect more than they show; large systems produce cluttered diagrams or many levels; notation varies between textbooks, which can confuse; do not show the user interface or data structure in detail; need skill to draw correctly, and errors such as unbalanced levels may mislead. Conclusion on overall usefulness, e.g. very effective for agreeing scope and data requirements when combined with other documentation. Levels: 5-6 balanced evaluation of strengths and limitations with a supported conclusion; 3-4 several points, mostly one-sided or limited conclusion; 1-2 basic points with little evaluation.

E Quick quiz

Circle one answer · 9 marks

1 In a DFD for a garage's booking system, a car owner sends booking requests and receives confirmations. What is the car owner? [1]

A A process
B A data store
C An external entity that is both a source and a sink ✓
D A data flow

2 How many processes does a context diagram contain? [1]

A One ✓
B Two
C None
D As many as needed

3 Which flow is invalid? [1]

A Customer → 1.0 Take order
B 1.0 Take order → D1 Orders
C D1 Orders → D2 Stock ✓
D D2 Stock → 2.0 Reorder

4 A good name for a process is... [1]

A Orders
B Customer
C Data
D Calculate invoice total ✓

5 Breaking the single process of a context diagram into its main sub-processes is described as... [1]

A normalising the diagram
B exploding (decomposing) the process ✓
C balancing the entities
D iterating the flows

6 What does 'balancing' mean for DFDs? [1]

A Each level has the same number of processes
B The external data flows at one level also appear at the next level ✓
C Every store has two flows
D Entities are drawn the same size

7 Which element is usually omitted from a context diagram? [1]

A External entities
B Data flows
C The process
D Data stores ✓

8 On a Level 1 DFD, what is new compared with the context diagram? [1]

A Sub-processes, data stores and internal flows ✓
B New external entities
C Removal of all flows
D Decisions and loops

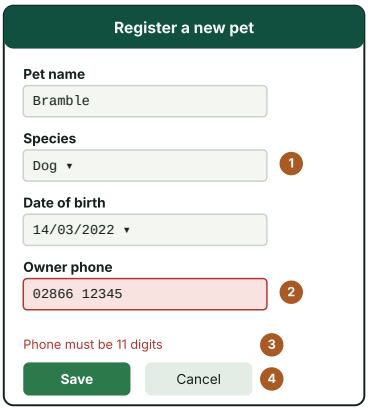
9 Which of these is a benefit of DFDs mentioned in mark schemes? [1]

A They show the exact program code
B They give a visual representation that non-specialists can understand ✓
C They replace the need for testing
D They show the timing of every event

System design: storyboarding, prototyping and data design

AP06 ANSWERS

I can: Describe what the design phase produces and how it leads into development and testing · Explain how storyboards and prototypes are used to get user feedback, with advantages and disadvantages of each · Apply HCI principles to evaluate and design input screens and outputs · Produce a data dictionary entry for a data item, including a precise validation rule · Distinguish between validation and verification with examples



HCI features

1 Drop-down list prevents typing errors
2 Field highlighted where the error is
3 Clear, helpful error message (feedback)
4 Buttons in the same place on every screen

Data dictionary (extract)

Field	Type	Validation
PetName	String(30)	Presence
Species	String(10)	Lookup list
DateOfBirth	Date	Range: not future
OwnerPhone	String(11)	Length, format

Use this diagram to help you: An annotated input screen design for registering a pet, with numbered HCI features, beside part of its data dictionary

A Key terms

Write the letter of the matching definition · 6 marks

A Design document

F Storyboarding

E Prototyping

C Stub

B Data dictionary

D Validation

A The output of the design phase, setting out the user interface, inputs, outputs and reports, data structures, validation and verification, algorithms and an initial test plan for the new system.

B A central record of every data item in a system, giving its name, data type, length, format, description, example and validation rules.

C A part of a prototype or program, such as a menu option or subroutine, that is present but has no working functionality yet.

D An automatic check by the system that data is reasonable and follows set rules before it is accepted; it cannot ensure the data is correct.

E Building an early, incomplete working version of a system so that users can experience its look and feel and give feedback on the interface and data requirements.

F A design technique in which each screen of a proposed interface is sketched, showing its contents, inputs and outputs and the navigation between screens, to obtain early feedback from users.

B Fill the gaps

Use the word bank · 9 marks

range check | storyboarding | design document | Validation | prototype | navigation | data dictionary | stubs | verification

The design phase produces a **design document** that describes how the system will meet the user requirements. In **storyboarding** each screen is sketched and arrows show the **navigation** between screens. A **prototype** is an early working model that gives users the look and feel of the system; options that appear but do nothing yet are called **stubs**. Every field's name, data type and validation rule is recorded in the **data dictionary**. **Validation** checks that data is sensible, for example a **range check** on a top-up amount, while **verification** checks that data was entered accurately, for example by typing an email address twice.

C True or false?

Tick one box · 9 marks

	True	False
Systems design can only properly begin once the analysis phase has identified the user requirements.	☑	☐
A storyboard lets users click through a working version of the software. *(A storyboard is a set of sketches; it is not interactive. A prototype gives users something they can try.)*	☐	☑
A storyboard identifies the inputs and outputs on each screen and the navigation between screens.	☑	☐
A prototype always contains full validation of every input. *(Prototypes are incomplete and often have little or no error checking; some options are only stubs.)*	☐	☑
Prototyping can help identify the tables and relationships needed in a database.	☑	☐
Validation guarantees that data entered is correct. *(Validation only checks data is reasonable and follows rules; valid data can still be wrong.)*	☐	☑
Entering a new password twice is an example of verification.	☑	☐
A data dictionary records the data type and validation rule for each field.	☑	☐
One risk of prototyping is that users judge it on appearance and overlook accuracy and security.	☑	☐

D Exam-style questions

38 marks

1 A Craigavon software company is designing a new booking system for a driving school. State four items that should be included in its design document. [4]

> Any four from: design of the user interface/screens and navigation (1) / input design (1) / output and report design (1) / specification of data structures/data dictionary (1) / validation and verification required (1) / algorithms for processes (1) / security or access levels (1) / a test plan (1)

2 Describe how storyboarding could be used when designing the screens of a mobile app for booking pitches at a GAA club. [3]

> Each screen of the app (e.g. login, choose pitch, choose time, confirm) is sketched as a frame showing its contents (1) / the inputs and outputs on each screen are identified, e.g. date selected, available slots shown (1) / arrows show navigation between screens, and the storyboard is walked through with club members to get feedback/spot awkward routes before coding (1)

3 Give two advantages and two disadvantages of using storyboarding to design a user interface. [4]

Advantages, two from: quick/cheap to create (1) / promotes early communication between designer and users (1) / no specialist knowledge needed to understand (1) / early feedback on functionality and navigation (1) / changes can be made early/cheaply (1). Disadvantages, two from: not interactive (1) / limited detail (1) / cannot show speed, responsiveness or usability accurately (1)

4 Explain how prototyping can be used in the design of (a) the user interface and (b) the data model of a new system. [4]

(a) An early version shows the look and feel, e.g. menu arrangements and colour schemes (1) / users give feedback and the client checks it meets the agreed requirements (1). (b) It focuses on what inputs are needed and what outputs must be produced, and the processes between them (1) / this identifies the data tables and the relationships needed (1)

5 State what is meant by a stub in a software prototype. [1]

A menu option, button or module that appears in the prototype but has no working functionality yet / a placeholder (1)

6 Distinguish between validation and verification, giving an example of each. [4]

Validation is an automatic check by the system that data is reasonable/follows rules (1), e.g. range check that a mark is 0–100 (1) / verification checks that data has been entered or copied accurately (1), e.g. double entry of a password or proof-reading against the source document (1)

7 A Derry hospital outpatient system records PatientNumber (three letters then seven digits), AppointmentDate and FollowUpDate. Name a suitable validation check for each field and give the exact rule. [3]

PatientNumber: format check – three letters followed by seven digits (1) / AppointmentDate: type/range check – a valid date, e.g. not in the past or within the next 12 months (1) / FollowUpDate: consistency (cross-field) check – must be after AppointmentDate (1). Accept presence or length checks with an exact rule.

8 State three items of information recorded about each field in a data dictionary. [3]

Any three from: field name (1) / data type (1) / length/size (1) / format (1) / description (1) / example value (1) / validation rule (1) / whether it is a key (1)

9 Describe two HCI principles that should be applied when designing an online renewal screen for Libraries NI members, explaining how each helps users. [4]

Two principles, each identified (1) and explained in context (1), e.g. consistency – the Renew button is in the same place on every page (1) so members learn the site quickly and make fewer mistakes (1) / feedback – a confirmation with the new due date is shown (1) so the member knows the renewal worked (1) / error prevention – choose from a list of current loans rather than typing book codes (1) / accessibility – large text and screen-reader labels (1) so visually impaired members can renew independently (1)

10 A Belfast leisure centre is developing a new membership and class-booking system. Evaluate the use of storyboarding and prototyping during the design of this system. [8]

Indicative content: Storyboarding – sketch membership, timetable, booking and payment screens; shows inputs/outputs and navigation; quick and cheap; staff and members need no technical knowledge; finds long booking routes early; but not interactive, limited detail, cannot show speed of booking at busy times. Prototyping – working model of class timetable/booking screen; members try look and feel; reveals data model (Member, Class, Booking, Instructor tables and relationships); better requirements, fewer misunderstandings; but cost and time, members may focus on colours not security of payment data, may expect quick delivery, early version may be accepted too soon. Sensible conclusion, e.g. storyboard first for navigation then a prototype of the booking process. Levels: 7–8 balanced evaluation of both techniques applied to the leisure centre with a justified conclusion and good technical language; 4–6 advantages and disadvantages of both with some context; 1–3 limited description of one technique.

E Quick quiz

Circle one answer · 9 marks

1 A designer sketches every screen of a new app, lists the inputs and outputs on each and draws arrows showing how users move between them. What technique is this? [1]

A Prototyping
B Storyboarding ✓
C Black-box testing
D Data conversion

2 Which is a disadvantage of prototyping rather than an advantage? [1]

A Problems are found earlier
B Users are more involved
C Users may expect the finished system to be ready very soon ✓
D Misunderstandings are reduced

3 In a prototype, a 'Print report' button that displays 'Coming soon' is an example of a... [1]

A stub ✓
B check digit
C storyboard
D patch

4 Which check ensures a field has not been left blank? [1]

A Range
B Format
C Type
D Presence ✓

5 Checking that a follow-up date is after the appointment date is a... [1]

A length check
B consistency (cross-field) check ✓
C check digit
D type check

6 Which HCI principle is shown by placing the Save button in the same place on every screen? [1]

A Feedback
B Accessibility
C Consistency ✓
D Verification

7 Proof-reading data on screen against the original paper form is... [1]

A verification ✓
B validation
C normalisation
D encryption

8 How can prototyping help design the data model? [1]

A It encrypts the tables
B It removes the need for a data dictionary
C It tests the hardware
D Focusing on inputs, outputs and processes reveals the tables and relationships needed ✓

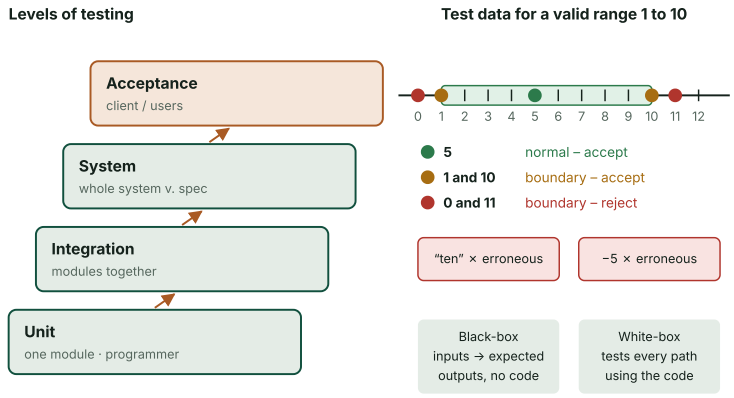
9 Which is a limitation of a storyboard? [1]

A It needs specialist knowledge to read
B It cannot show how responsive the system will be ✓
C It is expensive to produce
D It can only be made after coding

Testing and test plans

AP07 ANSWERS

I can: Explain why testing happens at every stage and why unit testing alone is not enough · Describe each stage of testing, who carries it out and where · Compare black-box and white-box testing using an example · Write a test plan with normal, boundary and erroneous data and specific expected outcomes



Use this diagram to help you: Levels of testing from unit to acceptance, and normal, boundary and erroneous test data on a number line for a valid range of 1 to 10

A Key terms

Write the letter of the matching definition · 6 marks

D Unit testing

F Integration testing

A System testing

E Alpha testing

C Beta testing

B Acceptance testing

A Testing the complete, integrated system against its agreed specification; essentially black-box testing of the whole system.

B The final test, by end users in their real working environment with genuine data at realistic volumes, to confirm the system meets their requirements before they sign it off.

C Testing of a pre-release version by a group of potential external users in a realistic environment, without developers present, who report any faults.

D Testing an individual module or subroutine on its own, usually in-house by the programmer, to check it works correctly in all circumstances.

E The final phase of in-house testing carried out by the developer, following a test plan, before the software is released to users.

F Testing units that have already been tested once they are combined, to check that they interact correctly and pass data between them as expected.

B Fill the gaps

Use the word bank · 10 marks

integration testing | beta testing | test plan | acceptance testing | expected outcome | alpha testing | Unit testing | white-box testing | black-box testing | boundary data

Unit testing checks each module on its own, and **integration testing** checks that tested modules work together. In **black-box testing** the tester cannot see the code, while **white-box testing** uses the code to

test every branch. The developer's final in-house phase is **alpha testing**; a pre-release version is then given to potential users for **beta testing** . In **acceptance testing** end users use real data before they sign off the system. A **test plan** records the test data and the **expected outcome** for each test, and values at the edges of a valid range are **boundary data**.

C True or false?

Tick one box · 9 marks

	True	False
If every unit passes its own tests, the integrated system is certain to work. *(Integration errors, such as units using different date formats, only appear when units are joined, so integration testing is still needed.)*	☐	☑
White-box testing is done without looking at the code. *(That is black-box testing; white-box testing uses knowledge of the code to test every path and branch.)*	☐	☑
System testing is essentially black-box testing applied to the whole system.	☑	☐
Alpha testing is carried out in-house by the developer.	☑	☐
Beta testers work alongside the developers in the developer's offices. *(Beta testers are external potential users who test a pre-release version in a realistic environment without the developers present.)*	☐	☑
Acceptance testing uses real volumes of data in a live environment and ends with users signing off the software.	☑	☐
The actual outcome column of a test plan is filled in before testing begins. *(The expected outcome is written in advance; the actual outcome and PASS/FAIL are completed by the tester during testing.)*	☐	☑
For whole-number marks from 0 to 100, the values 0 and 100 are boundary data that should be accepted.	☑	☐
Successful testing proves that a program contains no errors. *(Testing can show that errors are present but can never prove that none remain.)*	☐	☑

D Exam-style questions

39 marks

1 Explain why testing should take place throughout the development of a system rather than only at the end. [3]

> Any three from: errors are cheaper/easier to fix when found early, near where they were made (1) / each unit can be checked before being combined, narrowing the search for faults (1) / integration errors only appear when units are joined so must be tested at that stage (1) / the system must be checked against requirements before handover (1) / confidence/quality is built up at each stage (1)

2 Describe what happens during unit testing and integration testing. [4]

> Unit: each individual module/component is tested on its own (1) / in-house by the developer/programmer to check it works in all circumstances, using black-box or white-box methods (1). Integration: tested units are combined (1) / to check they work together and data passes correctly between them/at their interfaces (1)

3 Explain why a system made of units that have all passed unit testing may still fail integration testing. Use an example. [3]

> Units may expect data in different forms, so faults appear at the interfaces between them (1) / each unit's own tests would not reveal this (1) / example, e.g. one unit stores dates as dd/mm/yyyy and another reads

mm/dd/yyyy, so dates are misread when combined (1)

4 Compare black-box and white-box testing. [4]

Black box: tester has no access to/knowledge of the code (1) / tests inputs against expected outputs from the specification and can be done by non-technical testers (1). White box: tester has access to the code (1) / tests are designed to exercise every path/branch/loop, focusing on where errors may occur, needs programming knowledge (1)

5 State two differences between alpha testing and beta testing. [2]

Any two from: alpha is in-house, beta is external (1) / alpha is by the developer's staff, beta by potential users (1) / alpha comes before beta (1) / beta is in a realistic environment without developers present (1) / alpha may use simulated data (1)

6 Describe acceptance testing. [3]

Carried out when the software is ready to be handed over, after system testing (1) / by end users/the client in their real working environment using genuine data at realistic volumes (1) / to confirm it meets their requirements; problems are reported and resolved, then users sign off the software (1)

7 State four components that should be recorded for each test in a test plan. [4]

Any four from: test number (1) / part of the system being tested (1) / what is being tested/purpose/type of data (1) / test data to be used (1) / expected outcome (1) / actual outcome (1) / pass or fail (1)

8 A Fermanagh activity centre charges by age: Junior 5 to 11, Teen 12 to 17, Adult 18 or over. Ages are whole numbers. Suggest (a) one normal value for Teen, (b) both boundary values for Junior, (c) the boundary value for the lower limit of Adult and (d) two erroneous values, stating the expected outcome for (d). [6]

(a) any whole number 13–16, e.g. 14 (1) / (b) 5 (1) and 11 (1) / (c) 18 (1) / (d) two from: 4, a decimal such as 12.5, text such as 'ten', a negative number, blank (1) / expected outcome: rejected with an error message (1). Note: 12 and 17 are boundary values for Teen, so do not credit them as normal data.

9 A further education college in Newry is testing its new online enrolment system, and the systems analyst who wrote the specification remains on the project. Explain two ways the analyst could contribute to testing. [2]

The analyst could suggest test data, especially boundary values such as the minimum and maximum ages for enrolment (1) / and could compare the test results with the requirements in the specification to confirm the system does what was agreed (1)

10 A software house is building a new patient appointment system for a group of GP practices in Northern Ireland. Discuss the stages of testing that should take place before the system goes live, and why each is needed. [8]

Indicative content: unit testing of modules (booking, patient records, reminders) by programmers with black-box and white-box tests; integration testing to catch interface errors such as appointment times or dates passed in different formats between the booking and SMS reminder units; system testing of the whole system against the specification including performance at busy times and security of patient data; alpha testing in-house as final developer phase; possibly beta testing with one practice in a realistic setting without developers; acceptance testing by practice staff with real volumes of data in a live environment, checking it works with existing GP systems, ending in sign-off. Why: patient safety, confidentiality, errors cost more if found late, staff confidence. Levels: 7–8 all stages described accurately in context with clear reasons and good terminology; 4–6 most stages described with some context or reasons; 1–3 a few stages named or vaguely described.

E Quick quiz

Circle one answer · 9 marks

1 Two modules each pass their own tests, but when joined one reads dates as day/month/year and the other as month/day/year. Which stage of testing should find this? [1]

A Unit testing **B Integration testing ✓**

C Beta testing

D White-box testing of one module

2 A games studio releases a pre-release version to 2 000 volunteer players who play at home and report bugs. This is... [1]

A alpha testing

B acceptance testing

C beta testing ✓

D unit testing

3 Testing a single subroutine on its own is... [1]

A unit testing ✓

B system testing

C acceptance testing

D beta testing

4 Which testing approach designs tests so that every branch of an IF statement is run? [1]

A Black-box

B Beta

C Acceptance

D White-box ✓

5 Valid range 10 to 20 whole numbers. Which is erroneous data? [1]

A 15

B 10

C twelve ✓

D 20

6 Which test is carried out by end users with real volumes of data, before they sign off the system? [1]

A Acceptance ✓

B Alpha

C Integration

D Unit

7 Grades: Fail 0–39, Pass 40–59. Which pair are both boundary values? [1]

A 20 and 50

B 39 and 40 ✓

C 41 and 58

D −1 and 101

8 Who normally completes the 'Actual outcome' and 'PASS/FAIL' columns of a test plan? [1]

A The client's accountant

B The test designer, before testing

C The tester, while carrying out the tests ✓

D The beta testers' manager

9 System testing checks the complete system against... [1]

A competitors' products

B the agreed specification ✓

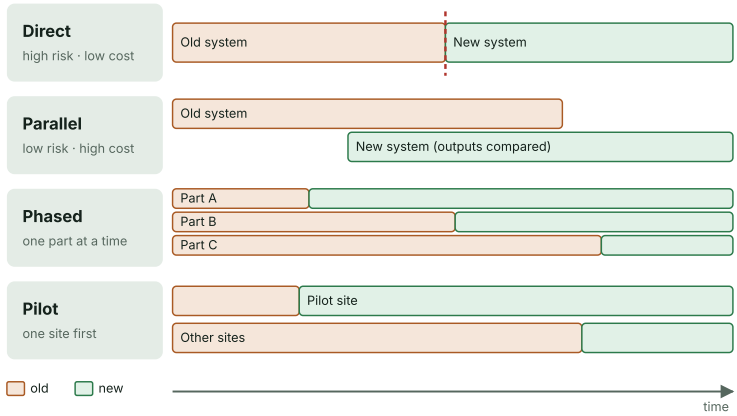
C the old system's source code

D the hardware warranty

Implementation: changeover, data conversion and training

AP08 ANSWERS

I can: Describe what implementation involves and why it must be planned · Describe the four changeover methods, with a benefit and a drawback of each · Choose and justify the most suitable changeover method for an organisation · Explain data conversion from a paper-based or older computer system · Compare methods of training users



Use this diagram to help you: Timelines comparing direct, parallel, phased and pilot changeover, showing when the old and new systems are in use

A Key terms

Write the letter of the matching definition · 6 marks

C Implementation

F Changeover

D Parallel changeover

E Direct changeover

A Pilot changeover

B Phased changeover

A A changeover method in which the whole new system is introduced in one branch, department or site before being extended to every other part of the organisation.

B A changeover method in which the new system is introduced one component or function at a time.

C The stage at which a new system is put into operation, including installation, data conversion, training, documentation and changeover from the old system.

D A changeover method in which the old and new systems run at the same time on the same transactions so that outputs can be compared until the new system is proven.

E A changeover method in which the old system stops and the new one takes over at a single moment, with no overlap.

F The process of moving from an old system to a new system, using a planned method: parallel, direct, pilot or phased.

B Fill the gaps

Use the word bank · 9 marks

direct changeover | phased changeover | verification | parallel changeover | big bang | Pilot changeover | Implementation | data conversion | cascade training

Implementation is the stage when the new system is put into operation. In **parallel changeover** the old and new systems run together and their outputs are compared. In **direct changeover**, also called the **big bang** approach, the old system stops and the new one starts on a set date. **Pilot changeover** introduces the whole system in one branch first, while **phased changeover** introduces one component at a time. Records from the old system are put into the new format by **data conversion**, and entered data is checked by **verification**. In **cascade training** a few staff are trained and then train their colleagues.

C True or false?

Tick one box · 9 marks

	True	False
Implementation should be planned while the system is still being developed.	☑	☐
Parallel changeover is the cheapest changeover method. *(It is the most expensive, because both systems must be resourced and every transaction is processed twice.)*	☐	☑
Direct changeover is the riskiest method because there may be nothing to fall back on if the new system fails.	☑	☐
Phased changeover means introducing the whole system in one location first. *(That is pilot changeover; phased changeover introduces one component of the system at a time.)*	☐	☑
In pilot changeover, staff who used the system first can help train colleagues elsewhere.	☑	☐
Data conversion from a paper system only involves scanning the documents; no checking is needed. *(The new database structure must be created, records keyed in or scanned, and the data then verified and validated.)*	☐	☑
Parallel running can lead to two sets of data that are not fully synchronised.	☑	☐
Phased changeover is usually the quickest way to complete a changeover. *(Phased changeover can take a long time because parts are introduced one after another; direct is quickest.)*	☐	☑
E-learning lets staff learn at their own pace but gives no immediate help with problems.	☑	☐

D Exam-style questions

35 marks

1 Describe the purpose of system implementation. [2]

> To put the new system into operation/replace the old system (1) / ensuring resources such as trained staff, hardware and software are in place and data is converted so the system can become fully operational with minimum disruption (1)

2 Describe parallel changeover, giving one benefit and one drawback. [3]

> Both systems operate at the same time, each transaction processed by both and outputs compared, until the new system is proven (1) / benefit: very safe, the old system is a backup if the new one fails / staff can get used to the new system (1) / drawback: expensive/time-consuming to run two systems / time needed to compare results / two sets of data may not be synchronised (1)

3 Describe direct changeover, giving one benefit and one drawback. [3]

> On a chosen date the old system stops and the new one takes over straight away, with no overlap (1) / benefit: no work is done twice / all staff switch at once / cheapest and quickest (1) / drawback: high risk, should the new

system break down, going back may be hard/impossible / there may be a period with no system available (1)

4 Explain the difference between pilot and phased changeover. [2]

Pilot: the whole new system is introduced in one section/branch first and then rolled out to the rest (1) / phased: one part/component of the system is introduced at a time across the organisation (1)

5 Identify the changeover method in each case: (a) a sports shop chain trials new tills in its Coleraine branch before every other branch; (b) a dentist keeps paper appointment cards for a month while also using new booking software; (c) a hotel introduces new room booking, then restaurant ordering, then billing, one at a time; (d) a library switches off its old catalogue on Friday and launches the new one on Monday. [4]

(a) pilot (1) / (b) parallel (1) / (c) phased (1) / (d) direct (1)

6 A Lisburn dog-grooming business keeps customer details in a notebook and takes bookings in a paper diary. Explain what data conversion would involve if it introduces a computerised booking system. [4]

Data conversion means transferring existing records into the structure and format the new system expects (1) / a database structure must be created, e.g. Customer, Dog and Booking tables (1) / customer details from the notebook and future bookings from the diary are keyed in (1) / the entered data must be verified against the paper records and validated, e.g. phone numbers (1). Credit: repeat customers stored once rather than rewritten each time (1). Max 4.

7 Explain one reason why direct changeover may be the only realistic choice for an organisation. [2]

Old and new systems cannot run at the same time, e.g. the change affects everyone at once / there are insufficient staff or resources to run both (1) / so the switch must happen at a single point, which should be carefully planned and timed for a quiet period (1)

8 Describe two methods of training staff to use a new system, giving an advantage of each. [4]

Two methods, each described (1) with an advantage (1), e.g. instructor-led course (1) – staff can ask questions and practise hands-on (1) / e-learning or video tutorials (1) – learn at own pace and revisit material, cheap per person (1) / cascade training (1) – cheap for a large organisation, creates local experts (1) / one-to-one (1) – tailored to the individual (1)

9 A supermarket chain with 25 stores across Northern Ireland is introducing a new stock control system. Justify the use of pilot changeover. [3]

Any three, applied to the context: the system can be tried in one store first (1) so any problems affect only that store and the other 24 continue on the old system (1) / the stores are similar, so success in one store is a good guide to the rest (1) / staff from the pilot store can help train staff in other stores (1) / avoids the cost of running parallel systems in all 25 stores (1)

10 The Northern Health and Social Care Trust is replacing the system that schedules operations in its hospitals. Evaluate the four changeover methods and recommend the most suitable one for this situation. [8]

Indicative content: Direct – cheapest and quickest but unacceptable risk; failure could cancel operations or endanger patients; no fallback. Parallel – safest, old system as backup, results compared; but double entry for busy clinical/admin staff, two schedules may conflict, expensive. Pilot – one hospital first (e.g. Antrim Area), problems contained, pilot staff train others; but sharing schedules between hospitals on different systems may be hard, extra resources. Phased – e.g. theatre booking first then staff rostering; staff learn in stages; long overall; parts must interface with old system. Recommendation should be justified, e.g. pilot in one hospital with parallel running there, or parallel throughout because of patient safety. Levels: 7–8 all four methods evaluated with benefits and drawbacks applied to the Trust and a clear justified recommendation; 4–6 most methods described with some evaluation and a recommendation; 1–3 methods described with little application.

E Quick quiz

Circle one answer · 9 marks

1 A bank installs a new system but also keeps running its old system for three months, comparing the results of both. Which changeover method is this? [1]

A Direct
B Parallel ✓
C Pilot
D Phased

2 A council introduces its new planning system in its Omagh office before all its other offices. Which method is this? [1]

A Phased
B Direct
C Pilot ✓
D Parallel

3 Which is a drawback of direct changeover? [1]

A There may be nothing to go back to if the new system fails ✓
B Every transaction must be entered twice
C It takes the longest of all methods
D Two systems run in different branches

4 A leisure centre refurbishes its gym, then its pool, then its reception, finishing each before starting the next. This is most like which changeover method? [1]

A Pilot
B Parallel
C Direct
D Phased ✓

5 Which is the best description of data conversion? [1]

A Training staff to use the new system
B Transferring existing records into the format the new system expects ✓
C Fixing bugs after the system goes live
D Running two systems together

6 In which situation is parallel running most justified despite its cost? [1]

A A café's menu display screen
B A school's lunchtime music playlist
C A payroll system where errors mean staff are paid wrongly ✓
D A personal blog

7 Which is a drawback of phased changeover? [1]

A The whole changeover can take a long time ✓
B Everyone changes at once
C No training is possible
D Data never needs converting

8 Training a few staff who then train their colleagues is called... [1]

A e-learning
B beta testing
C cascade training ✓
D parallel training

9 Why was Sweden's 1967 switch to driving on the right a direct changeover? [1]

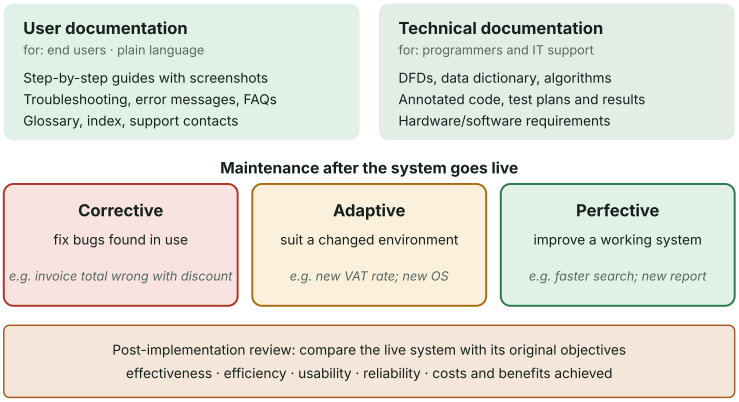
A It was the cheapest software option
B Driving on both sides at once was impossible ✓
C It was tested in one town first
D It was introduced one road at a time

AS 1: APPROACHES TO SYSTEMS DEVELOPMENT · SYSTEM IMPLEMENTATION

Documentation, maintenance and evaluation

AP09 ANSWERS

I can: Explain why documentation is produced throughout development · Compare user and technical documentation by audience and content · Classify maintenance tasks as corrective, adaptive or perfective and justify the choice · Explain who usually pays for each type of maintenance and what it reveals about the project · Describe how a system is evaluated after implementation



Use this diagram to help you: User and technical documentation compared, the three types of maintenance with examples, and the post-implementation review

A Key terms

Write the letter of the matching definition · 6 marks

C **User documentation**

F **Technical documentation**

A **System maintenance**

E **Corrective maintenance**

B **Adaptive maintenance**

D **Perfective maintenance**

A Any change applied to a system during its lifespan after it has been implemented.

B Changes made so that the system can still carry out its tasks after a change in circumstances, internal or external, such as new legislation, a new tax rate or a new business policy.

C Non-technical documentation for end users, such as an overview, installation instructions, user guide, troubleshooting, FAQs and backup procedures.

D Changes that improve the performance or usability of a system that already works, after initial delivery.

E Changes made to fix flaws, errors or weaknesses, including security flaws, discovered after the system is in use.

F Documentation for administrators and maintenance staff, including specifications, DFDs and data models, I/O and report designs, annotated code, test plans and results, and hardware/software configuration.

B Fill the gaps

Use the word bank · 9 marks

adaptive maintenance | perfective maintenance | User documentation | troubleshooting | annotated code | Technical documentation | software patch | corrective maintenance | post-implementation review

User documentation uses plain language and includes a user guide, a **troubleshooting** section and FAQs. **Technical documentation** is for maintenance staff and includes data flow diagrams, test plans

and **annotated code**. Fixing a flaw found after go-live is **corrective maintenance**, often supplied as a **software patch**. Changing the system because of a new VAT rate is **adaptive maintenance**. Making a working system faster or easier to use is **perfective maintenance**. A **post-implementation review** evaluates the system against its original objectives.

C True or false?

Tick one box · 9 marks

	True	False
Documentation should only be written once the system is finished. *(Documentation is produced at every stage, providing a starting point for the next stage and a record of work done.)*	☐	☑
Annotated source code belongs in the technical documentation.	☑	☐
A troubleshooting section and FAQs are typical parts of user documentation.	☑	☐
Adding a feature that users have requested to a working system is corrective maintenance. *(It is perfective maintenance; corrective maintenance fixes errors or weaknesses.)*	☐	☑
Updating a system for a new VAT rate is adaptive maintenance.	☑	☐
Corrective maintenance caused by programmer error is usually charged to the client. *(It is usually fixed free of charge by the developer; otherwise it may be covered by a maintenance contract.)*	☐	☑
The amount of corrective maintenance should decrease over time as errors are removed.	☑	☐
A large amount of adaptive maintenance soon after delivery may suggest the requirements were poorly analysed.	☑	☐
A post-implementation review takes place before the system goes live. *(It takes place after the system has been in use for a period, measuring it against the original objectives.)*	☐	☑

D Exam-style questions

38 marks

1 A software house in Banbridge is building a records system for a veterinary practice. Explain why the team should write documentation throughout the project rather than only at the end. [4]

> Any four from: each stage's documents give the next stage something to work from, e.g. programmers code from the design document (1) / they record what was done and decided at each stage (1) / they help later maintenance by showing how the records system was built and what has changed (1) / new staff can pick up the work if team members leave (1) / the finished system can be checked against what the practice agreed (1)

2 State three components of user documentation for an online appointment system used by patients of a Ballymena GP surgery. [3]

> Any three from: overview/introduction to the system (1) / hardware/software requirements and installation instructions (1) / user guide with step-by-step instructions/screenshots, e.g. how to book or cancel an appointment (1) / troubleshooting section/error messages explained (1) / FAQ section (1) / training materials/tutorials (1) / backup/maintenance procedures (1)

3 A Dungannon engineering firm's in-house IT technician has to maintain a stock system written by a contractor who has since left. State three components of technical documentation that the technician would need. [3]

> Any three from: system specification/user requirements (1) / data models, DFDs, ER diagrams, data dictionary (1) / input/output design (1) / report specifications (1) / query designs (1) / program code listings/annotated code (1) / test plans with data used and results (1) / detailed hardware and software configuration (1)

4 Explain why annotated program code is an important part of technical documentation. [2]

> Comments explain the purpose and logic of each part of the code (1) / so maintenance programmers, who may not have written it, can understand and change it quickly and safely, reducing cost and the risk of new errors (1)

5 A Newry credit union's members' system has been live for eighteen months. Explain what is meant by corrective, adaptive and perfective maintenance, giving an example of each that could apply to this system. [6]

> Corrective: fixing errors or weaknesses found while the system is in use (1), e.g. interest on savings calculated wrongly or a security flaw in member log-in patched (1) / Adaptive: changing the system because of a change in circumstances, internal or external (1), e.g. new financial regulation, a new branch opening, a new operating system (1) / Perfective: improving the performance or usability of a working system after delivery (1), e.g. faster member searches, a mobile-friendly loan application form, clearer statements (1)

6 A housing association in Derry/Londonderry uses a rent system. Classify each change as corrective, adaptive or perfective: (a) the system must handle a new rent-support scheme introduced by the government; (b) tenants with two-part surnames receive letters addressed to the wrong name; (c) managers ask for the arrears report to be shown as a chart; (d) after two associations merge, tenant records must also show which of the two former organisations owned each property. [4]

> (a) adaptive (1) / (b) corrective (1) / (c) perfective (1) / (d) adaptive (1)

7 Explain why adaptive maintenance is usually paid for by the client while corrective maintenance caused by programmer error is not. [2]

> Adaptive maintenance adds features beyond those in the original specification/agreed requirements, so the client pays (1) / corrective maintenance due to programmer error fixes something the developer should have delivered correctly, so the developer fixes it free (1)

8 Describe what is meant by a software patch. [2]

> A small piece of software/update used to fix, update or improve a program or its data (1) / usually downloaded and often installed automatically, e.g. to close a security vulnerability (1)

9 Describe the purpose of a post-implementation review and the evidence it might use. [4]

> Purpose: a formal evaluation after the system has been in use for some time (1) / to judge whether it meets the original objectives/requirements and identify problems, maintenance needs and lessons for future projects (1) / evidence, any two from: user questionnaires/interviews (1) / performance statistics, e.g. response times (1) / help-desk/error logs (1) / financial records compared with the cost–benefit forecast (1). Max 4.

10 A County Down estate agency has used a new property-management system for a year and has requested many changes. Evaluate how good documentation and the different types of maintenance affect the long-term cost and success of the system. [8]

> Indicative content: technical documentation (specifications, ER diagrams, annotated code, test results) lets new or different programmers make changes quickly and safely, lowering maintenance cost; out-of-date documentation increases cost and risk; user documentation reduces help-desk calls and training costs and supports staff turnover. Corrective maintenance: fixes errors, often patches; free if programmer error; should fall over time; many corrective changes suggest weak testing. Adaptive: changes in law (e.g. lettings regulation), new branches, new hardware; paid by client; many requests may show poor analysis of requirements. Perfective: improves usability or performance, e.g. mobile interface for agents, faster property search; improves benefit but costs money. Overall: maintenance often costs more than development; good documentation and modular design keep it affordable; regular review helps prioritise. Levels: 7–8 well-balanced evaluation of documentation and all three maintenance types applied to the estate agency, with a supported conclusion; 4–6 describes documentation and maintenance types with some application; 1–3 limited, generic description.

E Quick quiz

Circle one answer · 9 marks

1 A system is changed because the government introduces a new rate of VAT. Which type of maintenance is this? [1]

A Corrective
B Perfective
C Adaptive ✓
D Preventive testing

2 Who is technical documentation mainly written for? [1]

A Customers of the business
B Administrators and maintenance programmers ✓
C Shareholders
D Beta testers only

3 Making a slow report run much faster is... [1]

A perfective maintenance ✓
B corrective maintenance
C adaptive maintenance
D acceptance testing

4 Which item belongs in user documentation? [1]

A Data dictionary
B Annotated code listing
C Test results
D Troubleshooting guide ✓

5 Shortly after go-live, a system crashes whenever a surname contains an apostrophe. The fix is... [1]

A adaptive maintenance
B corrective maintenance ✓
C perfective maintenance
D data conversion

6 What is a software patch? [1]

A A small update that fixes or improves a program ✓
B A type of storyboard
C A changeover method
D A user guide

7 Which is NOT a reason for producing documentation at each stage? [1]

A To give the following stage something to build on
B To aid maintenance
C To replace the need for testing ✓
D To cope with changes of staff in the team

8 A post-implementation review compares the system with... [1]

A competitors' systems
B the source code
C the original objectives and requirements ✓
D the old hardware

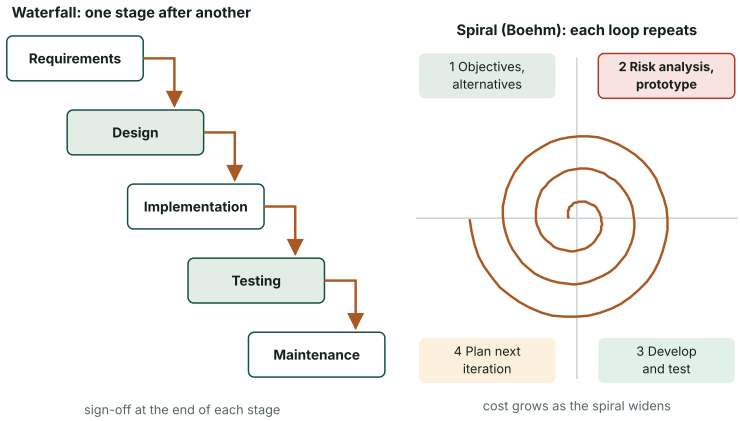
9 Who usually pays for adaptive maintenance? [1]

A The client organisation ✓
B The original programmer personally
C Nobody – it is always free
D The beta testers

The waterfall model, iterative and spiral development

AP10 ANSWERS

I can: Describe the five stages of the waterfall model and the deliverable from each · Explain when users are involved in a waterfall project and why that matters · Evaluate the advantages and disadvantages of the waterfall model · Describe iterative development and the four quadrants of the spiral model · Recommend a suitable development model for a given project and justify the choice



Use this diagram to help you: The waterfall model as a cascade of sequential stages beside the spiral model with its four quadrants and risk analysis in every loop

A Key terms

Write the letter of the matching definition · 6 marks

B Waterfall model

F Deliverable

D System specification

E Design document

C Handover

A Iterative development

A An approach in which a system is built through repeated cycles of planning, design, coding and testing, each producing a version that users review.

B A sequential development methodology in which distinct stages are completed in order, each producing a deliverable and being signed off before the next stage begins.

C The point at which work passes from one waterfall stage, and often one group of staff, to the next; misunderstandings introduced here may travel undetected through later stages.

D The deliverable of the requirements stage: a clear, unambiguous and complete document of the user requirements for the new system.

E The deliverable of the design stage, describing processes, inputs and outputs, software units, user interfaces and hardware precisely enough for programmers to code from.

F The end product of a stage of development, such as a system specification or design document, which is passed on as the input to the next stage.

B Fill the gaps

Use the word bank · 10 marks

system specification | risk analysis | waterfall model | deliverable | spiral model | acceptance testing | design document | iterative development | handover | prototype

The **waterfall model** is a sequential methodology in which each stage must be completed and signed off before the next begins. Each stage ends with a **deliverable**; for example the requirements stage produces the **system specification** and the design stage produces the **design document**. Misunderstandings can enter at each **handover** and may not be found until testing. Users are mainly involved during requirements and **acceptance testing**. By contrast, **iterative development** builds the system in repeated cycles, and the **spiral model** adds **risk analysis** to every loop, often using a **prototype**.

C True or false?

Tick one box · 9 marks

	True	False
In the waterfall model, a stage has to be finished and signed off before work on the following stage starts.	☑	☐
In the waterfall model, users are closely involved at every stage of development. *(Users are mainly involved during requirements gathering and acceptance testing; in between they usually see nothing working.)*	☐	☑
The deliverable of the requirements stage is the system specification.	☑	☐
Once a waterfall stage is signed off, it can never be revisited. *(If a later stage uncovers a mistake, the team may have to go back to an earlier stage, but this is slow and expensive.)*	☐	☑
A misunderstanding introduced at a handover between stages may not be detected until testing.	☑	☐
The waterfall model suits projects with stable, well-understood requirements.	☑	☐
The spiral model does not include risk analysis. *(Risk analysis in every loop is the spiral model's defining feature.)*	☐	☑
The spiral model is the most cost-effective choice for every small project. *(Its risk analysis adds cost and needs expertise that a small, low-risk project does not justify.)*	☐	☑
Iterative development delivers working versions early so that users can give feedback.	☑	☐

D Exam-style questions

34 marks

1 State two stages of the waterfall model and the deliverable produced by each. [2]

> Any two stage + deliverable pairs: requirements/analysis → system specification (1) / design → design document (1) / implementation → program code (1) / testing → test results / tested system (1) / maintenance → updated, supported system (1). Both stage and deliverable needed for each mark.

2 A software firm in Ballymena is using the waterfall model to build a payroll system for a haulage company. Describe the key characteristics of the way this project will be run. [4]

> Any four from: consists of a sequence of distinct stages (1) / e.g. analysis, design, implementation, testing, maintenance (1) / each stage must be completed/signed off before the next begins (1) / each stage produces a deliverable (1) / the deliverable is the input to the next stage (1) / the team may have to return to an earlier stage if a later stage reveals a mistake (1) / user mainly involved at analysis and testing (1).

3 Describe what happens during the design stage of the waterfall model. [3]

Any three from: starts from the system specification (1) / identifies the processes and the inputs and outputs of each (1) / designs the software modules and how they communicate with one another (1) / designs the user interface, e.g. storyboards (1) / identifies hardware needs (1) / produces a precise, unambiguous design document (1).

4 A Lisburn school's office staff will use a new attendance system built with the waterfall model. Describe when and how the staff would be involved during its development. [2]

Office staff are involved at the requirements/analysis stage, e.g. interviewed so the analyst can establish their requirements (1) / and at the testing stage, carrying out acceptance testing to check the system works as they intended (1).

5 A software house in Lisburn develops systems using the waterfall model. Explain two advantages to the company of using this model. [4]

Two explained advantages, e.g.: distinct stages with deliverables (1) so each can be signed off and the client invoiced in stages (1) / staff with specialist skills can work on one stage (1) and then move to the same stage of another project, using expertise efficiently (1) / thorough documentation at each stage (1) makes maintenance easier later (1) / clear milestones (1) make progress easy to monitor and costs easier to estimate (1).

6 Explain why an error made during the requirements stage of a waterfall project can be expensive to correct. [3]

The error is passed on in the system specification to the design stage (1) / design and code are built on the wrong requirement (1) / it may not be detected until testing / acceptance testing (1) / so specification, design, code and tests all have to be reworked, increasing cost and delay (1). Max 3.

7 Describe the four quadrants that make up each loop of the spiral model. [4]

Determine objectives, alternatives and constraints for the loop (1) / evaluate alternatives and identify and resolve risks, e.g. using a prototype (1) / develop and test the next level/version of the product (1) / review with the client and plan the next iteration (1).

8 Explain one benefit of iterative development for the users of a new system. [2]

Users see a working version early (1) so misunderstandings/errors are identified and corrected sooner / their feedback shapes the next version (1) / OR some functionality is delivered early (1) so users gain benefit before the whole system is finished (1).

9 Explain why the spiral model may not be suitable for a small, low-risk project. [2]

Risk analysis in every loop requires specialist expertise and adds cost/time (1) / a small, low-risk project has few risks to analyse, so the overhead is not justified (1).

10 The Northern Health and Social Care Trust plans a new system to schedule community nursing visits. Requirements are partly known, the system is large and errors could affect patient safety. Evaluate the waterfall model and the spiral model for this project and recommend one. [8]

Indicative content – Waterfall: clear stages and deliverables, formal sign-off and full documentation (valuable for patient safety and audit); easy to monitor and estimate; but requirements are only partly known, so changes later are costly; nurses would not see the system until testing; misunderstandings at handovers may surface late with safety implications. Spiral: risk analysis each loop tackles safety and scheduling risks early; prototypes let nurses and managers refine requirements; suits large, complex, high-risk systems; but expensive, needs risk expertise and is harder to manage and close. Recommendation: spiral (or another iterative approach) is likely more suitable because requirements are incomplete and the risks are high, provided budget and expertise exist. Levels: 7–8 balanced evaluation of both models applied to the scenario with justified recommendation and good use of terminology; 4–6 some evaluation of both, partly applied, recommendation may be weakly justified; 1–3 description of one or both models with little reference to the scenario.

E Quick quiz

Circle one answer · 9 marks

1 Which statement best describes the waterfall model? [1]

A Stages repeat in short cycles with constant user feedback

B A prototype is refined until it becomes the final system

C Distinct stages are completed in sequence, each signed off before the next begins ✓

D The team works in fixed-length sprints from a product backlog

2 During which two stages are users mainly involved in a waterfall project? [1]

A Requirements and acceptance testing ✓

B Design and implementation

C Implementation and maintenance

D Design and testing

3 What is the deliverable of the design stage? [1]

A The system specification

B The test results

C A design document ✓

D The product backlog

4 Why are errors in the requirements stage so costly in the waterfall model? [1]

A Programmers refuse to change code

B They may not be found until testing, after design and code are built on them ✓

C Requirements are never documented

D Testing is skipped in waterfall

5 The distinctive feature of the spiral model is... [1]

A no documentation

B pair programming

C direct changeover

D risk analysis in every loop ✓

6 Which project is best suited to the waterfall model? [1]

A A start-up app with unclear needs

B A payroll change for a known new tax rule ✓

C An experimental game with changing features

D A new product where users want weekly releases

7 Each repeated cycle in iterative development is called... [1]

A a milestone

B a deliverable

C an iteration ✓

D a handover

8 Which is an advantage of the waterfall model? [1]

A Clear milestones allow the client to be invoiced stage by stage ✓

B Requirements can change easily at any point

C Users see working software from the first week

D No documentation is needed

9 A disadvantage of the spiral model is that it... [1]

A cannot handle risk

B has no user involvement

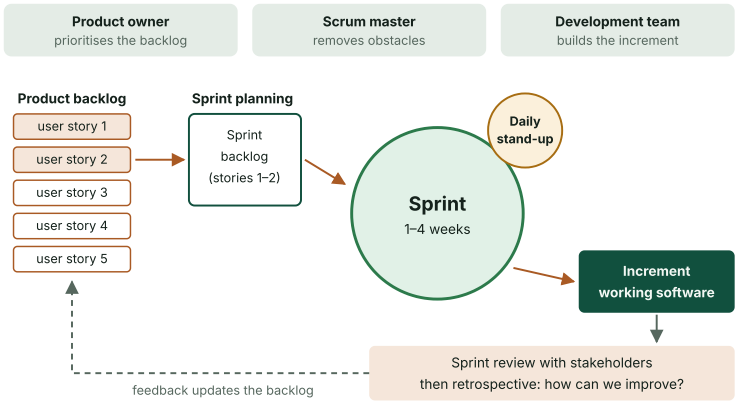
C never uses prototypes

D can be expensive and needs risk-analysis expertise ✓

Prototyping, RAD and Agile methods

AP11 ANSWERS

I can: Compare throwaway and evolutionary prototyping · Describe the four phases of RAD and explain how timeboxing and prioritised requirements work · Describe how an Agile project uses backlogs, sprints, Scrum roles, daily Scrums and burn down charts · Interpret a burn down chart to judge whether a sprint is on schedule · Evaluate waterfall, RAD and Agile for a given scenario and justify a recommendation



Use this diagram to help you: The Scrum framework: a product backlog feeds sprint planning, the team works in a 1–4 week sprint with daily stand-ups, producing an increment that is reviewed

A Key terms

Write the letter of the matching definition · 6 marks

D Prototype

F Throwaway prototyping

A Evolutionary prototyping

E Rapid application development (RAD)

B Timeboxing

C Product backlog

A An approach in which a prototype is repeatedly refined in response to user feedback until it becomes the final system.

B Fixing the time allowed for a project or part of it so that the deadline cannot be overrun; lower-priority features are dropped instead.

C In Agile, the prioritised wish list of all the features the product should eventually provide.

D An early model of all or part of a system, built quickly so that users can evaluate it and clarify their requirements.

E A methodology that delivers systems quickly through four phases (requirements, user design, construction, cutover), frequent prototypes, close user involvement and a strict timescale with prioritised requirements.

F An approach in which a prototype is used to establish requirements and is then discarded before the real system is built.

B Fill the gaps

Use the word bank · 9 marks

sprints | Scrum master | product backlog | Rapid application development | timeboxing | prototype | cutover | evolutionary prototyping | burn down chart

A **prototype** is an early model of a system that users can try. In **evolutionary prototyping** it is refined until it becomes the final system. **Rapid application development** has four phases, ending with

cutover, and uses **timeboxing** so that low-priority features are dropped rather than the deadline. In Agile, the **product backlog** is the prioritised wish list, and work is done in fixed periods called **sprints**. The **Scrum master** removes obstacles, and progress is shown on a **burn down chart**.

C True or false?

Tick one box · 10 marks

	True	False
A throwaway prototype is discarded once the requirements have been agreed.	☑	☐
In RAD, the deadline is extended until every requirement has been implemented. *(RAD uses a strict timescale; requirements are prioritised and desirable add-ons are dropped if time runs out.)*	☐	☑
The four phases of RAD are requirements planning, user design, construction and cutover.	☑	☐
Users are involved more continually in RAD than in the waterfall model.	☑	☐
The Agile Manifesto values comprehensive documentation over working software. *(The Agile Manifesto values working software over comprehensive documentation.)*	☐	☑
Features selected for the next release are moved from the product backlog into the release backlog.	☑	☐
On a burn down chart, an actual line above the ideal line means the team is ahead of schedule. *(Above the ideal line means more work remains than planned, so the team is behind schedule.)*	☐	☑
The sprint retrospective reviews the finished product with the product owner. *(The sprint review examines the product; the retrospective reflects on the team's process.)*	☐	☑
In test-driven development, tests are written before the code.	☑	☐
A possible disadvantage of Agile is that documentation may be less detailed than in traditional methods.	☑	☐

D Exam-style questions

37 marks

1 Explain the difference between throwaway and evolutionary prototyping. [2]

> Throwaway: the prototype is discarded once requirements are agreed and the real system is built separately (1) / Evolutionary: the prototype is refined repeatedly until it becomes the final system (1).

2 A Craigavon developer promises a hotel group a working room-booking system 'in twelve weeks, guaranteed' using Rapid Application Development (RAD). Describe the features of RAD that make such a promise possible. [4]

> Any four from: project divided into a series of builds (1) / strict timescale/timebox that cannot be overrun (1) / requirements ranked in priority order (1) / desirable add-ons dropped if time runs out (1) / prototypes produced frequently and delivered to users for feedback (1) / much more user involvement than waterfall (1) / four phases: requirements, user design, construction, cutover (1) / reuse of components / GUI builders (1).

3 Name the four phases of RAD and briefly describe what happens in the user design phase. [4]

> Requirements (planning), user design, construction, cutover (2 marks for all four in any order; 1 mark for two or three) / User design: users and analysts interact regularly (1) / prototypes/mock-ups show users what the system will look like and test alternative interfaces (1).

4 Reception staff at the hotel group will help develop the booking system using RAD. Describe how they would be involved, and explain one benefit of this involvement. [3]

> Users are involved continually/throughout development (1) / prototypes are produced frequently and delivered to users for feedback (1) / Benefit: misunderstandings and errors are identified at an early stage, so they are cheaper to fix / the final system closely matches user needs (1).

5 Describe the roles of the Scrum master and the product owner in an Agile project. [4]

> Scrum master: ensures the team functions appropriately / co-ordinates the team (1) / removes obstacles to progress / arranges meetings and monitors progress (1). Product owner: acts as the voice of the client (1) / prioritises tasks in the backlogs / ensures nothing unrealistic is added / accountable for delivering value to the client (1).

6 Explain the difference between the product backlog and the release backlog. [2]

> Product backlog: the initial prioritised wish list of everything the product should do (1) / Release backlog: the features selected from the product backlog for the next release, removed from the product backlog (1).

7 State the three questions each team member answers at a daily Scrum. [3]

> What did you do yesterday? (1) / What are you planning to do today? (1) / What obstacles are in your way? (1)

8 A team has 45 story points to complete in a 9-day sprint. Calculate the ideal amount of work remaining at the end of day 6. At the end of day 6 the actual work remaining is 20 points. State whether the team is ahead of or behind schedule and by how much. [3]

> Ideal rate = 45 ÷ 9 = 5 points per day (1) / ideal remaining after day 6 = 45 − 6 × 5 = 15 points (1) / actual 20 > 15, so the team is behind schedule by 5 points (1).

9 Explain two benefits of pair programming. [4]

> Two explained benefits, e.g.: the second programmer reviews code as it is written (1) so errors are spotted and fixed immediately, improving quality (1) / knowledge of the code is shared between two people (1) so the project is less affected if one leaves / less experienced programmers learn from more experienced ones (1) / two people discuss the design (1) giving better solutions to problems (1).

10 A Coleraine company is developing an online booking system for a chain of fitness studios. The owners know their core needs but expect to add features once members start using it, and they want a first version within three months. Evaluate the waterfall model, RAD and Agile for this project and recommend the most suitable approach. [8]

> Indicative content – Waterfall: clear stages and documentation, but owners expect to add features later, which is costly once stages are signed off; nothing usable until the end, so the three-month first version is at risk. RAD: fixed timebox suits the three-month deadline; prototypes of booking screens let owners and members give feedback; desirable add-ons can be dropped to hit the date; but needs skilled RAD practitioners and committed users, may be expensive. Agile: product backlog of booking features; first increment after early sprints; later features added from member feedback; product owner from the studios sets priorities; multi-skilled team; but time estimates may be inaccurate, documentation lighter and the client must stay involved. Recommendation: Agile or RAD, justified – e.g. Agile because requirements will keep evolving after launch. Levels: 7–8 all three evaluated against the scenario with a clear, justified recommendation; 4–6 at least two approaches evaluated with some application, recommendation present; 1–3 mainly descriptive, little reference to the scenario.

E Quick quiz

Circle one answer · 10 marks

1 In RAD, what happens if time is running out before the deadline? [1]

A The deadline is moved back

B Lower-priority 'desirable' requirements are dropped ✓

C The project returns to the requirements phase

D More prototypes are built

2 In an Agile project, who is responsible for removing obstacles that slow the team down? [1]

A The product owner

B The customer

C The Scrum master ✓ D The tester

3 Which is the correct order of the four RAD phases? [1]

A User design, requirements, cutover, construction B Requirements, construction, user design, cutover

C Requirements, user design, construction, cutover ✓ D Construction, user design, requirements, cutover

4 A prototype that is refined until it becomes the final system is... [1]

A evolutionary ✓ B throwaway

C a burn down chart D a release backlog

5 What does a burn down chart show? [1]

A The critical path through a project **B Work remaining in a sprint plotted against time ✓**

C Who made each change to the code D The cost of each feature

6 Who acts as the voice of the client and sets priorities in the backlog? [1]

A Scrum master B Tester

C Systems analyst **D Product owner ✓**

7 A short daily meeting where members say what they did, what they will do and any obstacles is the... [1]

A sprint retrospective B sprint review

C release planning meeting **D daily Scrum ✓**

8 Which is an extreme programming practice? [1]

A Pair programming ✓ B Parallel running

C Feasibility study D Direct changeover

9 Which is a recognised disadvantage of Agile? [1]

A Users see nothing until the end B Requirements can never change

C Task times can be hard to estimate accurately ✓ D There is no customer involvement

10 'As a member, I want to book a class online so that I don't have to phone' is a... [1]

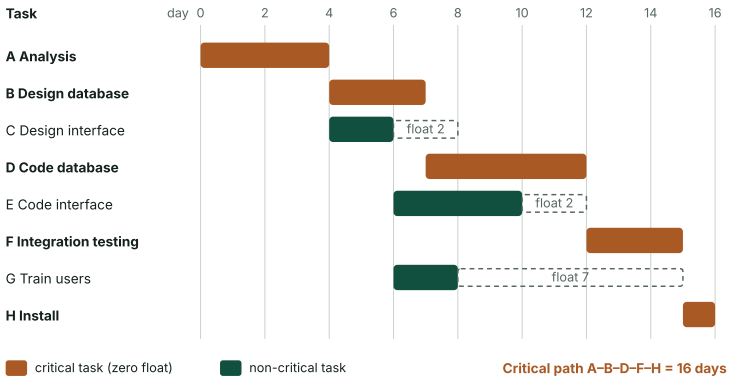
A test plan **B user story ✓**

C Gantt chart D data flow

Organising and managing a software project

AP12 ANSWERS

I can: Classify the resources of a software project and explain the constraints and risks that affect it · Explain why managing a software project differs from other projects and describe the project manager's role · Write SMART objectives for a software project · Draw a Gantt chart and carry out critical path analysis to find the critical path, project duration and float · Evaluate how Gantt charts and critical path analysis help a project manager



Use this diagram to help you: Gantt chart for an eight-task project showing the critical path A, B, D, F, H of 16 days and the float on tasks C, E and G

A Key terms

Write the letter of the matching definition · 6 marks

B Resources

F Elastic resource

D Constraint

A Risk

E Requirements shift

C Project manager

A Anything that could have a harmful effect on a project or cause it to fail, such as missed deadlines, requirements shift, staff turnover or low productivity.

B Anything a project can use to meet its objectives: human (staff), technical (e.g. processing power), financial (budget) or physical (e.g. accommodation).

C The person responsible for planning, scheduling, resourcing, monitoring and managing risk so that a project is delivered on time, within budget and to the required quality.

D A restriction on a project that the developer cannot change, usually concerning scope, time or resources.

E Changes to what the client wants after the project has started, often caused by inadequate user involvement during analysis.

F A resource whose amount can be increased or decreased, such as the budget or number of staff; an inelastic resource, such as time before a hard deadline, cannot.

B Fill the gaps

Use the word bank · 9 marks

risks | depleting | critical path analysis | constraints | SMART | inelastic | resources | Gantt chart | float

Every software project deploys **resources** within **constraints** while managing **risks**. A resource such as money that is used up over time is **depleting**, while the amount of an **inelastic** resource, such as time before a hard deadline, cannot be changed. Objectives should be **SMART**. A **Gantt chart** shows tasks as horizontal bars against a timeline. In **critical path analysis** the critical path is the longest chain of dependent tasks, and a non-critical task has **float**, so it can be delayed without delaying the project.

C True or false?

Tick one box · 9 marks

	True	False
Time and money are examples of depleting resources.	☑	☐
A shared resource is assigned only to one project for its entire duration. *(That is a dedicated resource; a shared resource also works on other projects.)*	☐	☑
'Deliver as cheaply as possible' is a well-stated constraint. *(Constraints should be explicit, e.g. a stated maximum budget, because vague wording can be interpreted differently.)*	☐	☑
Requirements shift is often caused by users not being involved enough during analysis.	☑	☐
In SMART, the A stands for Agreed: objectives are signed off by both client and developer.	☑	☐
The critical path is the shortest route through a project network. *(It is the longest chain of dependent tasks, which sets the minimum project duration.)*	☐	☑
In the forward pass of CPA, where several arrows meet at a node, the highest earliest start time is taken.	☑	☐
Tasks on the critical path have the most float. *(Critical tasks have zero float; any delay to them delays the whole project.)*	☐	☑
Software is intangible, which makes progress on a software project harder to judge.	☑	☐

D Exam-style questions

37 marks

1 State the four types of resource that may be available to a software project and give an example of each. [4]

> Human, e.g. programmer / database administrator (1) / Technical, e.g. processing power / network bandwidth (1) / Financial, e.g. the budget (1) / Physical, e.g. office accommodation / transport links (1). Type and valid example needed for each mark.

2 Explain the difference between an elastic resource and an inelastic resource, using an example of each. [2]

> Elastic: the amount can be increased or decreased, e.g. more money or staff can be assigned (1) / Inelastic: the amount cannot be changed, e.g. the time available before a fixed deadline (1).

3 Describe three constraints that may be placed on a software project. [3]

> Scope – characteristics of the product fixed by the client, e.g. number of simultaneous users, response time, browser compatibility (1) / Time – a fixed deadline by which the system must be delivered (1) / Resources – limits on budget, staff or equipment, e.g. a maximum budget (1).

4 A software company in Enniskillen is developing a stock control system for a local builders' merchant. Identify two risks to the project and describe how the project manager could reduce each. [4]

> Two risks each with a matching mitigation, e.g.: requirements shift / unclear requirements (1) – involve the merchant's staff fully during analysis and use change control (1) / staff turnover (1) – encourage collaboration, documentation and knowledge sharing (1) / missed deadline (1) – realistic schedule with milestones and regular monitoring (1) / low productivity (1) – split long tasks into smaller sections with their own deadlines (1).

5 Explain why managing a software project can be more difficult than managing the construction of a building. [3]

> Any three from: software is intangible, so progress is hard to see/judge (1) / software is usually bespoke, so experience from previous projects is less transferable (1) / problems are less likely to have been met before (1) / technology changes rapidly, so new tools may appear during the project and force changes (1).

6 A council's objective for its new planning portal reads: 'Improve the planning website soon.' The project manager says it should be SMART. State what each letter of SMART stands for. [5]

> Specific (1) / Measurable (1) / Agreed (1) / Realistic (1) / Timed (1).

7 Using the worked example (tasks A–H), state the critical path and the minimum project duration, and explain what would happen if task E took two days longer than planned. [4]

> Critical path A, B, D, F, H (1) / duration 16 days (1) / E has 1 day of float, so a 2-day delay delays the project by 1 day, to 17 days (1) / the critical path changes to A, C, E, F, H (1).

8 A project has these tasks: P (2 days, no dependency), Q (4 days, depends on P), R (3 days, depends on P), S (2 days, depends on R), T (3 days, depends on Q and S). Calculate the earliest start time of T, the project duration and the float on task Q, showing your working. [4]

> EST of T = max(end of Q, end of S) (1): Q ends 2 + 4 = 6; S ends 2 + 3 + 2 = 7, so EST of T = 7 (1) / duration = 7 + 3 = 10 days (1) / float of Q = latest finish 7 − earliest start 2 − duration 4 = 1 day (1).

9 Explain what is meant by float and why it is useful to a project manager. [2]

> Float is the amount of time a non-critical task can be delayed without delaying the whole project (1) / it lets the manager move resources from tasks with float to critical tasks, or reschedule tasks, without affecting the end date (1).

10 A Belfast software house is managing a large project to build an online appointments system for a group of GP practices. Evaluate the use of Gantt charts and critical path analysis to help the project manager deliver the project on time. [6]

> Indicative content – Gantt chart: shows each task's start, end and duration against a timeline; shows overlaps and parallel tasks; milestones and progress to date can be shown; easy to share with the GP practices; but on a large project dependencies are hard to follow and the critical path is not obvious. CPA: identifies the critical path and the minimum duration; shows float, so the manager knows which tasks can slip; identifies bottlenecks; helps reallocate staff when a delay occurs; resources can be scheduled for a task's EST and released at its LFT; but more complex to draw and read and relies on accurate estimates. Both: must be kept up to date; do not on their own manage quality or risk. Conclusion: used together they help deliver on time – Gantt for communication and monitoring, CPA for analysis of critical tasks. Levels: 5–6 balanced evaluation of both tools applied to the project with a conclusion; 3–4 benefits and some limitations, partly applied; 1–2 basic description of one or both tools.

E Quick quiz

Circle one answer · 9 marks

1 What is the critical path in a project? [1]

A The shortest chain of tasks

B The longest chain of dependent tasks, which sets the minimum project duration ✓

C The tasks with the most float

D The tasks that use the most staff

2 Which of these is a depleting resource for a software project? [1]

A A dedicated programmer
B A fixed server room
C The project budget ✓
D A shared graphic designer

3 In SMART objectives, what does the ‘M’ stand for? [1]

A Manageable
B Monitored
C Minimal
D Measurable ✓

4 A developer leaving mid-project and taking vital knowledge with them is which risk? [1]

A Staff turnover ✓
B Requirements shift
C Productivity
D Scope

5 In the backward pass of CPA, where two paths meet at a node, which latest finish time is used? [1]

A The highest
B The average
C The lowest ✓
D The first one calculated

6 Task D has latest finish 12, earliest start 7 and duration 5. Its float is... [1]

A 0 days ✓
B 5 days
C 7 days
D 12 days

7 Which is a limitation of a Gantt chart? [1]

A It cannot show task durations
B It cannot show overlapping tasks
C It cannot be shown to a client
D Dependencies and the critical path may be unclear on large projects ✓

8 Why is software project management different from managing a building project? [1]

A Software has no deadlines
B Software is intangible, so progress is hard to see ✓
C Software projects never use teams
D Software projects have no budget

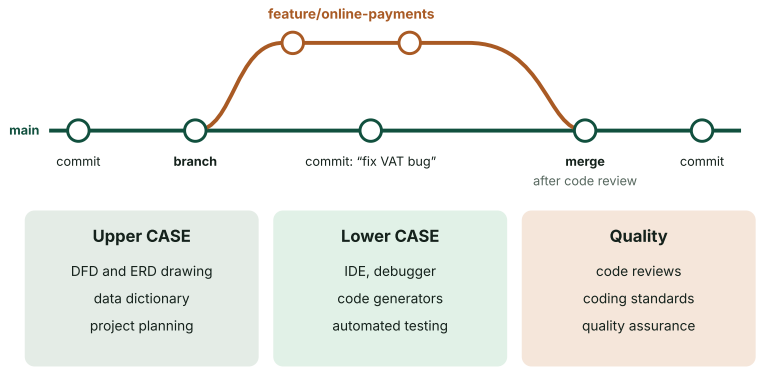
9 Which is a scope constraint? [1]

A The system must support 500 simultaneous users ✓
B The lead programmer may leave
C The budget is £80,000
D Coding starts after design

Version management and tools for software projects

AP13 ANSWERS

I can: Explain why version management is essential when a team works on shared code · Draw and interpret a version management graph showing parallel versions, merges and rollbacks · Describe what a version management system records and the conditions for it to work · Describe upper and lower CASE tools and collaboration tools and their benefits · Explain how code reviews, quality assurance and standards improve software quality



Use this diagram to help you: Version control: commits on the main branch, a feature branch that is merged back, and the CASE tools and quality techniques used in software projects

A Key terms

Write the letter of the matching definition · 6 marks

B Version management

F Repository

A Commit

C Branch

E Merge

D Rollback

A Saving a change to the repository as a new version, together with a message describing the change.

B The process by which a development team keeps track of changes to code files over time, recording who made each change, when and why, and allowing return to earlier versions.

C In version management, a separate line of development used to work on a feature or fix without affecting the main version.

D Returning to an earlier, stable version of the code and continuing development from that point.

E Combining two parallel versions or branches into a single version that contains both sets of changes.

F The central store in which a version management system holds all of a project's files and their complete history of changes.

B Fill the gaps

Use the word bank · 10 marks

merge | CASE tools | code review | Version management | repository | conflict | commit | roll back | branch | coding standards

Version management keeps track of every change to a project's code files. All files are kept in a **repository**, and each **commit** records who made the change, when, and a note describing it. A developer can create a **branch** to work on a feature separately and later **merge** it into the

main version; if two people change the same lines a **conflict** is flagged. If an error is found, the team can **roll back** to an earlier version. **CASE tools** support development, and in a **code review** other developers check code for errors and adherence to **coding standards**.

C True or false?

Tick one box · 9 marks

	True	False
A version management system records who made each change, when, and a note describing it.	☑	☐
Version management is simply another name for making a backup copy of files. *(It tracks every change with its author and description, supports branching and merging and allows rollback; a remote repository is only one way it also protects work.)*	☐	☑
For version management to work, all developers should work from the same version of each file.	☑	☐
When two developers change the same lines, a version management system silently keeps only the newest change. *(It flags a conflict so that a developer can decide how to combine the changes.)*	☐	☑
In a version management graph, two arrows entering one node show a merge.	☑	☐
After rolling back to an earlier version, any later faulty changes are excluded from the next version created.	☑	☐
Upper CASE tools include code generators and debuggers. *(Those are lower CASE tools; upper CASE tools support analysis and design, e.g. drawing DFDs and ERDs.)*	☐	☑
Quality assurance means testing the system once it is finished. *(QA is planned activity throughout the project, such as reviews, inspections and audits; testing is only one part.)*	☐	☑
Code reviews help to spread knowledge of the code across the team.	☑	☐

D Exam-style questions

37 marks

1 State three items of information a version management system records each time a change is made. [3]

> Who made the change (1) / when the change was made (1) / a note/message describing the change and any knock-on effects (1). Also accept: the new version number (1). Max 3.

2 Eight programmers at a Belfast software house are working on the same patient-records system at the same time. Explain why the company should use version management. [4]

> Any four from: many programmers work on a shared set of files (1) / without control, changes would overwrite each other / chaos (1) / every change is tracked with who, when and why (1) / the team can return to a previous working version if an error is found (1) / branches allow parallel development without affecting the working version (1) / merging combines work and flags conflicts (1) / provides an audit trail (1) / remote repository acts as a backup (1).

3 Explain the terms branch and merge in version management. [2]

> Branch: a separate line of development created from a version so that work, e.g. a new feature, does not affect the main version (1) / Merge: combining two lines of development/versions into a single version containing both sets of changes (1).

4 Describe three conditions that must be met for version management to work successfully. [3]

> A consistent method of tracking changes and recording all version numbers (1) / it must be possible to reverse

changes (1) / every developer works from the same current copy of each code file when editing (1).

5 Two developers at a Derry software company are updating a booking system. Niamh starts from v1 and creates v2 by adding online payments. Both Niamh and Rory then work from v2: Niamh creates v3 by redesigning the receipts and Rory creates v4 by adding a staff rota. v3 and v4 are merged to create v5. Testing shows Rory's rota work contains a serious error, so the team returns to v3 and Niamh creates v6 from it. Draw a version management graph for v1 to v6. [5]

v1 → v2 (1) / v2 → v3 and v2 → v4, drawn as parallel branches from v2 (1) / v3 → v5 and v4 → v5 showing the merge (1) / v3 → v6, with v6 drawn from v3 and not from v5 (1) / all six versions labelled, with arrows pointing from older to newer (1).

6 Using your graph, state which versions' work is included in v6 and explain why v4 and v5 are not. [2]

v6 includes the work of v1, v2 and v3 (1) / v4 contains the faulty rota work and v5 was built from v4, so the team rolled back to v3 and excluded them (1).

7 State what is meant by a CASE tool and give one example of an upper CASE tool and one of a lower CASE tool. [3]

Software that supports or automates parts of the software development process (1) / upper CASE example, e.g. a tool for drawing DFDs/ERDs or maintaining a data dictionary (1) / lower CASE example, e.g. an IDE, code generator, debugger or automated testing tool (1).

8 Explain two benefits of carrying out code reviews. [4]

Two explained benefits, e.g.: errors/security weaknesses are found before code is accepted (1) so they are cheaper to fix than after release (1) / knowledge of the code is shared across the team (1) so the project is less affected if a developer leaves (1) / checks adherence to coding standards (1) so code is consistent and easier to maintain (1) / less experienced programmers learn from feedback (1) improving future code (1).

9 Explain why a software company would require all its programmers to follow coding standards. [3]

Any three from: code is consistent across the team (1) / any programmer can read and understand another's code (1) / easier to maintain, debug and update later (1) / easier for new staff to join the project (1) / fewer errors, e.g. through consistent error handling (1) / supports code reviews and quality assurance (1).

10 A software house in Armagh has twelve developers, some working from home, building an online system for a regional bus company. Discuss how version management, collaboration tools and quality assurance could help the team deliver reliable software. [8]

Indicative content – Version management: shared repository with every change tracked (who, when, note); branches let developers build features in parallel without breaking the working version; merging and conflict detection prevent home and office workers overwriting each other; rollback to a stable version if a change breaks timetables or ticketing; audit trail; remote repository as a backup; needs everyone to work from the current version and commit regularly. Collaboration tools: issue tracker for bugs and tasks; messaging and video meetings for remote staff; task boards show who is doing what; continuous integration builds and tests after each commit, catching errors early; needs clear rules. Quality assurance: code reviews before merging; coding standards so twelve developers write consistent, maintainable code; inspections and testing throughout, not just at the end; accessibility standards for public users. Conclusion: together they reduce errors and lost work and make the software more reliable, but rely on discipline and training. Levels: 7–8 all three areas discussed in depth and applied to the scenario with a conclusion; 4–6 at least two areas with some application; 1–3 basic descriptions, little reference to the scenario.

E Quick quiz

Circle one answer · 9 marks

1 What is the main purpose of version management? [1]

A To compress code files so they take less storage

B To track every change to code files so the team can collaborate and return to earlier versions ✓

C To design the user interface

D To schedule tasks on a Gantt chart

2 Two developers both start work from v2 at the same time. On a version management graph this appears as... [1]

A two arrows entering v2

B a straight chain from v2

C v2 being deleted

D two arrows leaving v2 ✓

3 Which of these is NOT normally recorded when a change is committed? [1]

A Who made the change

B When it was made

C The developer's salary ✓

D A note describing the change

4 Combining two parallel versions into a single version is called... [1]

A merging ✓

B branching

C committing

D refactoring

5 v3 contains a serious error, so the team returns to v4 and creates v6 from it. Which statement is true? [1]

A v6 includes the work from v3

B v6 includes v4's work but not v3's faulty changes ✓

C v6 must be created from v5

D v4 is deleted

6 Which is a lower CASE tool? [1]

A A DFD drawing tool

B A data dictionary

C A project planning tool

D A debugger ✓

7 Which condition is needed for version management to work successfully? [1]

A All developers work from the same version of each file ✓

B Only one developer may use the system

C Changes can never be reversed

D Version numbers are optional

8 What is the main benefit of a code review? [1]

A It removes the need for testing

B It replaces version management

C Defects are found early, before code is accepted ✓

D It speeds up compilation

9 Quality assurance is best described as... [1]

A testing only at the end of the project

B planned activities that build quality into every stage ✓

C a type of version control

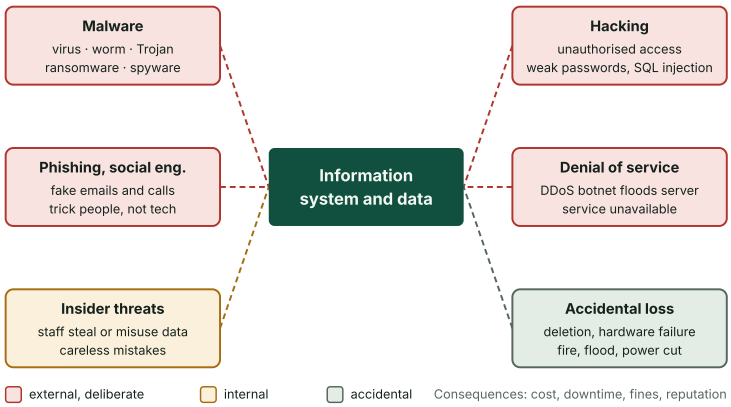
D a programming language standard

AS 1: APPROACHES TO SYSTEMS DEVELOPMENT · SECURITY ISSUES

Threats to data, archiving and the need for backup

AP14 ANSWERS

I can: Distinguish between types of malware by how they spread and what they do · Explain how hacking, social engineering and denial of service attacks work and the harm they cause · Describe internal and accidental threats and apply their consequences to a given organisation · Explain why inactive data is archived and weigh up the advantages and disadvantages · Explain why current data must be backed up and how a restore point is used to recover a system



Use this diagram to help you: Threats to an information system grouped as external deliberate attacks, insider threats and accidental loss

A Key terms

Write the letter of the matching definition · 6 marks

C Malware

F Worm

B Trojan

D Ransomware

E Phishing

A Social engineering

A Manipulating people into giving away confidential information or access, rather than attacking the technology directly.

B Malware disguised as legitimate software that the user installs; it carries a hidden harmful payload such as a back door and does not self-replicate.

C Malicious software, such as viruses, worms, Trojans, ransomware and spyware, written to damage or disrupt systems or to gain unauthorised access to data.

D Malware that encrypts a victim's files and demands payment in exchange for the decryption key.

E A social engineering attack using fraudulent emails, messages or websites that imitate trusted organisations to trick people into revealing personal details.

F Standalone malware that replicates itself across a network without needing a host file or any user action, usually by exploiting unpatched vulnerabilities.

B Fill the gaps

Use the word bank · 10 marks

Ransomware · backup · inactive data · index · Trojan · data archiving · restore point · phishing · worm · transaction log

A **worm** spreads across a network by itself, while a **Trojan** is disguised as useful software. **Ransomware** encrypts files and demands payment for the key. A **phishing** email tricks

people into revealing log-in details. Records that are rarely used are called **inactive data**, and moving them to secure long-term storage is **data archiving**. The archive is given an **index** so records can be found again. A **backup** is a copy of current data, and a **restore point** records the state of the system when that copy was made. Changes since then can be re-applied from a **transaction log**.

C True or false?

Tick one box · 9 marks

	True	False
A worm can spread across a network without the user running a host file.	☑	☐
A Trojan replicates itself to other computers on the network. *(A Trojan does not self-replicate; it relies on the user installing it because it looks legitimate.)*	☐	☑
Social engineering targets people rather than technical weaknesses.	☑	☐
The main purpose of a DDoS attack is to encrypt the victim's files. *(A DDoS attack floods a server so that it is unavailable; ransomware is the malware that encrypts files.)*	☐	☑
Archived data is deleted from the organisation's systems and cannot be retrieved. *(Archived data is moved to separate storage but is kept, indexed and can be retrieved when needed.)*	☐	☑
Archiving inactive records can make reports on the live system run faster.	☑	☐
Archiving and backing up are two names for the same process. *(Archiving moves inactive data out for long-term retention; backing up copies current data so it can be restored after a loss.)*	☐	☑
After restoring from the last restore point, any changes made since that backup are lost unless they were recorded elsewhere.	☑	☐
Accidentally emailing personal data to the wrong person can be a personal data breach.	☑	☐

D Exam-style questions

38 marks

1 Describe the difference between a virus and a worm. [2]

> A virus attaches to a host file and needs the user to run or share it to spread (1) / a worm is standalone and self-replicates across a network without user action (1).

2 A member of staff at a Lisburn engineering firm installs a free PDF tool that turns out to be a Trojan. Describe how the Trojan could harm the firm. [3]

> Any three from: it is disguised as legitimate software so the user installs it willingly (1) / it may open a back door giving an attacker remote access (1) / it may install spyware or a keylogger to steal passwords or designs (1) / it may download ransomware that encrypts the firm's files (1) / leading to loss of data, downtime or a data breach (1).

3 Explain what is meant by phishing and describe two features that may help a user recognise a phishing email. [3]

> Phishing is a fraudulent email or message pretending to come from a trusted organisation (1) to trick the user into revealing personal details such as passwords or bank details (1). Any one feature for 1 mark, max 3 total: urgent or threatening tone / generic greeting / sender address or link that does not match the real organisation / spelling and grammar errors / request for login or payment details / unexpected attachment.

4 A Belfast concert venue sells tickets online. Explain how a distributed denial of service attack works and its likely effect on the venue. [4]

> Many compromised devices (a botnet) controlled by the attacker (1) send huge numbers of requests to the venue's web server at once (1) so the server or its connection is overwhelmed and cannot respond to genuine users (1). Effect: customers cannot buy tickets, so sales are lost and the venue's reputation suffers / customers may buy elsewhere (1).

5 State two accidental threats to the data held by a GP surgery. [2]

> Any two from: staff deleting or overwriting files by mistake (1) / hardware failure such as a disk crash (1) / power cut or surge (1) / fire or flood (1) / sending patient data to the wrong recipient (1) / losing a laptop or USB stick (1).

6 The office manager at a Portadown veterinary practice says old client records should be 'archived'. Explain what this would involve. [2]

> Moving inactive data that is no longer needed for day-to-day use (1) to separate secure long-term storage from which it can be retrieved later, e.g. using an index (1).

7 A further education college in Enniskillen keeps the records of every student who has left in its live student database. Explain two advantages and one disadvantage to the college of archiving these records. [6]

> Advantages, two explained (2 marks each, 1 for point and 1 for applied expansion): live database holds fewer records (1) so timetabling, fee and attendance reports run faster (1) / backups of the live system are smaller (1) so they finish more quickly (1) / records are retained (1) for references, audits or legal requirements (1) / archive can use cheaper storage (1) saving the college money (1). Disadvantage, one explained (2): archived records are not immediately available (1) so if a former student asks for a reference staff must search the archive, which takes time (1) / access to archived data may be slow (1).

8 Explain two differences between archiving data and backing up data. [4]

> Any two differences, 2 marks each: archiving deals with inactive data (1) whereas backup deals with current data (1) / archiving removes data from the live system (1) whereas a backup is a copy and the original stays in place (1) / purpose of archiving is long-term retention for reference or compliance (1) whereas backup is for recovery after loss or corruption (1) / archiving is periodic, e.g. yearly (1) whereas backup is frequent, e.g. nightly (1).

9 A Cookstown builders' merchant's stock database fails at lunchtime. Its IT support firm says the system will be 'rolled back to last night's restore point and then brought up to date from the transaction log'. Explain what this means. [4]

> A restore point captures the state of the system at the moment a backup was taken (1). After a failure the system is returned to the most recent restore point by restoring that backup (1). Changes made after it would otherwise be lost (1), so transactions recorded in the log since the backup are re-applied in order to bring the data up to date (1).

10 A health and social care trust has suffered a ransomware attack that encrypted its patient administration system. Discuss the possible consequences for the trust, its staff and its patients, and the importance of backups in limiting the damage. [8]

> Indicative content: patient records, appointments and results unavailable, so operations and clinics are cancelled or delayed / clinical risk if staff cannot see allergies or test results / staff revert to slow paper processes, extra workload and stress / possible theft and publication of patient data, a personal data breach to report to the Information Commission within 72 hours (accept ICO) / cost of investigation, recovery and new security; reputational damage and loss of public trust / paying the ransom is discouraged and does not guarantee recovery / recent backups stored separately or offline allow systems to be restored without paying / backups must themselves be protected from encryption, and restores tested / a restore point means only changes since the last backup are lost, reduced further by transaction logs. Levels: 7–8 a balanced discussion covering trust, staff and patients with clear evaluation of the role of backups, applied to the context and with good use of technical terms; 4–6 several relevant consequences with some application and some mention of backups; 1–3 a few points, largely generic or listed.

E Quick quiz

Circle one answer · 9 marks

1 A sports club moves the records of members who left more than five years ago from its live database into separate, indexed storage. What is this process called? [1]

A Backing up
B Data archiving ✓
C Data recovery
D Encryption

2 Which type of malware spreads across a network on its own by exploiting unpatched vulnerabilities? [1]

A Trojan
B Spyware
C Worm ✓
D Adware

3 Which malware encrypts a victim's files and demands payment for the key? [1]

A Ransomware ✓
B Keylogger
C Worm
D Rootkit

4 A phone call claiming to be from the bank's fraud team asks for your PIN. This is... [1]

A a worm
B a DDoS attack
C a brute-force attack
D vishing (social engineering) ✓

5 Which UK law makes unauthorised access to computer material an offence? [1]

A Copyright, Designs and Patents Act 1988
B Freedom of Information Act 2000
C Computer Misuse Act 1990 ✓
D Equality Act 2010

6 Which is a disadvantage of archiving data? [1]

A Archived data is not immediately available and can be slow to access ✓
B The live system becomes slower
C Backups of the live system take longer
D Older data cannot be kept for legal reasons

7 What does a restore point record? [1]

A The list of users allowed to restore files
B The state of the system at the moment it was backed up ✓
C The time limit for recovering each system
D Which files are inactive

8 A shop backs up at 22:00 each night. The server fails at 14:00 the next day and there is no transaction log. How many hours of changes are lost? [1]

A 8
B 10
C 14
D 16 ✓

9 What is the key difference between backing up and archiving? [1]

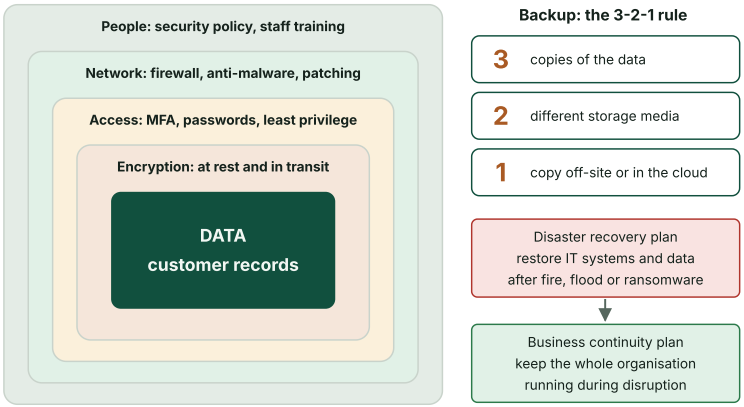
A Backups are only made once a year
B Archiving copies current data and leaves it in place
C Backing up copies current data so it can be restored; archiving moves inactive data out for long-term storage ✓
D Archives are always stored on the same server as the live data

AS 1: APPROACHES TO SYSTEMS DEVELOPMENT · SECURITY ISSUES

Protecting systems: security measures, backup methods and disaster recovery

AP15 ANSWERS

I can: Explain how a security policy, authentication, access rights, encryption and firewalls protect data · Describe how full, differential and incremental backups work and evaluate each one · Describe the steps needed to restore a system from each type of backup · Explain why an organisation needs a disaster recovery plan and describe its contents · Explain recovery point and recovery time objectives and how a plan is tested



Use this diagram to help you: Layers of protection around data – policy and training, network defences, authentication and access rights, encryption – alongside the 3-2-1 backup rule and disaster recovery

A Key terms

Write the letter of the matching definition · 6 marks

E Security policy

F Authentication

D Multi-factor authentication (MFA)

B Encryption

C Firewall

A Full backup

A A backup that copies every file on the system, whether or not it has changed, each time it runs.

B Converting plaintext into ciphertext using an algorithm and a key so that it cannot be understood without the correct key.

C Hardware or software that monitors network traffic and blocks data that does not meet a set of security rules.

D Authentication requiring two or more different types of evidence, such as a password and a code sent to a phone.

E A formal document setting out the rules, procedures and responsibilities an organisation uses to protect its systems and data.

F The process of proving that a user is who they claim to be, using something they know, have or are.

B Fill the gaps

Use the word bank · 9 marks

access rights | Multi-factor authentication | full backup | off-site backup | disaster recovery plan | differential backup | security policy | incremental backup | recovery time objective

A **security policy** sets out the rules staff must follow. **Multi-factor authentication** needs more than one type of evidence of identity, and **access rights** limit what each user can do. A **full backup** copies

every file, a **differential backup** copies files changed since the last full backup, and an **incremental backup** copies files modified after the most recent backup of either kind. Keeping an **off-site backup** protects against fire and flood. A **disaster recovery plan** identifies critical data, key personnel and an alternative site, and sets a **recovery time objective** for each system to be restored.

C True or false?

Tick one box · 9 marks

	True	False
Encryption prevents data from being stolen. *(Encryption makes stolen data unreadable without the key, but it does not stop data being copied or deleted.)*	☐	☑
Multi-factor authentication combines at least two different types of factor, such as something you know and something you have.	☑	☐
An incremental backup copies every file each time it runs. *(That describes a full backup; an incremental backup copies only files modified after the most recent backup of either kind.)*	☐	☑
A differential backup copies all files changed since the last full backup.	☑	☐
Restoring from incremental backups needs only the last full backup and the most recent incremental. *(It needs the last full backup and every incremental made since, applied in order; the two-backup restore describes differential.)*	☐	☑
Storing backups in the same building as the server protects the data from fire. *(A fire could destroy both; at least one backup should be stored off-site or in the cloud.)*	☐	☑
The purpose of a disaster recovery plan is to minimise the effects of a disaster, not to prevent disasters.	☑	☐
In a simulation test of a disaster recovery plan, working systems are disabled so that recovery can be tried for real.	☑	☐
Giving every user administrator rights improves security. *(Users should have only the minimum rights their job needs (least privilege).)*	☐	☑

D Exam-style questions

39 marks

1 State three items that would be included in an organisation's security policy. [3]

> Any three from: acceptable use of equipment and internet (1) / password rules (1) / access levels for different staff (1) / use of personal devices and removable media (1) / remote working rules (1) / procedure for reporting incidents (1) / backup procedures (1) / staff training (1) / disciplinary consequences of breaches (1).

2 A Northern Ireland bank asks customers to enter a password and then a code sent to their phone. Explain how this multi-factor authentication improves security. [2]

> It requires two different types of evidence, something the customer knows and something they have (1); so a criminal who has stolen or phished the password still cannot log in without the customer's phone (1).

3 Describe the difference between symmetric and asymmetric encryption. [2]

> Symmetric encryption uses the same key to encrypt and decrypt (1); asymmetric uses a public key to encrypt and a different, matching private key to decrypt (1).

4 Describe how a firewall protects a school network. [2]

> It examines incoming and outgoing traffic/packets against a set of rules, e.g. IP address, port or protocol (1) and blocks traffic that does not meet the rules, preventing unauthorised access to the school network (1).

5 A Larne dental practice is considering incremental backups for its patient records system. Describe how an incremental backup works and state one advantage and one disadvantage of the method for the practice. [4]

> Only files modified after the most recent backup, full or incremental, are copied (1); it starts from a full backup (1). Advantage, one of: fastest backup to create / least storage (1). Disadvantage, one of: slowest restore because the full backup and every incremental since are needed / if one incremental is corrupt the system cannot be fully restored (1).

6 A company has 500 GB of data. Each day about 20 GB of different files changes. Calculate the total storage needed for one week if it takes a full backup on day 0 followed by six daily differential backups, and compare this with six daily incremental backups. [4]

> Differentials: 20 + 40 + 60 + 80 + 100 + 120 = 420 GB (1); total with full = 500 + 420 = 920 GB (1). Incrementals: 6 × 20 = 120 GB, total 620 GB (1). Comparison: incremental uses 300 GB less because each differential repeats all changes since the full backup (1).

7 A Ballymena builders' merchant takes a full backup every Saturday at 23:00 and an incremental backup at 21:30 on each of the other six nights. Its server fails on Thursday at 09:45. Explain how the company would recover its data and state how much work could be lost without a transaction log. [5]

> Restore the full backup taken at 23:00 on Saturday (1). Then restore the incremental backups from Sunday, Monday, Tuesday and Wednesday at 21:30 (1) in date order, so later copies of a file replace earlier ones (1). The system is now as it was at 21:30 on Wednesday (1). Changes since then must be re-applied from a transaction log if one exists; otherwise 12¼ hours of work (21:30 Wednesday to 09:45 Thursday) must be re-entered (1).

8 A cyber attack has taken a Fermanagh district council's planning system offline for a week. Explain what is meant by disaster recovery, and identify four things the council's disaster recovery plan should have set out in advance. [5]

> Disaster recovery is how an organisation restores its IT systems and data and resumes work after a serious incident (1), whether accidental (fire), natural (flood) or deliberate (cyber attack) (1). The plan should identify, any four of: the critical data/systems (e.g. planning applications) (1) / key personnel and their responsibilities (1) / the backup and recovery processes (1) / an alternative site or cloud servers to run from (1) / priorities and time limits for recovery (RPO/RTO) (1). Max 5.

9 The Southern Health and Social Care Trust is updating its disaster recovery plan. Using examples from the trust, explain the terms recovery point objective and recovery time objective. [4]

> RPOs: the systems, ranked in priority, that need to be brought back after a disaster (1), e.g. patient administration system, pharmacy system, staff rostering (1). RTOs: the time limit for recovering each part (1), e.g. patient records within 4 hours but rostering within 48 hours (1). Also credit RPO explained as maximum acceptable data loss measured in time, with an example.

10 A family-run veterinary group with three surgeries in County Down keeps all its appointment and patient records on one server in its Downpatrick branch, and copies the nightly backup to an external drive kept in the same room. Evaluate this approach and recommend how the group should develop a disaster recovery plan. [8]

> Indicative content: regular backups are good, but keeping the drive in the same room means a fire, flood or theft could destroy both the server and the backups / ransomware could also encrypt backups connected to the network / recommend off-site or cloud backups, encrypted, with at least one offline copy / consider backup method: e.g. weekly full plus daily incremental for speed and storage, or differential for quicker restores, justified / DR plan: identify critical data (appointment and patient records), key personnel and their roles, backup and recovery processes, an alternative site or cloud-hosted servers / set RPOs and RTOs ranked by business importance, e.g. appointments and patient histories before management reports / test the plan through walk-through and simulation, and review it as the group grows / consider cost against risk. Levels: 7–8 a balanced

> evaluation of the current approach with clear, justified recommendations applied to the company; 4–6 relevant weaknesses and recommendations with some application; 1–3 limited, generic points.

E Quick quiz

Circle one answer · 10 marks

1 A business takes a full backup on Sunday and backs up only the files changed since the previous night's backup on each other night. Which backup method is used on those other nights? [1]

A Full
B Differential
C Mirror
D Incremental ✓

2 Which of these should a disaster recovery plan identify? [1]

A The colour scheme of the company website
B A fallback location where staff can keep working ✓
C The marketing budget for next year
D Each customer's favourite products

3 A password plus a code sent to your phone is an example of... [1]

A multi-factor authentication ✓
B biometrics
C encryption
D a firewall

4 Which backup method needs only the last full backup and one other backup to restore? [1]

A Incremental
B Differential ✓
C Transaction log
D Partial

5 Which backup method is fastest to create and uses the least storage? [1]

A Full
B Differential
C Incremental ✓
D All are the same

6 A full backup is taken on day 0 of 100 GB. 5 GB of different files changes each day. How large is the differential backup on day 4? [1]

A 5 GB
B 10 GB
C 20 GB ✓
D 105 GB

7 In disaster recovery planning, the recovery time objective is... [1]

A the time limit for recovering a part of the system ✓
B the time a backup takes to run
C the date the plan was written
D the time a firewall takes to block a packet

8 Which device or software filters traffic using rules on ports and IP addresses? [1]

A Backup server
B Anti-malware
C Router table
D Firewall ✓

9 Why should at least one backup be kept off-site? [1]

A It makes the backup run faster
B So that a fire or flood at the main site does not destroy the backups too ✓
C Because off-site backups cannot be encrypted
D So that the backup does not need testing

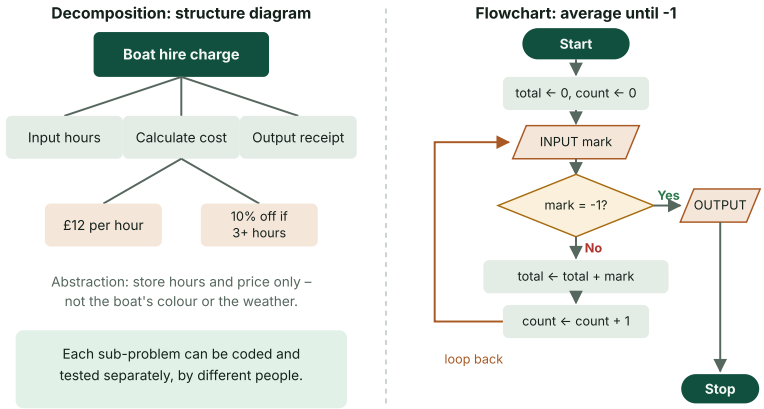
10 Staff talk through each step of the disaster recovery plan in a meeting without switching any systems off. This is a... [1]

A simulation test
B penetration test
C walk-through (theoretical) test ✓
D load test

Programs, algorithms and their design

AP16 ANSWERS

I can: Describe what a program is for and explain what makes a set of instructions an algorithm · Break a problem down and remove unnecessary detail before designing a solution · Express the same algorithm as pseudocode, a flowchart and C# using sequence, selection and iteration · Dry run an algorithm with a trace table to predict its output and locate logic errors



Use this diagram to help you: Decomposition of a boat-hire problem into a structure diagram, beside a flowchart of a sentinel-controlled averaging loop

A Key terms

Write the letter of the matching definition · 6 marks

E Computer program

F Application program

A Algorithm

C Finiteness

D Decomposition

B Abstraction

A A finite, ordered sequence of precise, unambiguous steps designed to solve a problem or complete a task.

B Removing unnecessary detail from a problem so that only the relevant details remain.

C The property that an algorithm must terminate after a limited number of steps.

D Breaking a complex problem into smaller sub-problems that can be solved independently.

E A set of instructions that a computer executes in order to perform a specific task.

F Software designed to perform a specific task for the user, such as a word processor or spreadsheet.

B Fill the gaps

Use the word bank · 10 marks

pseudocode | algorithm | finiteness | program | decomposition | flowchart | diamond | abstraction | unambiguous | trace table

A computer **program** is a set of instructions that a computer carries out to perform a task. It is based on an **algorithm**, an ordered set of precise, **unambiguous** steps that must also terminate, a property called **finiteness**. Splitting the problem into smaller sub-problems is called **decomposition**, and removing unnecessary detail is called **abstraction**. The design can be written as **pseudocode** or

drawn as a **flowchart**, in which a **diamond** represents a decision. Working through the design by hand while recording variable values in a **trace table** is called a dry run.

C True or false?

Tick one box · 9 marks

	True	False
An algorithm must be written in a programming language. *(An algorithm is language-independent; it can be written in English, pseudocode or as a flowchart and then coded in any language.)*	☐	☑
An algorithm must always terminate after a finite number of steps.	☑	☐
An algorithm must have at least one input. *(An algorithm can have zero or more inputs, but it must produce at least one output.)*	☐	☑
A parallelogram in a flowchart represents input or output.	☑	☐
Abstraction means splitting a problem into smaller parts. *(That is decomposition; abstraction removes unnecessary detail.)*	☐	☑
A FOR loop is suitable when the number of repetitions is known before the loop starts.	☑	☐
A flowchart decision symbol has only one exit. *(A decision has two exits, labelled Yes/No or True/False.)*	☐	☑
A dry run can reveal logic errors that a compiler would not detect.	☑	☐
Plain-English algorithms are harder for non-programmers to understand than pseudocode. *(Plain English is easier for non-programmers; pseudocode is more useful to the programmer.)*	☐	☑

D Exam-style questions

39 marks

1 A Year 8 pupil says that a program and the computer it runs on are 'the same thing'. Describe the purpose of a computer program, explaining why it is different from the hardware. [2]

> A set of instructions (1) / executed by the computer to carry out a specific task or solve a problem; it is software, separate from the physical hardware that runs it (1)

2 Before coding a lunch-ordering app for a Strabane primary school, a developer writes an algorithm. Explain what is meant by an algorithm. [2]

> A sequence of (step-by-step) instructions (1) / that are precise/unambiguous and in a specific order, designed to solve a problem/complete a task (1)

3 An algorithm is written to guide visitors around Castle Coole. Explain three properties that the algorithm must have to be useful. [6]

> Any three properties, each identified (1) and explained in context (1), e.g.: steps in a specific order (1) so visitors see rooms in the correct route (1) / each step precise (1) e.g. 'walk 20 metres to the staircase' not 'walk a bit' (1) / unambiguous (1) e.g. naming which door when there are several (1) / finite (1) the tour must end, e.g. at the exit (1) / has inputs and at least one output (1) e.g. visitor's start point in, route out (1) / correct and effective (1) deals with a closed room by giving an alternative (1) / efficient (1) no unnecessary doubling back (1)

4 State the flowchart symbol used for (a) a decision and (b) an input. [2]

> (a) Diamond/rhombus (1) / (b) Parallelogram (1)

5 Give one advantage of writing an algorithm in plain English rather than pseudocode, and one advantage of pseudocode over plain English. [2]

> Plain English: easier for non-programmers/clients to understand and check (1) / Pseudocode: closer to program code so easier/quicker for the programmer to convert into a program (1)

6 Explain two benefits of decomposing a problem before writing a program. [4]

> Any two benefits, each identified (1) and explained (1): sub-problems are simpler to solve (1) so the design is less error-prone (1) / modules can be shared among programmers (1) so development is faster (1) / each module can be tested separately (1) so errors are easier to locate (1) / modules can be reused in other programs (1) saving time (1)

7 An algorithm sets x ← 1 and y ← 0, then repeats y ← y + x and x ← x * 2 while x < 20. Complete a trace table and state the final values of x and y. [4]

> Trace: x 1, y 0 → y 1, x 2 → y 3, x 4 → y 7, x 8 → y 15, x 16 → y 31, x 32 (loop ends as 32 is not < 20). Marks: correct first two passes (1) / correct remaining passes (1) / final x = 32 (1) / final y = 31 (1)

8 A Coleraine tennis club charges an annual fee of £60 for adults and £25 for juniors. Write an algorithm, in pseudocode, that inputs the membership type for 20 members and outputs the total fees collected and the number of juniors. [6]

> Initialises total and junior count to 0 (1) / loop that runs 20 times, e.g. FOR i ← 1 TO 20 (1) / inputs membership type inside the loop (1) / IF type = junior THEN adds 25 and increments junior count (1) / ELSE adds 60 (1) / outputs total and junior count after the loop (1)

9 A pupil's algorithm to total five numbers sets total ← 0 inside the loop, before each number is input. Using the inputs 4, 7, 2, 5, 1, explain what the algorithm outputs and why, and state the type of error. [3]

> Outputs 1 (the last number) instead of 19 (1) / because total is reset to 0 on every pass so previous values are lost (1) / logic error (1)

10 A programmer is designing an algorithm to calculate parking charges at an Enniskillen car park. Discuss how decomposition, abstraction, the choice of representation and dry running would help produce a correct, efficient solution. [8]

> Indicative content: decomposition into sub-problems such as entering arrival/departure times, calculating duration, applying tariff bands, discounts (e.g. Blue Badge), payment and printing a ticket; modules developed and tested separately, reused. Abstraction: keep only what affects the charge (times, vehicle type, tariff) and ignore detail such as car colour or driver's name; simpler model and less data. Representation: flowchart helps the council/client check the tariff logic visually; pseudocode is closer to code and suits detailed logic; both language-independent. Dry running with trace tables using normal, boundary (e.g. exactly 2 hours) and erroneous data finds logic errors before coding, which is cheaper; checks finiteness and correctness. Efficiency: avoid repeated or unnecessary steps. Levels: 7–8 all four areas discussed in depth and applied to the car park, accurate terminology, well structured; 4–6 three areas with some application; 1–3 limited, generic points or definitions only.

E Quick quiz

`Circle one answer · 9 marks`

1 Which property of an algorithm is broken by the instruction 'keep stirring until the sauce looks right'? [1]

A Inputs
B Finiteness ✓
C Sequence
D Abstraction

2 A bus map shows only stops and routes, not the exact shape of each road. This is an example of... [1]

A abstraction ✓
B decomposition
C iteration
D validation

3 Which construct repeats a set of instructions? [1]

A Sequence
B Selection

C Iteration ✓ D Decomposition

4 Which flowchart symbol is used for 'total ← total + price'? [1]

A Diamond B Parallelogram

C Rounded rectangle **D Rectangle ✓**

5 What is a sentinel value? [1]

A The first value in an array **B A special input value that signals the end of the data ✓**

C A constant used for VAT D An error message

6 Which tool records variable values step by step during a dry run? [1]

A Structure diagram B Data dictionary

C Trace table ✓ D Gantt chart

7 Inputs 12, 4, 15, 0 are processed by the rainfall algorithm (count days with rain > 10). What is output for wetDays? [1]

A 1 **B 2 ✓**

C 3 D 4

8 Why might an algorithm be written in plain English first? [1]

A It runs faster B It is shorter than code

C A client with no programming knowledge can check it is correct ✓ D It cannot contain errors

9 A loop totals 5 numbers but outputs only the last number entered. What is the most likely cause? [1]

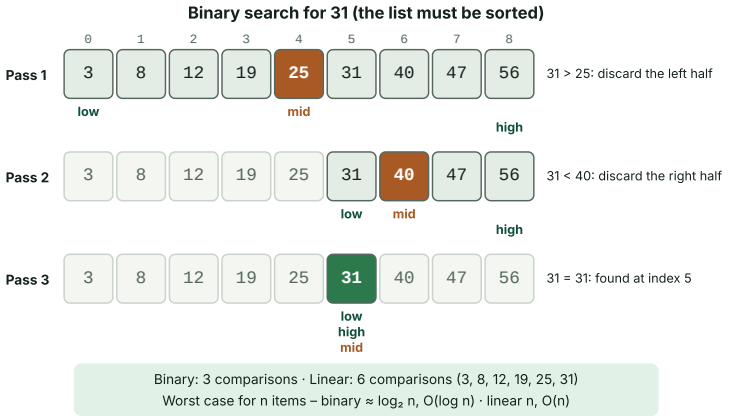
A total is reset to 0 inside the loop ✓ B The loop is a FOR loop

C total is output after the loop D The numbers are integers

Searching algorithms

AP17 ANSWERS

I can: Describe and code a linear search and a binary search in C# · Trace a binary search, recording low, high and mid and the items compared · Calculate the maximum number of comparisons each search needs for a list of a given size · Justify which search to use for a given scenario



Use this diagram to help you: Binary search for 31 in a sorted array of nine items: each pass compares the middle item and discards half, finding it in 3 comparisons compared with 6 for a linear search

A Key terms

Write the letter of the matching definition · 6 marks

A Linear search

F Binary search

D Divide and conquer

C Time complexity

B Big-O notation

E Worst case

A A search that checks each item in turn from the start until the target is found or the end of the list is reached.

B Notation that describes the order of growth of an algorithm's time or space, usually in the worst case, as input size increases, e.g. O(n).

C A measure of how the number of steps an algorithm takes grows as the size of its input grows.

D An approach that solves a problem by repeatedly reducing it to smaller sub-problems, as when binary search halves the list each time.

E The input that makes an algorithm do the most work, such as the target being last or absent in a linear search.

F A search of a sorted list that repeatedly compares the target with the middle item and discards the half that cannot contain it.

B Fill the gaps

Use the word bank · 9 marks

integer division · comparisons · linear search · sorted · middle · divide and conquer · binary search · log n · n

A **linear search** checks every item in turn and works on unsorted data; its worst case for n items is n **comparisons**. A **binary search** needs the data to be **sorted** and compares the target with the **middle** item, discarding half of the remaining items each time, which is a **divide and conquer**

approach. The middle position is found using **integer division**. Binary search has a time complexity of O(**log n**), whereas linear search is O(**n**).

C True or false?

Tick one box · 8 marks

	True	False
A binary search can be carried out on an unsorted list. *(The list must be sorted, or discarding half of it could throw away the target.)*	☐	☑
Linear search has a best case of one comparison.	☑	☐
Doubling the number of items roughly doubles the worst-case comparisons of a binary search. *(It adds only about one extra comparison; doubling applies to linear search.)*	☐	☑
O(n) means the time grows in proportion to the number of items.	☑	☐
Binary search is an example of divide and conquer.	☑	☐
A linear search can be used on records read one at a time from a serial file.	☑	☐
A binary search stops when the middle item is zero. *(It stops when the target is found or when low becomes greater than high (no items remain).)*	☐	☑
For a list of 10 items, binary search is always much faster than linear search. *(For very small lists the difference is negligible, and if the target is first a linear search needs only one comparison.)*	☐	☑

D Exam-style questions

36 marks

1 State one condition that must be met before a binary search can be used. [1]

> The list/data must be sorted (into order) (1)

2 Describe how a linear search finds an item in a list. [3]

> Start at the first item (1) / compare each item in turn with the target (1) / stop when the item is found or the end of the list is reached (target not present) (1)

3 Describe how a binary search finds an item in a list. [5]

> List must be sorted (1) / find/compare the middle item with the target (1) / if equal, the item is found and the search stops (1) / if the target is less than the middle item, discard the upper half (or if greater, discard the lower half) (1) / repeat with the remaining half until found or no items remain (1)

4 The sorted list 4, 9, 15, 22, 30, 37, 41 is searched for 37 using a binary search. List the items compared, in order. [2]

> 22 (1) / 37 (1)

5 A Ballymena library holds 15 book codes in ascending order: 11, 17, 23, 28, 34, 42, 49, 55, 61, 68, 74, 80, 87, 93, 99. Using positions 0 to 14 and mid = (low + high) DIV 2, complete a trace table for a binary search for 50 and state the result. [5]

> Pass 1: low 0, high 14, mid 7, item 55, 50 < 55 so high ← 6 (1) / Pass 2: low 0, high 6, mid 3, item 28, 50 > 28 so low ← 4 (1) / Pass 3: low 4, high 6, mid 5, item 42, low ← 6 (1) / Pass 4: low 6, high 6, mid 6, item 49, low ← 7 (1) / low (7) > high (6) so 50 is not in the list, after 4 comparisons (1)

6 A list contains 1 024 sorted items. State the maximum number of comparisons needed for (a) a linear search and (b) a binary search. [2]

> (a) 1 024 (1) / (b) 11 (1) – $1\,024 = 2^{10}$ so $\lfloor \log_2 1024 \rfloor + 1 = 11$

7 Explain two reasons why a linear search might be chosen instead of a binary search. [4]

> Any two, each identified (1) and explained (1): data is unsorted (1) and sorting first would take extra time (1) / list is small (1) so the speed difference is negligible and the simpler algorithm is quicker to write and less error-prone (1) / data is read serially, e.g. from a file (1) so the middle item cannot be accessed directly (1) / data changes frequently (1) so keeping it sorted is costly (1)

8 Explain what is meant by the statement that binary search has a time complexity of O(log n). [2]

> The (maximum) number of comparisons/steps grows logarithmically with the number of items (1) / e.g. doubling the number of items adds only one more comparison (1)

9 Write a C# method that performs a linear search of an array of strings and returns the position of the target, or -1 if it is not found. [4]

> Correct method header with string array and string target parameters returning int (1) / loop through every index from 0 to Length − 1 (1) / compare each element with the target and return the index when equal (1) / return -1 after the loop if not found (1)

10 The Western Health and Social Care Trust stores 200 000 patient records and adds new records daily. Evaluate the use of linear search and binary search to find a record by patient number. Refer to efficiency in your answer. [8]

> Indicative content: linear search works on unsorted records so new records can simply be appended; but worst case 200 000 comparisons, average about 100 000; O(n); slow when searched frequently by many staff. Binary search requires records sorted by patient number; worst case $\lfloor \log_2 200\,000 \rfloor + 1 = 18$ comparisons; O(log n); dramatically faster at this scale; needs direct access to records. Daily additions mean the list must be kept sorted – insert new records in order or re-sort, which has a cost; but patient numbers may be issued in increasing order, so appending keeps order. Searches far outnumber insertions in a hospital, so the sorting cost is outweighed. Conclusion: binary search (or an index) is justified. Levels: 7–8 both algorithms evaluated with accurate efficiency figures, the impact of daily additions discussed and a justified conclusion in context; 4–6 sound comparison with some efficiency reference and limited context; 1–3 descriptions of the searches with little evaluation.

E Quick quiz

Circle one answer · 9 marks

1 A sorted list holds 1 000 items. What is the maximum number of comparisons a binary search needs? [1]

A 10 ✓ B 500
C 100 D 1 000

2 Why must the list be sorted before a binary search? [1]

A So the middle item can be calculated B So the program uses less memory
C So the half that cannot contain the target can be safely discarded ✓ D So that every item is checked

3 What is the worst-case number of comparisons for a linear search of 500 items? [1]

A 9 B 250
C 500 ✓ D 1 000

4 What is the big-O time complexity of binary search? [1]

A O(1) **B O(log n)** ✓
C O(n) D $O(n^2)$

5 In a binary search, low = 4 and high = 9. What is mid? [1]

A 5 B 7
C 6.5 **D 6** ✓

6 When does a binary search stop if the target is not present? [1]

A When mid = 0 **B When low > high** ✓

C After n comparisons

D When high = mid

7 Which is a good reason to choose a linear search? [1]

A The list is huge and sorted

B The list is sorted and searched constantly

C It is O(log n)

D The list is short and unsorted ✓

8 The list 3 8 12 19 25 31 40 47 56 is searched for 31. Which items are compared? [1]

A 25, 40, 31 ✓

B 25, 31

C 3, 8, 12, 19, 25, 31

D 40, 31

9 A data set grows from 1 000 to 2 000 items. Roughly what happens to the worst-case comparisons of a linear search? [1]

A Stays the same

B Increases by one

C Halves

D Doubles ✓

AS 1: APPROACHES TO SYSTEMS DEVELOPMENT · ALGORITHMS

Sorting algorithms

AP18 ANSWERS

I can: Trace bubble sort and insertion sort pass by pass, showing every comparison and swap · Write both sorts in C# using arrays, nested loops and a temporary variable · Describe how merge sort and quick sort divide and recombine a list · Choose and justify a sorting algorithm for a scenario using big-O and practical factors



Use this diagram to help you: Bubble sort and insertion sort of 5 1 4 2 8 shown pass by pass side by side

A Key terms

Write the letter of the matching definition · 6 marks

A Bubble sort

F Insertion sort

C Merge sort

B Quick sort

D Divide and conquer

E Pass

A A sort that repeatedly compares adjacent items and swaps them if they are in the wrong order until a pass makes no swaps.

B A divide-and-conquer sort that partitions items around a pivot and then recursively sorts each partition.

C A divide-and-conquer sort that splits a list into single items and then merges sub-lists back together in order.

D An approach that solves a problem by repeatedly breaking it into smaller sub-problems of the same type and combining their solutions.

E One complete traversal through the list (or the unsorted part of it) by a sorting algorithm.

F A sort that takes each item in turn and inserts it into its correct position within an already sorted section of the list.

B Fill the gaps

Use the word bank · 9 marks

bubble sort | adjacent | merge sort | temporary variable | pivot | swapped flag | Quick sort | insertion sort | O(n²)

A **bubble sort** compares **adjacent** items and swaps them if they are in the wrong order, using a **temporary variable** to avoid losing a value; a **swapped flag** lets it stop when a pass makes no swaps.

An **insertion sort** takes each item in turn and places it into the correct position in the sorted part of the list. A **merge sort** splits the list into single items and then merges them. **Quick sort** partitions the

list around a **pivot**. Bubble and insertion sort have a worst-case time complexity of **$O(n^2)$**, whereas merge sort is always O(n log n).

C True or false?

Tick one box · 9 marks

	True	False
After the first pass of a bubble sort, the largest item is in its final position (ascending sort).	☑	☐
Merge sort needs no extra memory. *(It needs extra space to hold the sub-lists while merging.)*	☐	☑
Insertion sort is efficient for a nearly sorted list.	☑	☐
Quick sort always takes O(n log n) time. *(Its worst case is $O(n^2)$, for example when a first-item pivot is used on already sorted data.)*	☐	☑
Merge sort and quick sort are divide-and-conquer algorithms.	☑	☐
A bubble sort with a swapped flag stops when a pass makes no swaps.	☑	☐
A swap of two array items can be done safely with two assignments and no extra variable. *(With only two assignments one value is overwritten; a temporary variable is needed.)*	☐	☑
After partitioning in quick sort, the pivot is in its final sorted position.	☑	☐
Doubling the size of a list roughly doubles the worst-case work of a bubble sort. *(Bubble sort is $O(n^2)$, so doubling n roughly quadruples the work.)*	☐	☑

D Exam-style questions

38 marks

1 Show the list 6, 3, 8, 2 after each pass of a bubble sort into ascending order. [3]

> Pass 1: 3 6 2 8 (1) / Pass 2: 3 2 6 8 (1) / Pass 3: 2 3 6 8 (1)

2 Show the list 6, 3, 8, 2 after each pass of an insertion sort into ascending order. [3]

> Pass 1: 3 6 8 2 (1) / Pass 2: 3 6 8 2 (1) / Pass 3: 2 3 6 8 (1)

3 Explain why a temporary variable is needed when swapping two items in an array. [2]

> When the first value is overwritten by the second, the original first value would be lost (1) / so it is copied to a temporary variable first and then copied into the second position (1)

4 Explain how a Boolean flag improves the efficiency of a bubble sort. [2]

> The flag records whether any swap was made during a pass (1) / if a pass makes no swaps the list is sorted so the algorithm stops early, avoiding unnecessary passes/comparisons (1)

5 A Derry gym records class sizes 12, 7, 15, 9, 20. State the number of comparisons and the number of swaps made by a bubble sort with a swapped flag, showing the list after each pass. [4]

> Pass 1: 7 12 9 15 20 (1) / Pass 2: 7 9 12 15 20 (1) / Pass 3: no swaps, so it stops (1) / 9 comparisons (4 + 3 + 2) and 3 swaps (1)

6 Describe how a merge sort sorts a list. [4]

> The list is repeatedly split/divided in half (1) / until each sub-list contains one item (1) / sub-lists are merged in pairs by comparing the front items and taking the smaller (1) / repeated until a single sorted list remains (1)

7 Using the first item as the pivot, show the first partition of the list 5, 8, 2, 9, 1, 6 in a quick sort and explain what has been achieved. [3]

> Pivot 5: items smaller (2, 1) placed to its left and larger (8, 9, 6) to its right, e.g. 2 1 5 8 9 6 (1) / the pivot 5 is now in its final sorted position (1) / the two sub-lists can now be sorted separately/recursively (1)

8 Write C# code that sorts an integer array called scores into ascending order using an insertion sort. [5]

> Outer loop from index 1 to scores.Length − 1 (1) / store scores[i] as key and set j = i − 1 (1) / while loop with j >= 0 and scores[j] > key (1) / shift scores[j] to scores[j + 1] and decrement j (1) / place key at scores[j + 1] after the while loop (1)

9 Compare merge sort and quick sort in terms of time efficiency and memory use. [4]

> Time: merge sort is always O(n log n) (1) whereas quick sort is O(n log n) on average but $O(n^2)$ in the worst case with a poor pivot (1) / Memory: merge sort needs extra memory for the sub-lists (1) whereas quick sort sorts in place, needing little extra memory (1)

10 A Belfast logistics company sorts 2 million parcel records each night; a small app sorts a pupil's 15 test scores, and a sports club adds two new members each week to an alphabetical list of 300. Evaluate which sorting algorithms would be suitable in each case. [8]

> Indicative content: 2 million records – $n^2 \approx 4 \times 10^{12}$ comparisons for bubble/insertion, unacceptable; $n \log_2 n \approx$ 42 million for merge/quick; merge sort guarantees O(n log n) and suits overnight batch processing (can also sort data too large for memory), but needs extra memory; quick sort usually fastest and in place but risk of $O(n^2)$ with poor pivot (use middle/random pivot). 15 scores – any algorithm is effectively instant; bubble or insertion sort is simplest to code, test and maintain; overhead of recursive sorts not justified. 300 members with two additions – list is already sorted, so insertion sort is ideal (close to O(n) on nearly sorted data, inserts the new names into place); bubble sort with a flag also finishes in few passes; full merge sort unnecessary. Levels: 7–8 all three scenarios evaluated with justified choices using efficiency and memory, accurate figures and terminology; 4–6 sensible choices for at least two scenarios with some justification; 1–3 descriptions of algorithms with little application.

E Quick quiz

`Circle one answer · 9 marks`

1 What is the list 4 2 7 1 after ONE pass of a bubble sort into ascending order? [1]

A 2 4 1 7 ✓ B 1 2 4 7
C 2 1 4 7 D 2 4 7 1

2 Which sorting algorithm repeatedly splits a list in half and then merges sorted sub-lists? [1]

A Bubble sort B Insertion sort
C Merge sort ✓ D Quick sort

3 Which sort places each item into the correct position in a growing sorted section? [1]

A Bubble sort B Merge sort
C Quick sort **D Insertion sort ✓**

4 What is the average time complexity of merge sort? [1]

A O(n) B $O(n^2)$
C O(n log n) ✓ D O(log n)

5 What does quick sort use to partition the list? [1]

A A pivot ✓ B A flag
C A sentinel D A temporary list of single items

6 Which is a disadvantage of merge sort? [1]

A It is $O(n^2)$ **B It needs extra memory ✓**
C It only works on sorted lists D It cannot sort strings

7 In an insertion sort of 14 9 22 5 17 11, what is the list after the third pass? [1]

A 9 14 22 5 17 11 **B 5 9 14 22 17 11 ✓**

C 5 9 11 14 17 22

D 9 14 5 22 17 11

8 A bubble sort with a swapped flag is given a list that is already sorted. How many passes does it make? [1]

A 1 ✔

B 0

C n − 1

D n

9 Why is quick sort's worst case $O(n^2)$? [1]

A It always needs extra memory

B It cannot handle duplicates

C A poor pivot can leave one partition almost as large as the list ✔

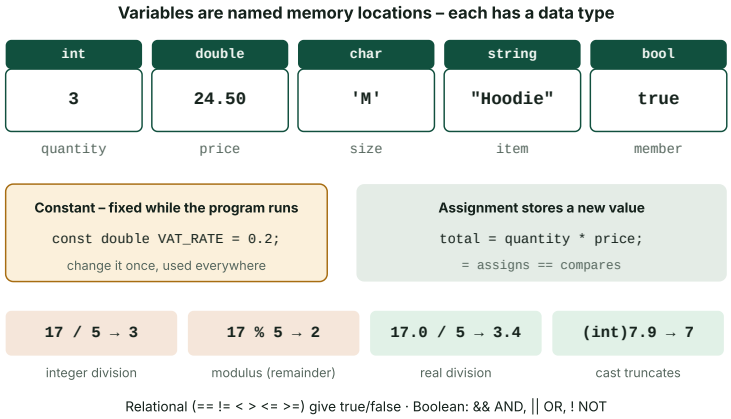
D It compares only adjacent items

AS 1: APPROACHES TO SYSTEMS DEVELOPMENT · FUNDAMENTALS OF PROGRAMMING

IDEs, translators, data types, variables and operators

AP19 ANSWERS

I can: Describe how the tools in an IDE help a programmer write, translate and debug code · Compare assemblers, compilers and interpreters and say when each is useful · Choose the most suitable C# data type for a value and justify the choice · Declare, initialise and assign variables and constants using rules and conventions for identifiers · Evaluate expressions and conditions using arithmetic, relational and Boolean operators



Use this diagram to help you: Variables as labelled memory boxes with their C# data types, a locked constant, and examples of integer division and modulus

A Key terms

Write the letter of the matching definition · 6 marks

A Data type

F Variable

E Constant

D Identifier

C Syntax

B Integrated development environment (IDE)

A A classification of data that determines the values a variable can hold, the memory it needs and the operations that can be performed on it.

B A software application that provides the tools needed to develop programs, such as an editor, translator and debugging tools, in one place.

C The set of rules that define how valid statements must be written in a programming language.

D The name given to a variable, constant or subroutine so that it can be referred to in a program.

E A named value that is set when the program is written and cannot change while the program runs.

F A named location in memory that stores a single value which can change while the program runs.

B Fill the gaps

Use the word bank · 9 marks

identifier | source code | interpreter | compiler | constant | Boolean | variable | IDE | modulus

Programmers write **source code** in an **IDE**, which provides an editor, a translator and debugging tools. A **compiler** translates the whole program before it runs, whereas an **interpreter** translates and executes one statement at a time. A named memory location whose value can change is a **variable**, and its name is called its **identifier**. A value that cannot change while the program

runs is a **constant** . A data type holding only true or false is **Boolean** . In C#, 17 % 5 gives 2 because % is the **modulus** operator.

C True or false?

Tick one box · 10 marks

	True	False
An interpreter produces an executable file that can be run later without the interpreter. *(An interpreter produces no object code, so the source code and the interpreter are needed every time the program runs.)*	☐	☑
A compiler reports errors after translating the whole program, and the program cannot run until they are corrected.	☑	☐
Each assembly language instruction is normally translated into one machine code instruction.	☑	☐
In C#, 9 / 2 evaluates to 4 when both values are integers.	☑	☐
A telephone number should be stored as an integer. *(It is never used in arithmetic and may have a leading zero or spaces, so it should be a string.)*	☐	☑
The value of a constant can be changed by an assignment later in the program. *(A const cannot be reassigned; attempting it is a compile-time error.)*	☐	☑
&& returns true only if both conditions are true.	☑	☐
A C# identifier may contain a space if it is meaningful, e.g. ticket price. *(Identifiers cannot contain spaces; use camelCase (ticketPrice) or an underscore.)*	☐	☑
Console.ReadLine() returns a string, even when the user types a number.	☑	☐
= and == mean the same thing in C#. *(= assigns a value to a variable; == compares two values and gives true or false.)*	☐	☑

D Exam-style questions

42 marks

1 A Belfast software start-up is choosing an IDE for its new apprentices. Describe three features of an IDE that would help the apprentices write correct source code. [6]

> Any three features, each with description of how it helps (2 marks each): syntax highlighting / colour coding (1) makes keywords and structure easier to read (1) / automatic indentation (1) shows which statements belong to which block (1) / syntax checking as you type (1) underlines errors such as a missing semicolon so they are fixed before compiling (1) / code completion or IntelliSense (1) suggests identifiers, reducing spelling mistakes (1) / preferences or auto-formatting (1) keep a team's layout consistent (1) / debugging tools such as breakpoints and stepping (1) let the programmer inspect variables to locate errors (1)

2 An apprentice says the C# file they typed 'is the program the CPU runs'. State what is meant by translation, and why it is needed before the program can run. [2]

> Converting a program from the language it is written in (source code, e.g. C#) (1) into machine code / object code that the processor can execute, because the CPU cannot run high-level source code directly (1)

3 Explain two differences between a compiler and an interpreter. [4]

> Any two differences, two marks each: a compiler translates the whole program before execution (1) whereas an interpreter translates and executes one statement at a time (1) / a compiler produces an executable/object code (1) whereas an interpreter produces no object code so it is needed each time (1) / a compiler reports all errors after translation (1) whereas an interpreter stops at the first error encountered (1) / compiled code runs faster (1) because an interpreter retranslates each statement every time it runs (1)

4 Explain what is meant by a variable and why its data type is declared before it is used. [3]

> A named location in memory (1) that holds a single value which can change while the program runs (1) / the data type is declared so that the correct amount of memory can be reserved / so the compiler can check that only valid operations are used (1)

5 A leisure centre in Newry stores the following for each booking. State the most appropriate data type for each and give a reason for (d): (a) the number of people in the booking (b) the booking fee (c) a court letter such as 'C' (d) the customer's mobile number (e) whether the booking has been paid. [6]

> (a) integer (1) (b) real / decimal / currency (1) (c) character (1) (d) string (1) (e) Boolean (1) / reason for (d): it may begin with a zero, which would be lost as a number / it is not used in calculations / it may contain spaces (1)

6 Given int x = 23, int y = 4 and double price = 12.5, state the value of: (a) x / y (b) x % y (c) x > 20 && y > 5 (d) (double)x / y (e) !(price * 2 > 20) || y == 4 [5]

> (a) 5 (1) (b) 3 (1) (c) false (23 > 20 is true, 4 > 5 is false) (1) (d) 5.75 (1) (e) true: price * 2 = 25, 25 > 20 is true, !true is false, y == 4 is true, false || true is true (1)

7 Explain two benefits of declaring the hourly parking rate in a car park program as a constant. [4]

> The value cannot be changed accidentally while the program runs (1) so calculations always use the correct rate (1) / if the rate changes it only needs to be updated in one place (1) reducing the chance of missing an occurrence / saving maintenance time (1) / the name (e.g. RATE_PER_HOUR) makes the code easier to read (1) than an unexplained number (1). Max 4.

8 Write a C# Boolean expression that is true when age is from 12 to 18 inclusive and hasPass is true. [3]

> age >= 12 (1) && age <= 18 (1) && hasPass (or hasPass == true) (1). Accept (age >= 12 && age <= 18) && hasPass with brackets.

9 Explain why the line double average = total / count; may give an incorrect result when total and count are both int, and show how to correct it. [3]

> Both operands are integers so integer division is performed (1) and the fractional part / remainder is discarded, e.g. 7 / 2 gives 3 (1) / correction: cast one operand, e.g. double average = (double)total / count; (1)

10 A software company in Derry~Londonderry is developing a booking app for a local tour operator. Discuss how the company might use an interpreter and a compiler during development and when distributing the finished app. [6]

> Indicative content: during development an interpreter allows code to be run and tested immediately without recompiling the whole program; errors are reported as soon as the faulty line is reached, helping debugging; but interpreted code runs more slowly and errors may only appear after part of the program has run, e.g. after a booking has been saved. For distribution a compiler produces a single optimised executable that runs faster on customers' devices; the source code is not distributed so the company's code is protected and cannot easily be altered; users do not need an interpreter installed; all syntax errors must be fixed before an executable exists, giving more confidence; but each change needs a full recompile. Conclusion: interpreter (or an IDE's quick build/run) during development, compiler for release. Levels: 5-6 balanced discussion of both translators in context with a justified conclusion; 3-4 accurate points about both with some context; 1-2 limited or one-sided points.

E Quick quiz

`Circle one answer · 10 marks`

1 Which translator converts the whole of a high-level program into object code before any of it is run? [1]

A Interpreter

B Assembler

C Compiler ✓

D Editor

2 A program stores a customer's mobile number, such as 07700 900123. Which data type is most suitable? [1]

A int
B double
C string ✓
D bool

3 What is the value of 29 % 6? [1]

A 4
B 5 ✓
C 4.83
D 0

4 Which data type is most suitable for a single grade letter such as 'B'? [1]

A string
B int
C bool
D char ✓

5 What does (int)8.9 evaluate to in C#? [1]

A 9
B 8.9
C 8 ✓
D An error

6 Which expression is true when x is outside the range 1 to 10? [1]

A x < 1 || x > 10 ✓
B x < 1 && x > 10
C x >= 1 && x <= 10
D !(x < 1)

7 Which keyword declares a constant in C#? [1]

A static
B const ✓
C final
D fixed

8 What is output by Console.WriteLine("Total: " + 3 + 4);? [1]

A Total: 7
B Total: 34 ✓
C An error
D Total: 3 + 4

9 Which IDE feature suggests variable and method names after the programmer types a few letters? [1]

A Code completion ✓
B Breakpoint
C Compiler
D Error report

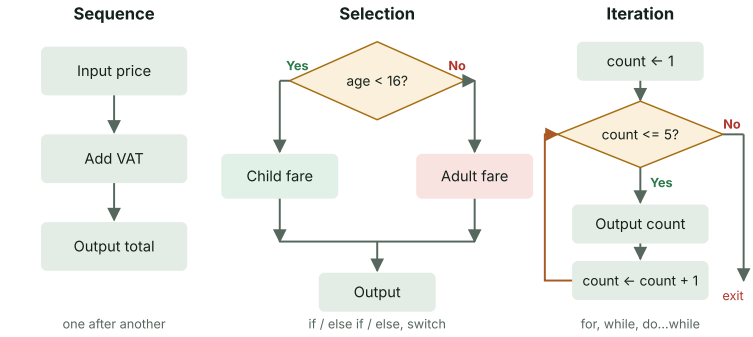
10 Which statement about an interpreter is correct? [1]

A It produces a standalone executable file
B It reports every syntax error before running
C It normally runs a program faster than compiled code
D It translates and executes one statement at a time ✓

Program control: sequence, selection and iteration

AP20 ANSWERS

I can: Explain sequence, selection and iteration with reference to a given algorithm · Choose between if...else if and switch for a selection problem and write the C# · Select the right type of loop for a task and justify the choice · Write and trace programs that use nested loops, sentinel values and nested selection



Use this diagram to help you: The three control structures as flowcharts: sequence, selection with two branches, and iteration with a loop back

A Key terms

Write the letter of the matching definition · 6 marks

D Sequence

F Selection

B Iteration

C Count-controlled loop

A Condition-controlled loop

E Nested loop

A A loop that repeats while or until a condition is met, so the number of repetitions need not be known in advance.

B Repeating a set of instructions, either a set number of times or while/until a condition is met.

C A loop that repeats a fixed number of times known before it starts, e.g. a for loop.

D Executing instructions one after another in the order in which they are written.

E A loop placed inside the body of another loop; the inner loop completes fully for each pass of the outer loop.

F A construct in which a condition determines which of two or more paths through the program is followed.

B Fill the gaps

Use the word bank · 9 marks

iteration | sentinel | while | for | infinite | nested | do...while | selection | sequence

Carrying out statements in the order they are written is called **sequence**. Choosing between paths using a condition is called **selection**, and repeating code is called **iteration**. A **for** loop is count-controlled because the number of repetitions is known in advance. A **while** loop tests its condition at the start, so it may run zero times, whereas a **do...while** loop always runs at least once. A dummy value such as -1 that marks the end of the data is a **sentinel** value. A loop

inside another loop is a **nested** loop, and a loop whose condition never becomes false is an **infinite** loop.

C True or false?

Tick one box · 9 marks

	True	False
A while loop always runs at least once. *(Its condition is tested at the start, so it may run zero times; a do...while loop always runs at least once.)*	☐	☑
In an else if chain, only the first branch whose condition is true is executed.	☑	☐
In an if...else statement, it is possible for both blocks to run. *(Exactly one of the two blocks runs, depending on whether the condition is true or false.)*	☐	☑
A switch statement uses a default case for values that match no case label.	☑	☐
A nested loop with 4 outer passes and 6 inner passes runs the inner body 10 times. *(It runs 4 × 6 = 24 times, because the inner loop completes fully for each outer pass.)*	☐	☑
A for loop is a count-controlled loop.	☑	☐
A sentinel value should be added to the running total like any other input. *(The sentinel only marks the end of the data; it must not be processed, so it is checked before being added.)*	☐	☑
for (int i = 0; i < 5; i++) runs the body five times.	☑	☐
REPEAT ... UNTIL x > 10 keeps looping while x > 10 is true. *(UNTIL stops the loop as soon as x > 10 becomes true; it keeps looping while the condition is false.)*	☐	☑

D Exam-style questions

39 marks

1 State the three basic programming constructs. [3]

> Sequence (1) / selection (1) / iteration or repetition (1)

2 A Ballymena sports shop's loyalty app uses this algorithm: points ← 0; FOR purchase ← 1 TO 5: INPUT spend; points ← points + spend DIV 10; NEXT purchase; IF points >= 50 THEN OUTPUT "Voucher earned" ENDIF; OUTPUT points. By referring to the algorithm, explain the terms (a) sequence (b) selection (c) iteration. [6]

> (a) Sequence: instructions are executed one after another in the order written (1) e.g. points is set to 0 before the loop, and the voucher check and final OUTPUT points happen only after all five spends have been processed (1) (b) Selection: a condition decides which path is followed / whether a block runs (1) e.g. "Voucher earned" is output only IF points >= 50 (1) (c) Iteration: a set of instructions is repeated (1) e.g. INPUT spend and the points update are repeated once for each of the 5 purchases by the FOR loop (1)

3 Describe the difference between a while loop and a do...while loop. [2]

> A while loop tests the condition at the start so it may run zero times (1) / a do...while loop tests the condition at the end so it always runs at least once (1)

4 A Strabane swimming club's program reads the times of the 8 swimmers in each race, then keeps asking the coach for extra notes until the coach types 'done'. Explain why a count-controlled loop suits the first task and a condition-controlled loop suits the second. [4]

> Count-controlled: repeats a number of times known before the loop starts (1) / there are always 8 swimmers, so a for loop that runs 8 times fits (1) / condition-controlled: repeats while/until a condition is met, so the number of repetitions need not be known (1) / the coach may add any number of notes (even none), so the loop continues until 'done' is entered (1)

5 State what is meant by a sentinel value and explain why a while loop is suitable for processing data that ends with one. [3]

> A dummy/rogue value that marks the end of the data and is not processed as data (1) / a while loop tests before each pass (1) so if the sentinel is entered first the body runs zero times / the number of values is unknown in advance (1)

6 How many times is the inner statement executed? for (int a = 0; a < 5; a++) { for (int b = 0; b < 3; b++) { Console.Write("*"); } } [1]

> 15 (5 × 3) (1)

7 Complete a trace table for the following code and state the final output: int n = 1, t = 0; while (t < 20) { t = t + n * 3; n++; } Console.WriteLine(n + " " + t); [4]

> Trace: t=0,n=1 → t=3,n=2 → t=9,n=3 → t=18,n=4 → t=30,n=5; loop stops because 30 is not < 20 (1 per correct pair of rows, max 3) / output: 5 30 (1)

8 Give one advantage and one disadvantage of a switch statement compared with an else if chain. [2]

> Advantage: clearer/more readable when one variable is matched against many exact values / less repetition / default handles all other values (1) / disadvantage: classic switch cannot test ranges or combined conditions / only compares one variable / needs break in each case (1)

9 A school in Coleraine records exam marks. Write C# code that repeatedly asks for a mark until a value from 0 to 100 is entered, then outputs Distinction for 70 or more, Pass for 40 to 69 and Fail otherwise. [6]

> Sample: int mark; do { Console.Write("Mark: "); mark = int.Parse(Console.ReadLine()); } while (mark < 0 || mark > 100); if (mark >= 70) Console.WriteLine("Distinction"); else if (mark >= 40) Console.WriteLine("Pass"); else Console.WriteLine("Fail"); / Marks: input read and converted to integer (1) / loop that repeats (1) / correct condition mark < 0 || mark > 100 (1) / Distinction for >= 70 (1) / Pass for 40-69 with correct boundaries (1) / Fail otherwise and correct else if structure (1)

10 A Fermanagh outdoor centre wants a program to calculate the cost of a group booking. The group size must be between 1 and 12, each person's age is entered, and prices depend on age band. Discuss how sequence, selection and iteration would be used in the program, justifying the choice of each type of loop and selection statement. [8]

> Indicative content: sequence – read the prices for each band, then the group size, then each age, then calculate and output the total; order matters because the total depends on all inputs. Iteration – a condition-controlled do...while loop to read and validate group size, because the user must enter it at least once and may need several attempts; a count-controlled for loop for the ages, because the group size is known before the loop starts; a running total updated inside the loop. Selection – an if...else if chain to choose the price band for each age because bands are ranges; switch unsuitable for ranges; an if inside the validation loop to show an error message; possibly a discount if total exceeds a threshold. Nesting – selection inside the for loop. Levels: 7-8 all three constructs explained in context with justified choice of loop types and selection statement; 4-6 all three constructs identified with some justification; 1-3 basic descriptions with little context or justification.

E Quick quiz

`Circle one answer · 9 marks`

1 A program must show a menu and keep showing it until the user chooses Quit. Which loop is most suitable? [1]

A for
B do...while ✓
C foreach
D None – use switch

2 What is the key difference between a count-controlled and a condition-controlled loop? [1]

A Only condition-controlled loops use a counter
B Count-controlled loops can never be nested
C A count-controlled loop repeats a number of times known before it starts ✓
D A condition-controlled loop always runs at least once

3 What is printed by for (int i = 2; i < 10; i += 3) Console.Write(i);? [1]

A 258 ✓ B 2581

C 25811 D 36912

4 What happens if a while loop's condition never becomes false? [1]

A A syntax error B It runs once

C An infinite loop ✓ D The program skips it

5 Which statement is most suitable for choosing between menu options 1, 2, 3 and 4? [1]

A while B for

C do...while **D switch ✓**

6 In REPEAT ... UNTIL x > 10, when does the loop stop? [1]

A When x > 10 becomes true ✓ B When x > 10 is false

C Before the first pass D Never

7 Scores are entered until -1 is typed. If the user types -1 first, what does a correctly written while loop do? [1]

A Adds -1 to the total B Runs the body once

C Crashes **D Does not run the body at all ✓**

8 Given if (age < 5) ... else if (age < 16) ... else if (age >= 60) ... else ..., which branch runs when age is 3? [1]

A The first only ✓ B The first and second

C All three if branches D The else

9 An outer loop runs 3 times and its inner loop runs 4 times per pass. How many times does the inner body run? [1]

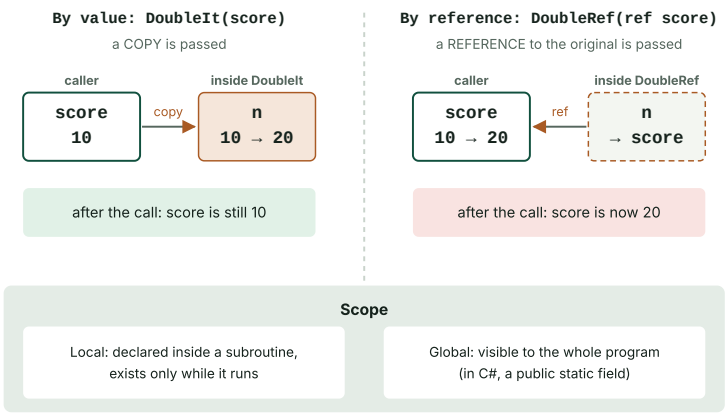
A 7 **B 12 ✓**

C 4 D 3

Subroutines: procedures, functions, parameters and scope

AP21 ANSWERS

I can: Write C# procedures and functions that take parameters and return values · Distinguish between parameters and arguments and match them in number, order and type · Predict the effect of passing an argument by value and by reference (ref) · Explain local and global scope and the advantages of a modular design



Use this diagram to help you: Passing by value copies the argument into the parameter so the original is unchanged; passing by reference makes the parameter refer to the original variable, so it changes

A Key terms

Write the letter of the matching definition · 6 marks

C Subroutine

F Procedure

B Function

D Parameter

A Argument

E Passing by value

A The actual value or variable supplied to a subroutine when it is called.

B A subroutine that performs a task and returns a single value to the point where it was called.

C A named, self-contained block of code that performs a specific task and can be called from elsewhere in a program.

D A variable in a subroutine's header that receives a value (argument) passed in when the subroutine is called.

E Passing a copy of an argument to a subroutine so that the original variable cannot be changed by it.

F A subroutine that performs a task but does not return a value (void in C#).

B Fill the gaps

Use the word bank · 10 marks

function · procedure · local · value · arguments · modular · reference · parameters · scope · subroutine

A named block of code that performs a task is a **subroutine** . One that returns a value is a **function** ; one that does not is a **procedure** . Values are received by the subroutine's **parameters** , while the values supplied in the call are the **arguments** . Passing by **value** sends a copy, whereas passing by **reference** lets the subroutine change the original variable. The

part of the program in which a variable can be used is its **scope** ; a **local** variable exists only inside the subroutine where it is declared. Dividing a program into self-contained subroutines is called **modular** programming.

C True or false?

Tick one box · 9 marks

	True	False
In C#, a procedure has the return type void.	☑	☐
Passing by value allows a subroutine to change the original variable. *(By value passes a copy; only by reference (ref) can change the caller's original variable.)*	☐	☑
A local variable can be accessed from any subroutine in the program. *(It can only be accessed within the subroutine or block where it is declared.)*	☐	☑
A function call can be used as part of an expression, e.g. double t = AddVat(40, 0.2);	☑	☐
Arguments must match the parameters in number, order and data type.	☑	☐
Modular programs are easier to test because each module can be tested on its own.	☑	☐
Using subroutines always makes a program run faster. *(Each call adds a small overhead; the benefits are in development, testing and maintenance, not speed.)*	☐	☑
A global variable exists for the whole time the program is running.	☑	☐
In C#, ref only needs to appear in the subroutine header, not in the call. *(ref must be written in both the header and the call, which makes the by-reference passing visible.)*	☐	☑

D Exam-style questions

36 marks

1 State the difference between a procedure and a function. [1]

> A function returns a value (to the calling code) whereas a procedure does not (1)

2 Explain the difference between a parameter and an argument, using an example. [3]

> A parameter is a variable in the subroutine header that receives a value (1) / an argument is the actual value/variable passed in the call (1) / valid example identifying both, e.g. in AddVat(double price) price is the parameter and in AddVat(40) 40 is the argument (1)

3 Write the header of a C# function called CalculateFare that takes the number of zones (an integer) and whether the passenger is a student (Boolean) and returns the fare as a real number. [3]

> static double CalculateFare(int zones, bool isStudent) – return type double (1) / name and int parameter (1) / bool parameter (1). Accept decimal/float for double; any sensible parameter names.

4 Explain the difference between passing a parameter by value and by reference. [4]

> By value: a copy of the variable/value is passed (1) so changes inside the subroutine do not affect the original (1) / by reference: the address of / a reference to the original variable is passed (1) so changes made in the subroutine change the original (1)

5 State the values output by this code and explain your answer: static void Change(int a, ref int b) { a = a + 5; b = b + 5; } int p = 1, q = 1; Change(p, ref q); Console.WriteLine(p + " " + q); [3]

> Output 1 6 (1) / p is passed by value so only the copy a is changed (1) / q is passed by reference so b refers to q and q becomes 6 (1)

6 Explain what is meant by a local variable and give two advantages of using local variables. [3]

> A variable declared inside a subroutine that can only be accessed within that subroutine / exists only while it runs (1) / any two advantages: the same name can be reused in other subroutines without conflict (1) / the subroutine is self-contained so can be tested and reused independently (1) / memory is freed when the subroutine ends (1) / cannot be changed accidentally by other parts of the program (1)

7 Give two disadvantages of using global variables. [2]

> Any two: can be changed from anywhere so errors are hard to trace (1) / makes modules dependent on each other, reducing reuse (1) / name clashes are more likely (1) / memory is held for the whole run (1)

8 A bowling alley in Bangor charges £6.50 per game for adults and £4.50 for under-16s. Write a C# function called GameCost that takes the age and number of games and returns the total cost, and show how it would be called to output the cost of 3 games for a 12-year-old. [6]

> Sample: static double GameCost(int age, int games) { double price; if (age < 16) price = 4.50; else price = 6.50; return price * games; } Console.WriteLine($"£{GameCost(12, 3):F2}"); / Marks: correct header with return type double (1) / two int parameters (1) / selection on age < 16 (1) / correct prices (1) / returns price × games (1) / correct call with output, giving £13.50 (1)

9 The function Divide(int a, int b) sets count to 0, then while a >= b it subtracts b from a and adds 1 to count, and finally returns count. Trace the call Divide(20, 4) and state the value returned. [3]

> a = 20, count = 0 → a = 16, count = 1 → a = 12, count = 2 → a = 8, count = 3 → a = 4, count = 4 → a = 0, count = 5 → 0 >= 4 is false so loop ends (2 for complete correct trace, 1 if partly correct) / returns 5 (1)

10 A team of programmers is developing a booking system for an Enniskillen leisure centre. Discuss the benefits and drawbacks of a modular approach using subroutines with parameters, including the use of local and global variables. [8]

> Indicative content – benefits: modules such as CheckAvailability, CalculatePrice and TakePayment can be developed by different programmers at the same time; each can be unit tested so errors are found more easily; code is reused (e.g. one price function for all activities), so less duplication; maintenance in one place, e.g. a price change; parameters let a module work on any booking; return values pass results back cleanly; libraries of tested modules can be reused in other centres' systems; main program is readable. Local variables keep modules independent and avoid name clashes; global variables (e.g. a shared opening-hours constant) may be justified but could be changed by any module, causing hard-to-trace errors. Drawbacks: time needed to plan modules and agree interfaces; integration testing still needed; passing many parameters can be cumbersome; small call overhead. Levels: 7-8 well-developed discussion of benefits and drawbacks in context, including parameters and scope, with a conclusion; 4-6 several accurate points with some context; 1-3 limited list of benefits.

E Quick quiz

Circle one answer · 9 marks

1 What is the main difference between a function and a procedure? [1]

A A function always has parameters
B A function returns a value to the calling code ✓
C A procedure can only be called once
D A procedure must use global variables

2 What does passing a parameter by value mean? [1]

A The subroutine receives a reference to the original
B The subroutine can only read global variables
C The subroutine receives a copy, so the original cannot be changed ✓
D The value is returned to the caller

3 What does the C# keyword ref do in a parameter list? [1]

A Makes the parameter read-only
B Passes the argument by reference ✓
C Returns a value
D Declares a global

4 Which is a valid call of static double Area(double w, double h)? [1]

A Area(3.5);
B double a = Area(3.5, "2");
C double a = Area(3.5, 2); ✓
D Area = (3.5, 2);

5 Where can a local variable be used? [1]

A Only inside the subroutine or block where it is declared ✓

B Anywhere in the program

C Only in the main program

D In any class

6 Which is NOT a benefit of modular programming? [1]

A Code reuse

B Easier testing

C Team working

D Programs always run faster ✓

7 What is printed? static void Add(int x) { x = x + 1; } int n = 4; Add(n); Console.WriteLine(n); [1]

A 5

B 1

C An error

D 4 ✓

8 In static int Max(int a, int b), what are a and b? [1]

A Arguments

B Global variables

C Return values

D Parameters ✓

9 Why might a programmer choose to pass a variable by reference? [1]

A So the subroutine can update the caller's variable ✓

B To protect the original from change

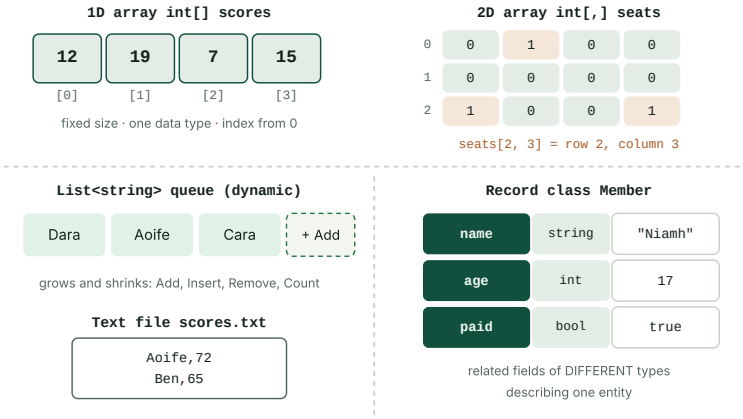
C To make the variable global

D Because functions cannot have parameters

Data structures: arrays, lists, records and files

AP22 ANSWERS

I can: Declare, populate, traverse and search 1D and 2D arrays in C# · Choose between an array and a list for a scenario and justify the choice · Define a record structure and process a collection of records · Save records to a comma-separated text file and load them back into a list



Use this diagram to help you: Four data structures: a one-dimensional array with indexes, a two-dimensional array addressed by row and column, a growing list, and a record with fields of different types

A Key terms

Write the letter of the matching definition · 6 marks

C Data structure

F Array

E Index

A Two-dimensional array

B Static data structure

D Dynamic data structure

A An array whose elements are arranged in rows and columns and are accessed using two indexes, [row, column].

B A data structure whose size is fixed when it is created and cannot change while the program runs, such as an array.

C A way of organising and storing a collection of related data items under one identifier so that they can be processed efficiently.

D A data structure whose size can grow or shrink while the program is running, such as a list.

E The position number used to access an element of an array or list; in C# the first index is 0.

F A static data structure that holds a fixed number of elements of the same data type under one identifier, each accessed by an index.

B Fill the gaps

Use the word bank · 9 marks

index · record · dynamic · static · fields · file · delimiter · two-dimensional array · array

An **array** holds a fixed number of elements of the same data type, and each element is reached by its **index**, which starts at 0. Because its size cannot change once it is created, an array is described as a **static** structure. Data arranged in rows and columns can be stored in a **two-dimensional array** and processed with nested loops. A List is a **dynamic** data structure

because items can be added and removed while the program runs. A **record** groups related **fields** that can have different data types. To keep data after the program closes it is written to a text **file**, often with fields separated by a comma **delimiter**.

C True or false?

Tick one box · 9 marks

	True	False
The first element of a C# array has index 1. *(Indexes start at 0, so the first element is at index 0 and the last at Length - 1.)*	☐	☑
A List<T> can grow as items are added while the program runs.	☑	☐
All the fields of a record must have the same data type. *(A record can group fields of different data types; it is arrays and lists that hold a single type.)*	☐	☑
In int[,] g = new int[3, 5]; the expression g.GetLength(1) returns 5.	☑	☐
Data stored in a list is kept after the program closes. *(A list is held in RAM and is lost; data must be written to a file for persistent storage.)*	☐	☑
Opening an existing file with new StreamWriter(name, false) adds new lines to the end of it. *(false means overwrite: the existing contents are replaced. Passing true appends to the end.)*	☐	☑
Values read from a text file must be converted before arithmetic is done on them.	☑	☐
Accessing an element beyond the end of an array causes a syntax error. *(The code compiles; it fails while running with an IndexOutOfRangeException, a run-time error.)*	☐	☑
A comma-separated file normally stores one record per line.	☑	☐

D Exam-style questions

33 marks

1 Define the term array. [2]

> A data structure that stores a fixed number of elements/values (1) / of the same data type under one identifier, each accessed by an index (1).

2 Explain the difference between a static and a dynamic data structure, giving a C# example of each. [4]

> Static: size is fixed when the structure is created and cannot change while the program runs (1) / e.g. int[] marks = new int[30]; (1) / Dynamic: size can grow or shrink at run time as items are added or removed (1) / e.g. List<string> names = new List<string>(); (1).

3 A youth theatre in Derry~Londonderry has 8 rows of 12 seats. Write a C# declaration of a suitable data structure to record whether each seat is booked, and a statement that marks the third seat in the first row as booked. [3]

> bool[,] seats = new bool[8, 12]; (accept int[,] or char[,] with explanation) - correct type and 2D syntax (1) / correct dimensions 8 and 12 (1) / seats[0, 2] = true; (1).

4 A sports centre in Newry stores each customer's name, age, membership number and whether they have a family pass. Explain why a record is more suitable than an array for one customer's details. [2]

> The details are of different data types (string, int, int/string, bool) (1) / an array can only store one data type, whereas a record groups related fields of different types for one customer (1).

5 State what is output by: int[] n = { 6, 2, 9, 4 }; Console.WriteLine(n[n.Length - 2] + n[0]); [1]

> n.Length - 2 = 2, n[2] = 9; n[0] = 6; output 15 (1).

6 A hospital ward stores hourly temperature readings for a patient during one day. The number of readings is always 24. State, with a reason, whether an array or a list is more suitable. [2]

> Array (1) / because the number of readings is known in advance and fixed at 24, so a static structure is suitable / no need to add or remove items (1).

7 Write C# code that outputs the total of every value in a 2D integer array called sales with 4 rows and 7 columns, without using the numbers 4 and 7 in the loops. [4]

> int total = 0; (1) / outer loop for (int r = 0; r < sales.GetLength(0); r++) (1) / inner loop for (int c = 0; c < sales.GetLength(1); c++) (1) / total += sales[r, c]; and Console.WriteLine(total) after the loops (1). Accept foreach (int v in sales) total += v; for 3 marks maximum if total initialised and output.

8 Describe the steps a program must take to read every record from a comma-separated text file of pupil names and marks and calculate the average mark. [4]

> Any four from: check the file exists / open the file for reading (1) / set a total and a counter to 0 (1) / loop until the end of the file, reading one line at a time (1) / split each line at the comma into fields (1) / convert the mark field to an integer and add it to the total, incrementing the counter (1) / close the file (1) / divide total by counter (as real) to output the average, checking the counter is not 0 (1).

9 Explain the difference between opening a text file for writing and opening it for appending, and state one situation in which each would be used. [3]

> Writing creates a new file or overwrites/replaces any existing contents (1) / appending adds new data to the end of the file, keeping existing contents (1) / suitable situations, one each, e.g. writing to save a complete updated member list; appending to add a new entry to a log or a new booking (1).

10 A rugby club in Enniskillen wants a program to manage its membership: members join and leave throughout the season, each member has several items of data of different types, and the data must be available every time the program is run. Discuss the data structures and file handling the programmer should use, justifying each choice. [8]

> Indicative content: a list for the members because membership changes during the season, so the structure must grow and shrink (Add/Remove) and an array would need a fixed maximum size and waste memory; a record (struct/class) for each member because name, age, membership number, fees paid etc. are related but of different data types, keeping one member's data together and avoiding parallel arrays; a List<Member> behaves like a table; a text (CSV) file for persistent storage because RAM is cleared when the program closes; one record per line with comma-delimited fields; load at start by reading lines, splitting and converting types; save on exit by overwriting the file with the updated list, or append when a new member joins; close files; check the file exists or use exception handling. Levels: 7-8 marks: detailed, well-justified choices for list, record and file handling linked to the club with accurate technical terms; 4-6 marks: appropriate choices with some justification, may omit one area; 1-3 marks: structures named with little or no justification.

E Quick quiz

Circle one answer · 9 marks

1 An array is declared as int[] a = new int[10]. What is the index of its last element? [1]

A 10 — **B 9 ✓**

C 1 — D 11

2 A café takes a varying number of online orders each day. Which structure best stores the orders while the program runs? [1]

A A list, because it can grow as orders arrive ✓ — B A 1D array of fixed size

C A constant — D A 2D array

3 What does grid.GetLength(1) return for int[,] grid = new int[5, 8]? [1]

A 5 — B 40

C 8 ✓ D 13

4 What is the value of "Ben,65".Split(',')[1]? [1]

A "Ben" B 65 as an int

C "," **D "65" as a string ✓**

5 Why must a file be closed after writing to it? [1]

A To delete the temporary copy **B So buffered data is written to disk and the file is released ✓**

C To convert it to binary D It does not need to be closed

6 Which statement best describes a record? [1]

A A fixed set of related fields that may have different data types ✓ B A list of values that must all be the same type

C A file stored on secondary storage D A loop that processes every element

7 A program needs to add a new line to an existing log file without losing earlier entries. Which should it use? [1]

A new StreamReader("log.txt") B new StreamWriter("log.txt", false)

C File.ReadAllLines("log.txt") **D new StreamWriter("log.txt", true) ✓**

8 int[] r = { 4, 0, 12, 7 }; What is the value of r[r.Length - 1] - r[1]? [1]

A 4 B 12

C 7 ✓ D -4

9 Which loop header correctly visits every element of an array called data? [1]

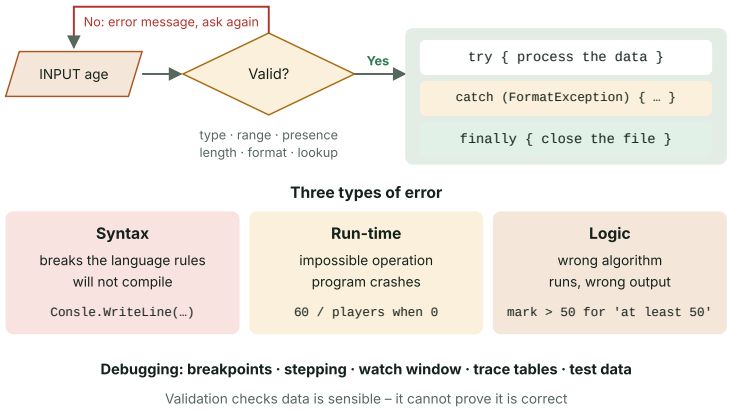
A for (int i = 1; i <= data.Length; i++) **B for (int i = 0; i < data.Length; i++) ✓**

C for (int i = 0; i <= data.Length; i++) D for (int i = 1; i < data.Length; i++)

Robust programs: validation, exceptions, errors, testing and debugging

AP23 ANSWERS

I can: Write validation loops in C# using presence, length, range, type, format and lookup checks · Use try...catch...finally to handle exceptions so a program does not crash · Classify programming errors and design a test plan with suitable test data · Use IDE tools such as breakpoints, stepping and watches to locate logic errors · Explain when each kind of error is detected when a program is compiled or interpreted



Use this diagram to help you: A robust input process: validation loop rejects bad data, try and catch handle run-time errors, with the three error types summarised

A Key terms

Write the letter of the matching definition · 6 marks

B Robust program

F Validation

E Verification

D Exception

C Syntax error

A Semantic error

A A statement that follows the syntax rules but has no valid meaning, such as assigning text to an integer variable.

B A program that copes with invalid input and unexpected events without crashing or producing incorrect results.

C An error that breaks the grammatical rules of the programming language, so the program cannot be compiled or run.

D An object raised when a run-time error occurs, which stops the program unless it is handled in a try...catch structure.

E A check that data has been entered or transferred accurately, for example by double entry or a visual check.

F An automatic check that input data is reasonable and follows set rules before it is processed.

B Fill the gaps

Use the word bank · 10 marks

catch · syntax · verification · extreme · try · interpreter · breakpoint · logic · range · validation

Checking automatically that input data is sensible is called **validation**, and a **range** check makes sure a value lies between limits. Typing a password twice so the two copies can be compared is

a form of **verification** . Code that might cause a run-time error is placed in a **try** block, and the error is handled in a **catch** block. A program that runs but gives the wrong answer contains a **logic** error, while a missing semicolon is a **syntax** error. Testing with values at the very edge of the allowed range uses **extreme** data. A **breakpoint** pauses the program at a chosen line, and an **interpreter** translates and runs one statement at a time.

C True or false?

Tick one box · 9 marks

	True	False
Validation guarantees that the data entered is correct. *(Validation only checks data is reasonable and follows rules; a valid value can still be wrong.)*	☐	☑
A syntax error stops the program from being compiled.	☑	☐
Dividing an integer by zero is an example of a logic error. *(It is a run-time (execution) error: the program crashes while it runs.)*	☐	☑
A finally block runs whether or not an exception occurs.	☑	☐
int.TryParse crashes the program if the text is not a whole number. *(TryParse returns false instead; it is int.Parse that throws a FormatException.)*	☐	☑
For a value allowed from 1 to 24, the numbers 1 and 24 are extreme (boundary) test data.	☑	☐
When a C# program is compiled, a missing semicolon is reported before any of the program runs.	☑	☐
An interpreter detects a syntax error on line 50 before it runs line 1. *(An interpreter only finds the error when it reaches line 50; lines 1 to 49 will already have run.)*	☐	☑
Logic errors are usually reported by the compiler with a line number. *(Logic errors produce no error message; the program runs but gives wrong results, so they are found by testing.)*	☐	☑

D Exam-style questions

38 marks

1 Describe the difference between validation and verification. [2]

> Validation checks that data is reasonable / sensible / follows rules (e.g. within a range) (1) / verification checks that data has been entered or copied accurately, matching the source (e.g. double entry) (1).

2 A GP surgery in Ballymena is building an online booking form. State a suitable validation check for each of: (a) the patient's age (b) a 10-digit Health and Care Number (c) the appointment type, which must be 'Doctor', 'Nurse' or 'Phone'. [3]

> (a) Range check, e.g. 0 to 120 (accept type check) (1) / (b) Length check of exactly 10 characters (accept type check for digits only / format check) (1) / (c) Lookup check against the list of allowed values (1).

3 Identify the type of error in each case: (a) int x = 10 / count; when count is 0 at run time (b) Console.WriteLine("Hi") with no semicolon (c) average = (a + b) * 2; when the average of a and b was required. [3]

> (a) Run-time (execution) error (1) / (b) Syntax error (1) / (c) Logic error (1).

4 A bike-hire scheme in Belfast lets customers hire a bike for between 1 and 24 whole hours. Give one item of valid, extreme and invalid test data for the number of hours, stating the expected outcome of each test. [4]

> Valid: e.g. 5 - accepted and processed (1) / Extreme: 1 or 24 - accepted (1) / Invalid: e.g. 0, 25, -3 or 'two' - rejected with an error message and asked again (1) / correct expected outcomes stated for all three (1).

5 Explain how a breakpoint and a watch window can be used together to find a logic error. [3]

> A breakpoint is set at/before the line where the error is suspected so that the program pauses there (1) / a watch is set on the relevant variables to display their current values (1) / the programmer steps through line by line, comparing actual values with expected values to find where they first go wrong (1).

6 Write C# code that asks a user for the number of tickets and repeats until a whole number from 1 to 6 is entered. [4]

> Loop structure that repeats while the input is invalid, e.g. do...while (1) / prompt and read input inside the loop (1) / type check, e.g. int.TryParse, or try/catch around int.Parse (1) / range check >= 1 && <= 6 with an error message (1).

7 Explain the purpose of the try, catch and finally blocks in exception handling. [3]

> try: contains the code that might cause a run-time error/exception (1) / catch: runs only if an exception of that type occurs, handling it, e.g. displaying a message, so the program does not crash (1) / finally: always runs whether or not an exception occurred, e.g. to close a file (1).

8 A Newry taxi firm's booking program crashes with an IndexOutOfRangeException inside a subroutine, GetFare, which is called from three different places. Describe how two debugging tools in an IDE could help the programmer find the cause. [4]

> Any two tools, each identified (1) and described in context (1): call stack (1) shows which of the three calls led to GetFare when the exception occurred (1) / breakpoint (1) pauses execution at the start of GetFare or the failing line so its state can be inspected (1) / watch or locals window (1) shows the index value compared with the array's Length, revealing it is out of range (1) / stepping (1) runs GetFare one line at a time to see where the index goes wrong (1).

9 A pupil's program has a missing semicolon on line 40 and a division by zero that happens on line 12 for some inputs. Explain when each error would be detected if the program were compiled, and how this would differ if it were run by an interpreter. [4]

> Compiled: the missing semicolon (syntax error) is reported by the compiler before the program runs, and no executable is produced until it is fixed (1) / the division by zero is a run-time error, detected only when line 12 executes with those inputs (1). Interpreted: lines 1–39 would run before the syntax error on line 40 is found (1) / so the division by zero on line 12 could happen first / data may already have been changed before the program stops (1).

10 A health trust in Northern Ireland is developing a program to record patients' blood pressure and temperature on hospital wards. Evaluate the techniques the developers should use to make the program robust and as free from errors as possible before it is used on the wards. [8]

> Indicative content: validation on every input - range checks on temperature and blood pressure, type checks with TryParse, presence checks on patient ID, format/length checks on Health and Care Number, lookup checks on ward names; verification - double entry or visual confirmation before saving, important because a valid but wrong reading could harm a patient; exception handling with try/catch for conversion errors, missing data files or network failures so the program does not crash and lose data, finally to close files; systematic test plan with valid, extreme (limits of each range) and invalid/null data and recorded expected/actual results; regression testing after fixes; use of IDE debugging tools (breakpoints, stepping, watches) to locate logic errors; evaluation of limitations - validation cannot guarantee correctness, testing cannot prove the absence of all errors, need for user testing with nurses. Levels: 7-8 marks: wide range of techniques, each justified in the ward context with clear evaluation of strengths and limitations; 4-6 marks: several relevant techniques with some justification; 1-3 marks: limited list of techniques with little context.

E Quick quiz

`Circle one answer · 10 marks`

1 An online form rejects a mark of 120 when marks must be out of 100. Which validation check has been applied? [1]

A Presence check
B Format check
C Range check ✓
D Length check

2 A program outputs the wrong total but never crashes. What type of error does it contain? [1]

A Logic error ✓
B Syntax error
C Run-time error
D Compilation error

3 Which exception does int.Parse("ten") throw? [1]

A DivideByZeroException
B FormatException ✓
C IndexOutOfRangeException
D FileNotFoundException

4 Which IDE feature shows how the values of variables change as a program runs? [1]

A Syntax highlighting
B Error list
C Code completion
D Watch window ✓

5 Why is a do...while loop often used for input validation? [1]

A It runs faster than a for loop
B It may run zero times
C The input must be requested at least once before it is checked ✓
D It catches exceptions

6 Valid ages are 0 to 120. Which value is extreme (boundary) test data? [1]

A 120 ✓
B 45
C -5
D abc

7 A program is run by an interpreter and line 30 contains a syntax error. What happens? [1]

A Nothing runs until the error is fixed
B The interpreter skips line 30 and carries on
C The error is reported as a logic error
D Lines 1 to 29 run, then execution stops at line 30 ✓

8 A programmer writes int total = "twelve"; Which type of error is this? [1]

A Logic error
B Semantic (type) error ✓
C Run-time error
D Boundary error

9 Which describes verification by double entry? [1]

A Checking a value is in a list
B Entering data twice and comparing the two copies ✓
C Checking data has the right number of characters
D Running the program twice

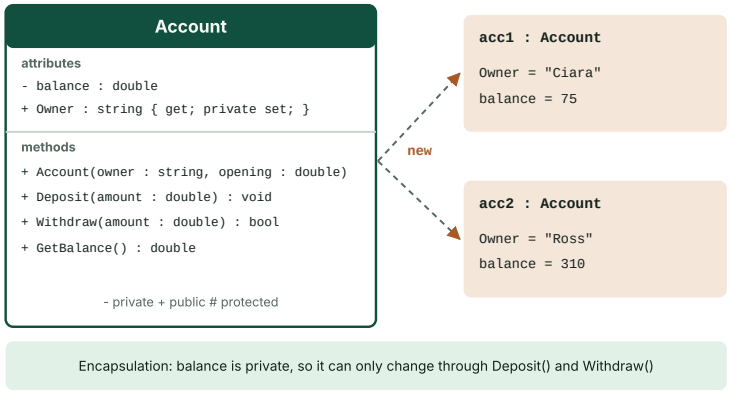
10 What does the C# compiler produce from source code? [1]

A Common Intermediate Language, later compiled to machine code by the JIT compiler ✓
B Assembly language mnemonics
C Python bytecode
D Nothing; C# is interpreted line by line

Classes, objects and encapsulation

AP24 ANSWERS

I can: Define class, object, attribute, method and constructor, and explain how a class relates to its objects · Explain encapsulation and decide which attributes need get methods, set methods or both · Write a C# class with private attributes, a constructor and methods, then instantiate and use objects · Trace object-oriented code to predict its output · Draw and interpret a class diagram using +, - and # visibility symbols



Use this diagram to help you: Class diagram of an Account class with private attribute and public methods, and two objects instantiated from it with their own state

A Key terms

Write the letter of the matching definition · 6 marks

B Class

F Object

D Instantiation

E Attribute

C Method

A Constructor

A A special method with the same name as the class and no return type that runs when an object is created to initialise its attributes.

B A template or blueprint that defines the attributes and methods (behaviours) shared by all objects of one type.

C A subroutine defined in a class that describes an action an object can perform or that can be performed on it.

D Creating a new object from a class, in C# using the new keyword, which calls the constructor.

E A data item (field) declared in a class whose values describe the state of each object.

F A specific instance of a class, with its own values for the class's attributes.

B Fill the gaps

Use the word bank · 9 marks

object | attributes | methods | encapsulation | parameter list | constructor | instantiation | private | class

A **class** is a template that defines the attributes and methods of a type of thing. An **object** is a specific instance of a class, created by **instantiation** using the new keyword. The data an object holds is stored in its **attributes**, and its behaviours are defined by its **methods**. A **constructor** runs automatically to give a new object its starting values. Keeping attributes

private and allowing access only through get and set methods is called **encapsulation**. The extra information a method needs in order to work is given in its **parameter list**.

C True or false?

Tick one box · 9 marks

	True	False
By convention, a C# class name starts with a capital letter.	☑	☐
An object is a template from which classes are made. *(It is the other way round: a class is the template and an object is a specific instance of it.)*	☐	☑
A constructor must have a return type of void. *(A constructor has the same name as the class and no return type at all.)*	☐	☑
Private attributes can be changed directly by code outside the class. *(Private attributes can only be accessed inside the class; other code must use its public methods or properties.)*	☐	☑
A set method can check that a new value is valid before storing it.	☑	☐
A method that needs no extra information is written with an empty pair of brackets, such as GetName().	☑	☐
If an attribute has a get method but no set method, other classes cannot read its value. *(They can read it through the getter; what they cannot do is change it.)*	☐	☑
In a class diagram, # shows a public member. *(# shows protected; + shows public and - shows private.)*	☐	☑
A static field is shared by all objects of a class.	☑	☐

D Exam-style questions

37 marks

1 State the two things that a class definition specifies for each of its objects. [2]

> The attributes / properties (data) of the objects (1) / the methods / behaviours that can be performed on the objects (1)

2 A school library in Dungannon is building a loans system in C#. Using examples from this system, explain the relationship between a class and an object. [3]

> A class is a template / blueprint defining attributes and methods (1) / an object is a specific instance / member of the class with its own attribute values (1) / suitable example, e.g. class Book; objects are individual books such as one copy with accession number A0041 (1)

3 State the purpose of a constructor and describe two ways in which a constructor differs from other methods in C#. [3]

> Purpose: to initialise a new object / set the starting values of its attributes when it is created (1) / Any two from: it has the same name as the class (1) / it has no return type, not even void (1) / it runs automatically when new is used (1)

4 A Lisburn credit union's Account class keeps each customer's balance. Explain how the class uses encapsulation to protect the balance. [3]

> The balance attribute and the methods that act on it (e.g. Deposit, Withdraw, GetBalance) are combined in one unit, the Account class (1) / balance is declared private, so code outside the class cannot read or change it directly (1) / it can only be read or changed through public methods such as GetBalance and Withdraw, which can validate changes, e.g. refusing a withdrawal larger than the balance (1)

5 A leisure centre in Omagh stores member details in a Member class. Each member's membership number must never change once it has been allocated. Explain how encapsulation can be used to make sure of this. [3]

> Make the membershipNumber attribute private so it cannot be changed directly (1) / set its value once in the constructor when the Member object is created (1) / provide a get method so it can be read, but no set method, so it cannot be changed (1)

6 A Customer class contains the method header public void SetPhone(string newPhone). Describe what the parameter list and the return type tell a programmer about this method. [2]

> Parameter list: the method needs one item of data, a string, to be supplied when it is called – the new phone number (1) / return type void: the method carries out an action (changes the phone attribute) and returns no value (1)

7 A class Counter has a private integer attribute total, set to 0 by its constructor. Its method Add(int n) adds n to total only if n is greater than zero, and GetTotal() returns total. An object c is created, then c.Add(5); c.Add(-2); c.Add(3); are executed. State the value returned by c.GetTotal() and explain your answer. [2]

> 8 (1) / Add(5) makes total 5, Add(-2) is ignored because -2 is not greater than zero, Add(3) makes total 8 (1)

8 Draw a class diagram for a Locker class used by a school. It stores a locker number (integer, which must not change) and the name of the pupil using it (string). It has a constructor, a get method for each attribute, a set method for the pupil's name, and a method IsFree that returns true if no pupil is assigned. [5]

> Class name Locker in top compartment (1) / - lockerNumber : int and - pupilName : string with minus signs (1) / + Locker(number : int, name : string) constructor with no return type (1) / + GetLockerNumber() : int, + GetPupilName() : string, + SetPupilName(newName : string) : void (1) / + IsFree() : bool (1). No set method for lockerNumber.

9 A hospital ward system uses a Patient class with private attributes hospitalNumber (string), name (string) and ward (string). Write C# code for the class, including a constructor, a get method for each attribute, a set method for ward only, and a method PrintDetails that displays all three attributes. Then write statements that create one Patient object and move the patient to a different ward. [8]

> class Patient with three private string attributes (1) / constructor public Patient(string num, string newName, string newWard) (1) assigning all three attributes (1) / three get methods returning the correct attribute with string return type (1) / SetWard(string newWard) with void return type assigning ward; no setter for hospitalNumber or name (1) / PrintDetails using Console.WriteLine to output all three attributes (1) / object created with new and suitable arguments, e.g. Patient p = new Patient("H1234567", "Seán Burns", "Ward 4"); (1) / p.SetWard("Ward 7"); (1)

10 A GP surgery is developing an appointment booking system in C#. One developer suggests making all attributes of the Appointment class public 'to save time'. Discuss whether the attributes should instead be private and accessed through methods or properties. [6]

> Indicative content – for private: encapsulation protects data from accidental or unauthorised change / setters can validate, e.g. reject an appointment time outside surgery hours or a negative duration / identifiers such as appointment number can be made read-only / side effects such as sending a confirmation text can be built into a setter / internal representation can change without breaking other code / easier debugging because only the class can change its data / supports team working through a clear interface. For public: quicker to write initially / less code / slight performance gain (negligible). Conclusion: private with methods/properties is better for a system holding patient data; C# auto-properties remove most of the extra effort. Levels: 5–6 well-argued, both sides, applied to the surgery, justified conclusion; 3–4 some points developed with limited context; 1–2 isolated points.

E Quick quiz

Circle one answer · 8 marks

1 In object-oriented programming, what is the relationship between a class and an object? [1]

A An object is a template from which classes are built

B A class is a template, and an object is a specific instance created from it ✓

C A class and an object are two names for the same thing

D A class stores data while an object stores only methods

2 A Patient class has a private attribute hospitalNumber with a get method but no set method. What is the effect? [1]

A Other code can never see the hospital number

B The program will not compile without a setter

C Other code can read the hospital number but cannot change it ✓

D Any code can change the hospital number directly

3 Which keyword is used to create an object in C#? [1]

A class

B static

C void

D new ✓

4 In a class diagram, what does '- battery : int' show? [1]

A A private attribute called battery of type integer ✓

B A public method that returns an integer

C A protected attribute

D A constructor parameter

5 Which statement correctly describes a constructor in C#? [1]

A A method that must return void

B A method with the same name as the class and no return type, which initialises a new object ✓

C Any method that changes the value of an attribute

D A method that deletes an object when it is no longer needed

6 What do the empty brackets in GetBattery() show? [1]

A The method is private

B The method returns nothing

C The method needs no parameters ✓

D The method is a constructor

7 Which statement best describes encapsulation? [1]

A Bundling data with the methods that act on it and hiding the data so it is only accessed through those methods ✓

B Creating a new class from an existing one

C Giving several methods the same name

D Storing all data in public variables so it is easy to reach

8 Two Bike objects, b1 and b2, are created from the same class. Which statement is true? [1]

A They share one battery attribute

B Changing b1's dock also changes b2's dock

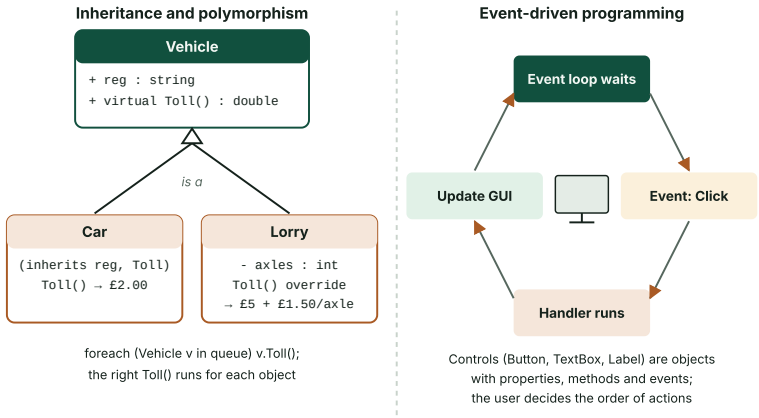
C Only the first object created can call Ride()

D Each object has its own values for the instance attributes ✓

Inheritance, polymorphism, evaluating OOP and event-driven programming

AP25 ANSWERS

I can: Explain inheritance and work out which attributes and methods a subclass object has · Write a C# subclass that calls the base constructor, adds members and overrides a virtual method · Explain how polymorphism lets one method call behave differently for different objects · Evaluate the usefulness of object-oriented programming for a given application · Describe how event-driven GUI programs respond to events using handlers and controls



Use this diagram to help you: Inheritance hierarchy with Car and Lorry deriving from Vehicle and Lorry overriding Toll, beside an event-driven cycle of event loop, event, handler and update

A Key terms

Write the letter of the matching definition · 6 marks

D Inheritance

F Base class

C Derived class

E Protected

B Abstract class

A Polymorphism

A The ability of objects of different classes to respond to the same method call in different ways.

B A class that cannot be instantiated; objects must be created from its subclasses, which must implement any abstract methods it declares.

C A more specialised class that inherits from a base class.

D The mechanism by which a new, more specialised class (subclass) takes on the attributes and methods of an existing, more general class (superclass) and can add or modify its own.

E An access modifier that makes a member visible inside its own class and any subclasses but hidden from other code; shown as # in a class diagram.

F A general class whose attributes and methods are inherited by other classes.

B Fill the gaps

Use the word bank · 10 marks

derived class | polymorphism | abstract | virtual | base class | handler | control | override | protected | inheritance

When a subclass takes on the attributes and methods of a superclass, this is called **inheritance**. The general class is also called the **base class** and the specialised class the **derived class**. A member

marked **protected** can be used by subclasses but is hidden from other code. A class that cannot be instantiated is an **abstract** class. When the same method call behaves differently for different classes, this is **polymorphism**. In C#, a base-class method that may be replaced is marked **virtual** and the replacement is marked **override**. In an event-driven program, a subroutine that runs when an event occurs on a **control** is an event **handler**.

C True or false?

Tick one box · 9 marks

	True	False
A subclass can add its own attributes and methods as well as using inherited ones.	☑	☐
A method defined only in one subclass can also be used by objects of a sibling subclass. *(Sibling subclasses are independent; only members of the shared superclass are available to both.)*	☐	☑
Overloading means a subclass replaces an inherited method. *(That is overriding; overloading is several methods with the same name but different parameter lists.)*	☐	☑
An abstract class can be instantiated with new as long as it has a constructor. *(An abstract class can never be instantiated; objects must be created from a non-abstract subclass.)*	☐	☑
A protected member can be accessed by subclasses.	☑	☐
Only new or overridden methods in a subclass need fresh testing if the superclass has already been tested.	☑	☐
In an event-driven program, the order of execution depends on the events that occur.	☑	☐
OOP always produces smaller and faster programs than procedural code. *(OOP often adds overhead; its benefits are realistic modelling, reuse and maintainability.)*	☐	☑
A subclass must always add at least one new attribute. *(It usually does, but it is not essential; it may only add or override methods.)*	☐	☑

D Exam-style questions

36 marks

1 A Fermanagh rugby club's system has a Member class and a new Junior class built from it. Explain what is meant by inheritance, using this example. [2]

> A new, more specialised class (subclass / derived class, here Junior) takes on / inherits the attributes and methods of an existing, more general class (superclass / base class, here Member) (1) / and can add its own attributes and methods, e.g. a guardian's phone number, or modify inherited ones (1)

2 A council's pitch-booking system has a superclass Booking with the attributes bookingRef, bookingDate and customerName. The subclass PitchBooking adds the attributes pitchNumber and floodlights. State how many attributes a PitchBooking object has and explain why. [2]

> 5 (1) / it inherits all 3 attributes of Booking and adds 2 of its own (1)

3 Explain what is meant by polymorphism, using an example. [3]

> The same method call / message (1) / produces different behaviour depending on the class of the object it is called on (1) / example, e.g. Fee() returns £120 for an Adult object but £40 for a Junior object because Junior overrides it (1)

4 State the purpose of each of the C# keywords virtual, override and abstract when used on a method. [3]

> virtual: marks a base-class method that subclasses may override (1) / override: marks a subclass method that replaces the inherited version (1) / abstract: declares a method with no body that every (non-abstract) subclass must override (1)

5 Describe the difference between overriding and overloading. [2]

> Overriding: a subclass redefines an inherited method with the same name and parameters (1) / overloading: methods in the same class have the same name but different parameter lists (1)

6 A cinema in Enniskillen sells Standard and Premium seats. Write C# code for a class Seat with a private attribute seatNumber (string), a constructor, a get method and a virtual method Price() that returns 7.50. Then write a subclass Premium that adds a private bool attribute includesSnack, calls the base constructor, and overrides Price() to return 11.00 if includesSnack is true and 9.50 otherwise. [7]

> class Seat with private string seatNumber (1) / constructor assigning seatNumber (1) / GetSeatNumber() returning string (1) / public virtual double Price() returning 7.50 (1) / class Premium : Seat with private bool includesSnack (1) / constructor with : base(number) and assigning includesSnack (1) / public override double Price() returning 11.00 if includesSnack else 9.50 (1)

7 Explain why an abstract class might be used as the superclass in a class hierarchy. [2]

> The superclass represents a general idea and no object of it should exist on its own / every object must be one of the specific subclasses (1) / declaring it abstract prevents it being instantiated, and abstract methods force each subclass to implement them (1)

8 Describe how an event-driven program responds when a user clicks a 'Calculate' button on a form. [3]

> The click is detected as an event (1) / the event loop / system passes it to the event handler attached to that button's Click event (1) / the handler code runs, e.g. reads the text boxes, calculates and displays the result in a label, then the program waits for the next event (1)

9 Explain two reasons why less testing may be needed when a new subclass is added to an existing, well-tested class hierarchy. [4]

> The inherited attributes and methods of the superclass have already been tested and are reused unchanged (1) / so they do not need to be retested for the new subclass (1) / only the new methods and any overridden methods in the subclass need to be tested (1) / encapsulation means the subclass cannot accidentally change the superclass's private data, so existing behaviour is unaffected (1)

10 A sports club in Newry is choosing between an object-oriented and a procedural approach for a new membership and fixtures system. Assess how well an object-oriented design would suit this system, considering both its benefits and its drawbacks. [8]

> Indicative content – advantages: realistic modelling of members, teams, fixtures, venues as classes / inheritance, e.g. Member → Junior, Adult, Coach, reuses code / less testing as base classes already tested / encapsulation protects personal data such as contact details, with validation in setters / modular, easier for more than one developer and for later maintenance or extension (e.g. new membership type added as a subclass) / polymorphism, e.g. Fee() for each member type in one loop. Disadvantages: more design time and complexity for what may be a small system / developers need OOP skills / extra memory and processing overhead (usually insignificant) / poor initial class design costly to fix / some parts (e.g. producing a simple report) may be equally well done procedurally. Conclusion: likely suitable as the system contains clear real-world entities and may grow. Levels: 7–8 balanced, developed, applied to the club, justified conclusion; 4–6 several points with some development and context; 1–3 limited, list-like points.

E Quick quiz

`Circle one answer · 8 marks`

1 What is meant by inheritance in object-oriented programming? [1]

A Hiding an object's data so it can only be changed through methods

B A new, more specialised class taking on the attributes and methods of an existing class ✓

C Several methods in one class sharing the same name

D Creating an object from a class using new

2 A Smartwatch subclass inherits three attributes from a Device superclass and adds heartRateSensor and strapSize. How many attributes does a Smartwatch object have? [1]

A 2

B 3

C 5 ✓

D 6

3 Which statement shows Lorry inheriting from Vehicle in C#? [1]

A class Vehicle : Lorry

B class Lorry extends Vehicle

C class Lorry inherits Vehicle

D class Lorry : Vehicle ✓

4 Which keyword allows a base-class method to be overridden but does not force it? [1]

A virtual ✓

B static

C private

D abstract

5 What is polymorphism? [1]

A Creating many objects from one class

B Hiding data in a class

C The same method call behaving differently depending on the object's class ✓

D Having several constructors

6 A credit union system has an Account superclass, and every real account is either a SavingsAccount or a LoanAccount. Why might Account be declared abstract? [1]

A So that its attributes are public

B So that every object must be created as a specific type of account, such as SavingsAccount or LoanAccount ✓

C So that it runs faster

D So that subclasses cannot inherit its methods

7 Which is a disadvantage of the object-oriented approach? [1]

A Code reuse through inheritance

B More accurate modelling of real-world entities

C Easier maintenance

D It does not suit very abstract problems that have no obvious real-world objects ✓

8 Which is an example of an event? [1]

A A button being clicked ✓

B A variable being declared

C A class being written

D A constant being defined